

Canonical GHZ configurations - cluster P3 - helicity flipping

species: electron
polarization cluster: P3
source cluster configurations: 87
canonical configurations collected: 43

canonical display criterion: exactly 2 retained final-state components
retained-component cutoff: fraction $\geq 2.0\%$
normalized amplitude display: bar length = $|A_{\text{tilde}}|$, bar color/text = phase

helicity-mechanism subset: helicity flipping
classification: complete-chain R_flip/conserves > 1
per-page labels: isolated flip/conserves and T/L powers; coherent cross terms are reported separately

cluster retained-component count distribution:

- 2 components: 53 configurations
- 4 components: 8 configurations
- 5 components: 4 configurations
- 6 components: 19 configurations
- 7 components: 3 configurations

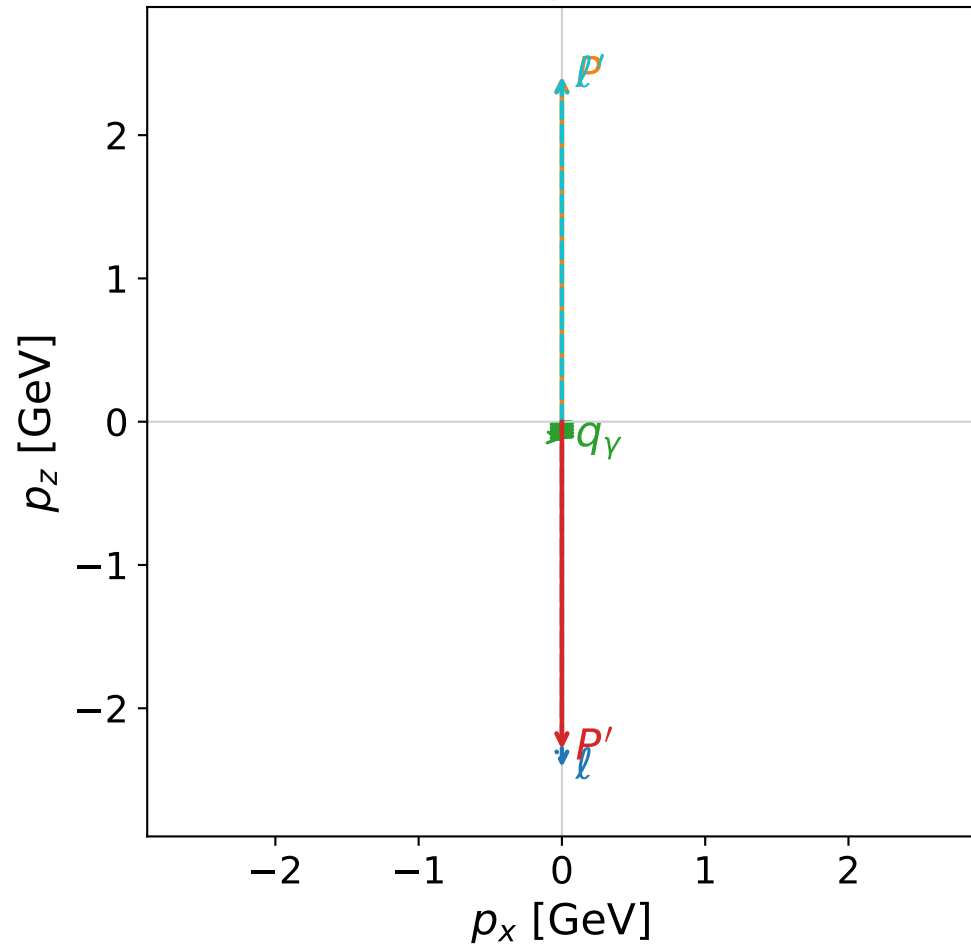
CSV index:

Output\GradientPhaseSpaceScan\electron\Data\GHZ\dghz\polarization_cluster_03\combined\
helicity_flipping\canonical_dghz_polarization_cluster_03_helicity_flipping_configurati
ons_electron.csv

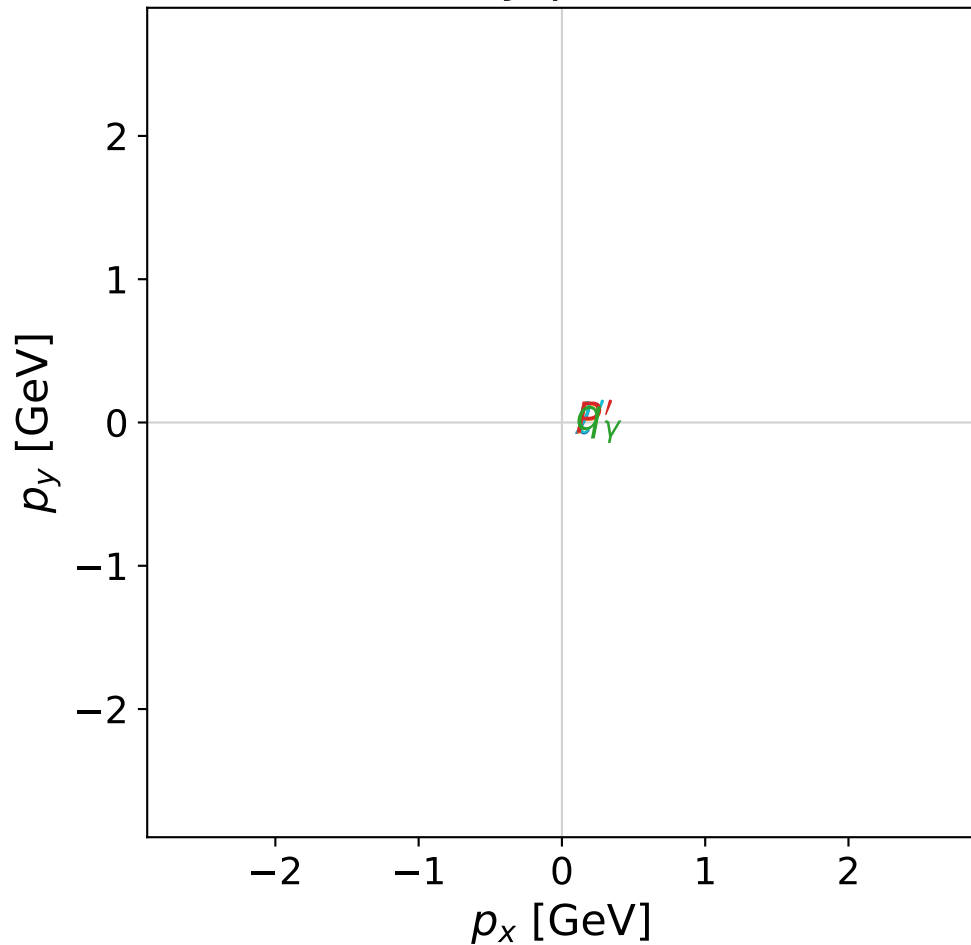
Each following page is one complete momentum, kinematic-summary, and normalized-amplitude configuration.

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



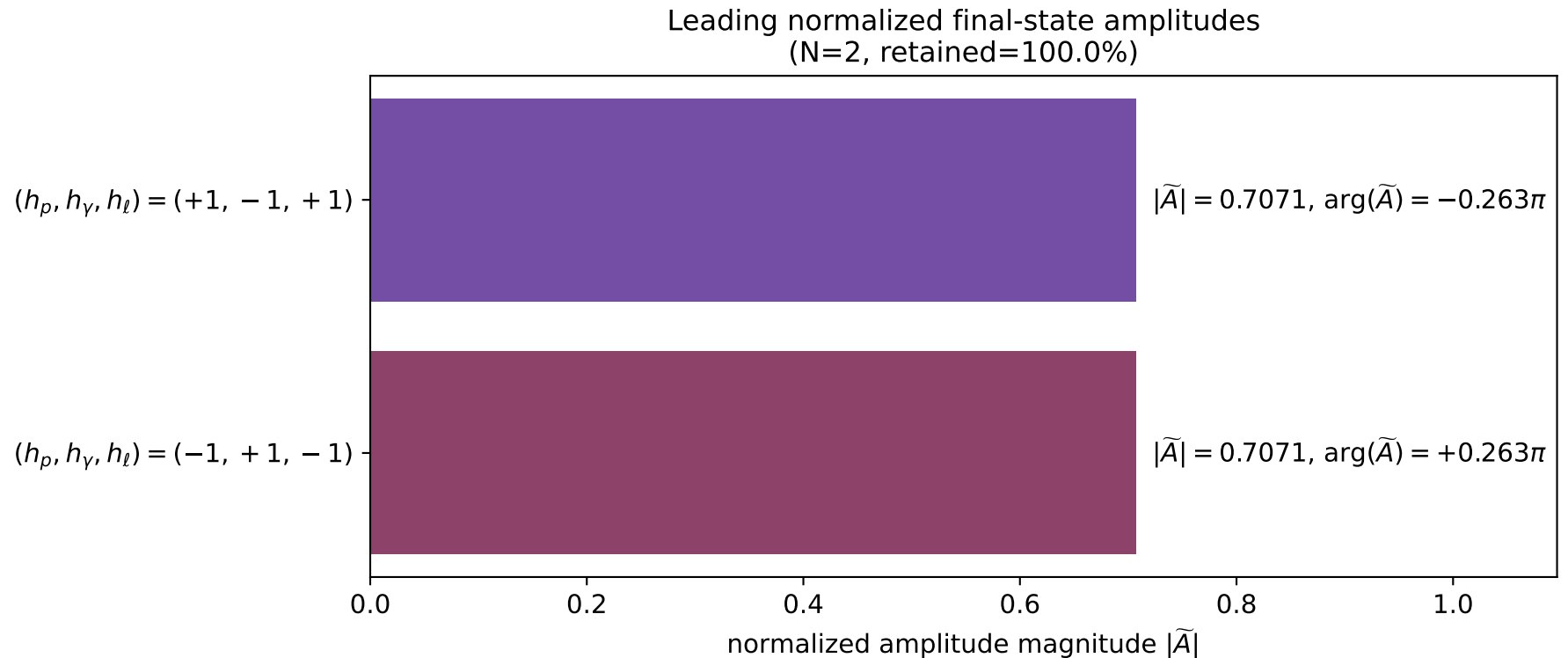
dghz_mixing_angles_polarization_03_local_minimum_1
 $d_{\{\rm GHZ\}}=2.28069\text{e-}08$
 region: polarization_cluster_03_local_minimum_1
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0001
 lepton mixing angle: $\alpha_e=2.4389177$ rad
 incoming lepton: $-0.763116|+\rangle +0.646261|-\rangle$
 proton mixing angle: $\alpha_p=0.70266188$ rad
 incoming proton: $0.763125|+\rangle +0.646251|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.28696$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=8.73394\text{e-}09$, $\phi_\gamma=2.31425$
 $E_\gamma=0.120598$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.18183$
 $Q^2=23.2283$, $x_B=0.998084$, $t=-22.0669$, $W^2=0.924438$, $y=0.964871$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

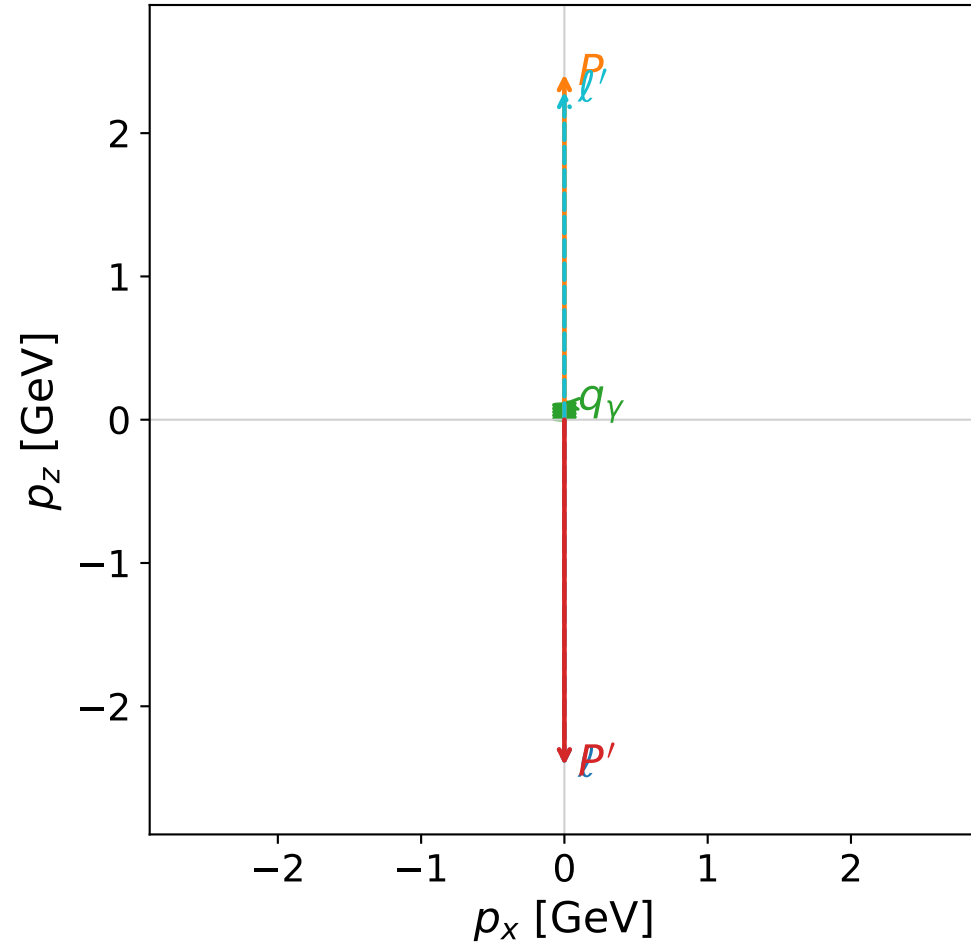
ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.4076$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.4076$
 P' $E= 2.4718$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2870$
 q_γ $E= 0.1206$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.1206$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=3.730\text{e+}21$; $I_h=-2.483\text{e-}12$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_{L/T}=5.239\text{e-}26$; $I_{TL}=+1.469\text{e-}17$

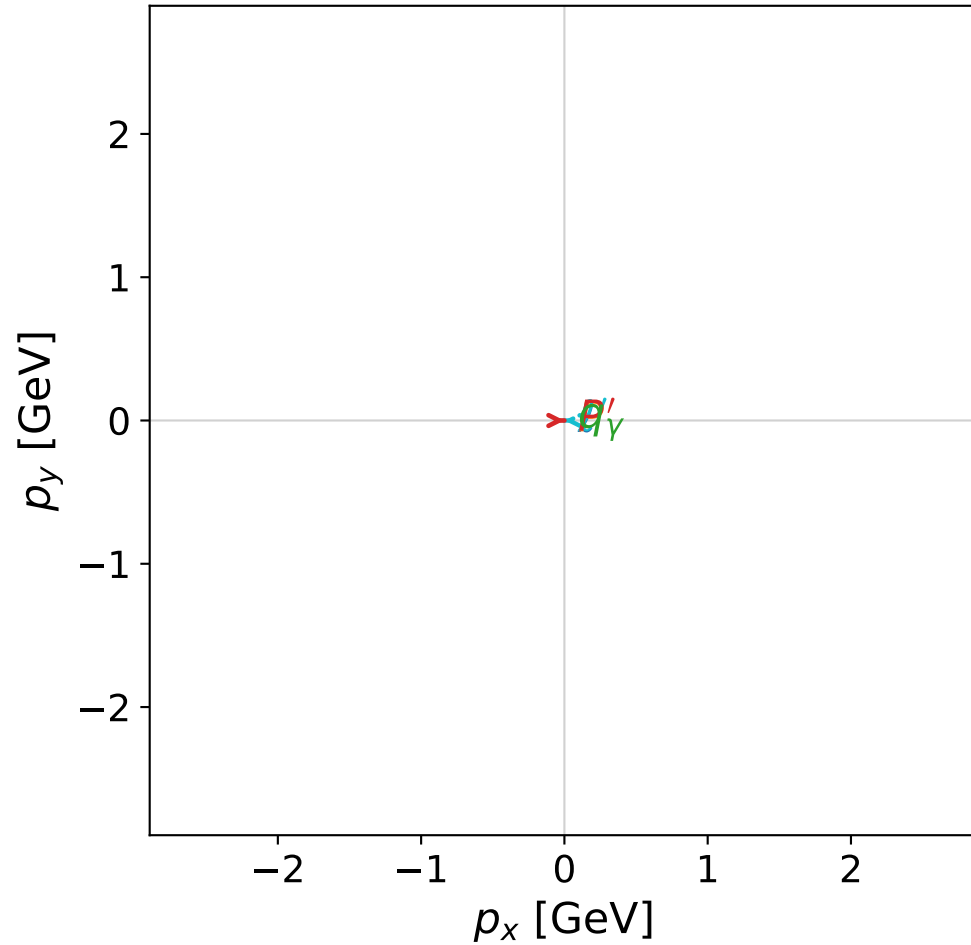


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



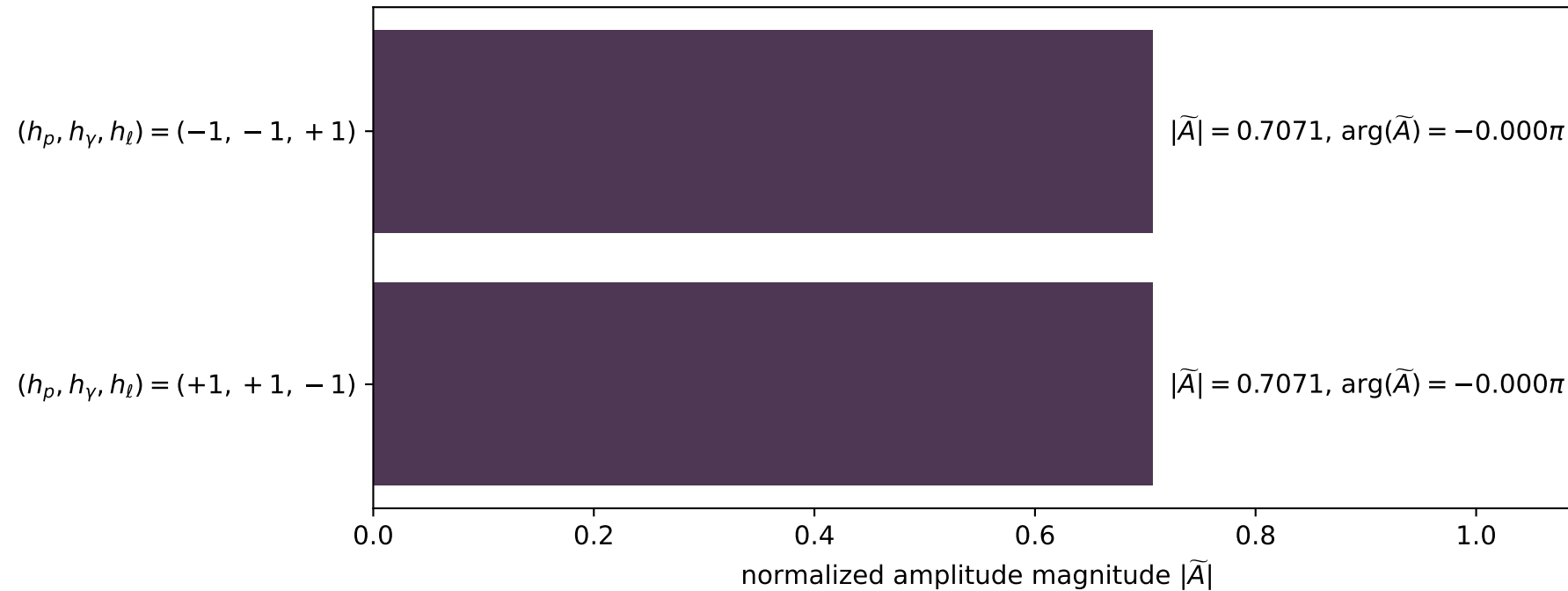
dghz_mixing_angles_polarization_03_local_minimum_5
 $d_{\text{GHZ}}=2.39007\text{e-}08$
 region: polarization_cluster_03_local_minimum_5
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0005
 lepton mixing angle: $\alpha_e=2.2809494$ rad
 incoming lepton: $-0.65195|+\rangle +0.758262|-\rangle$
 proton mixing angle: $\alpha_p=0.71016018$ rad
 incoming proton: $0.758257|+\rangle +0.651955|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.41202$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=6.28319$, $\phi_\gamma=0$
 $E_\gamma=0.120596$, $\theta_{l'}=0$, $\phi_{l'}=3.14159$
 $Q^2=22.1078$, $x_B=0.948273$, $t=-23.2713$, $W^2=2.0858$, $y=0.966566$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.2914$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.2914$
 P' $E= 2.5880$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.4120$
 q_γ $E= 0.1206$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1206$

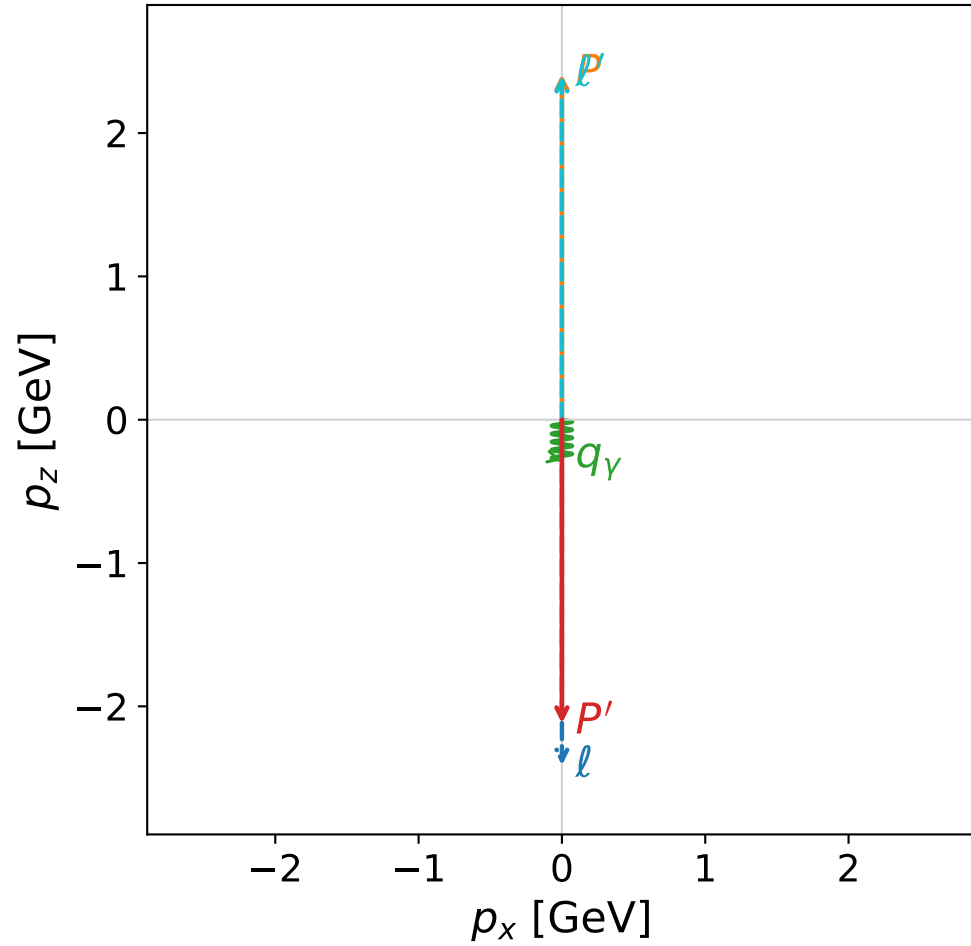
GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=3.932\text{e+}21$; $I_h=-2.607\text{e-}14$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=3.477\text{e-}28$; $I_{TL}=-2.605\text{e-}14$

Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

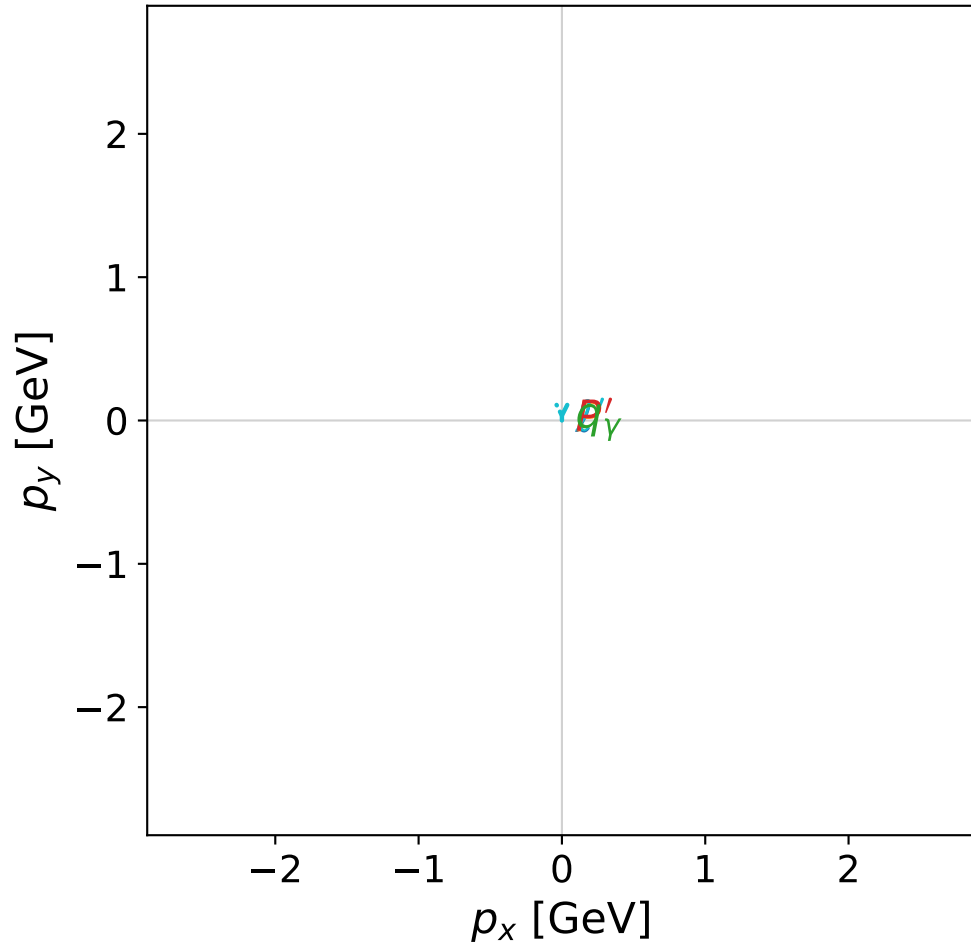


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

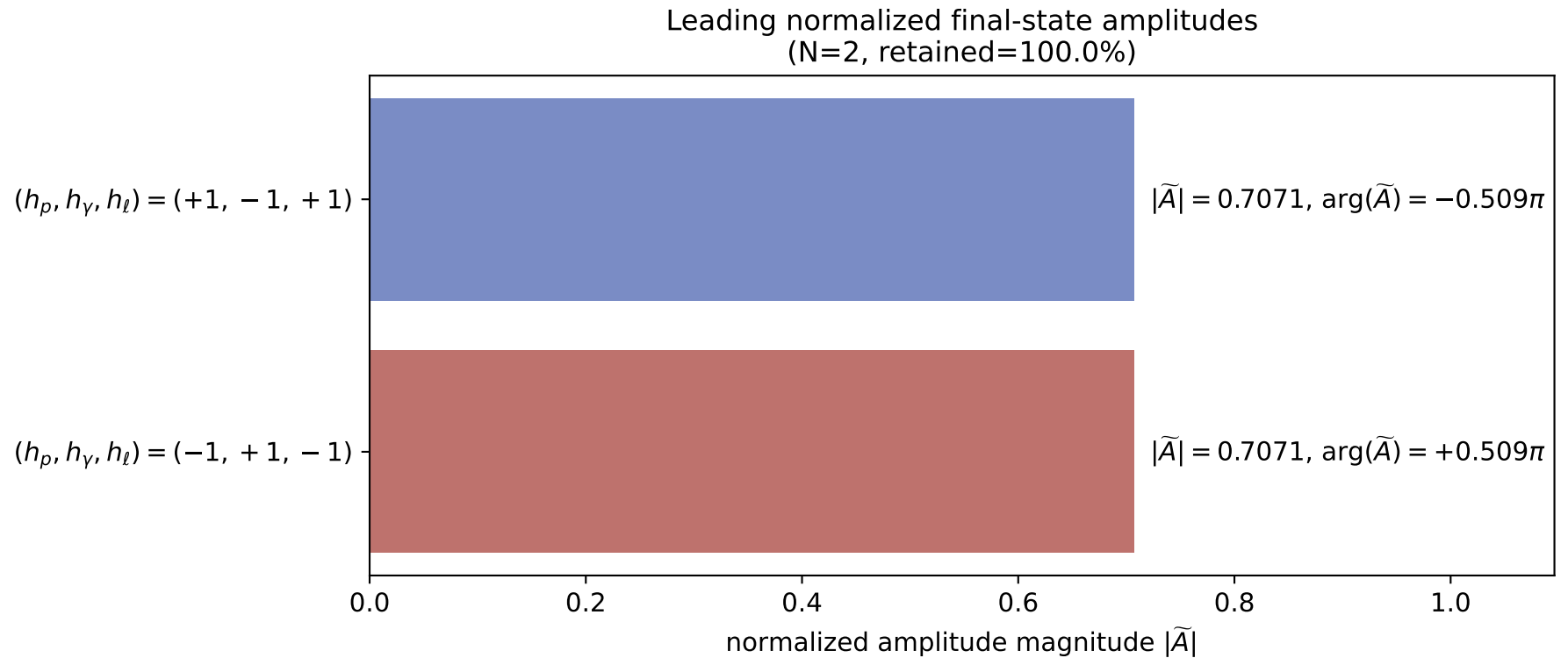


dghz_mixing_angles_polarization_03_local_minimum_12
 $d_{\{\rm GHZ\}}=2.46147\text{e-}08$
 region: polarization_cluster_03_local_minimum_12
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0012
 lepton mixing angle: $\alpha_e=2.1502889$ rad
 incoming lepton: $-0.547599|+\rangle +0.836741|-\rangle$
 proton mixing angle: $\alpha_p=0.99129539$ rad
 incoming proton: $0.547606|+\rangle +0.836736|-\rangle$

$s=24.9869$, $\sqrt{s}=4.99869$, $|\vec{P}|=2.41134$, $|\vec{P}'|=2.11911$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=1.71126$, $\phi_\gamma=1.54366$
 $E_\gamma=0.281071$, $\theta_{\ell'}=0$, $\phi_{\ell'}=4.83329$
 $Q^2=23.1506$, $x_B=0.995208$, $t=-20.4521$, $W^2=0.991326$, $y=0.964951$

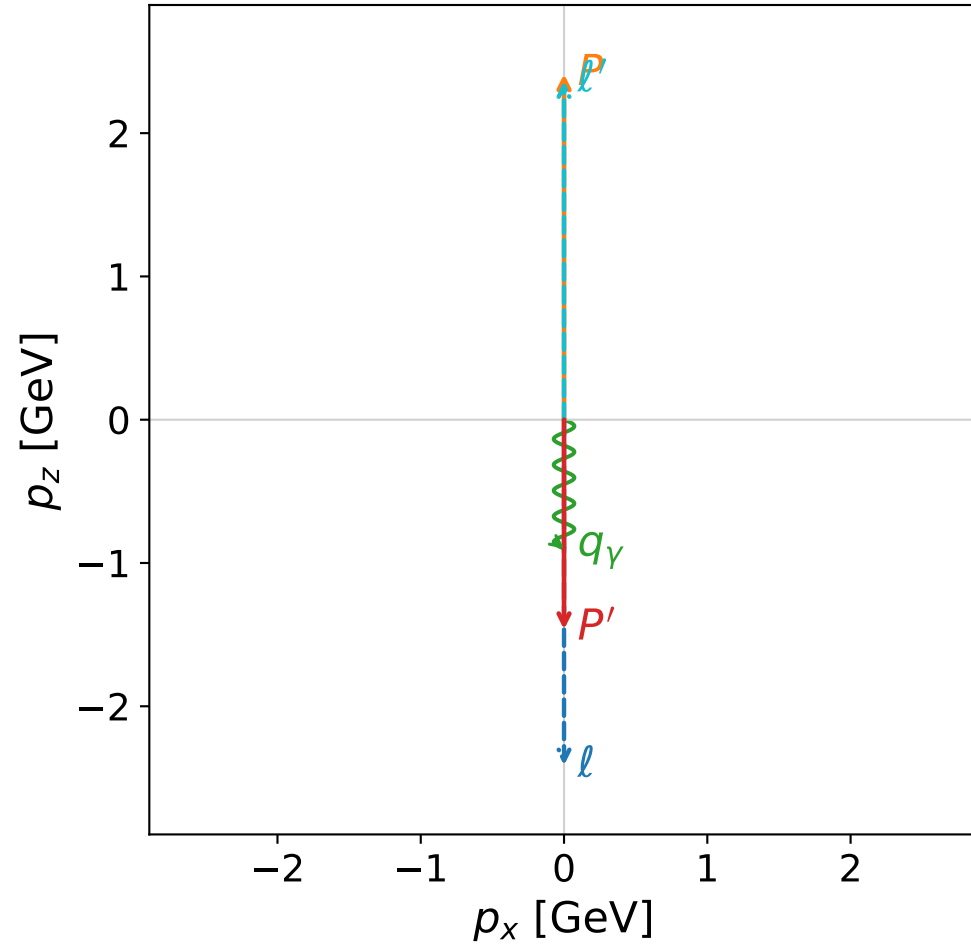
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4113$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4113$
 P $E= 2.5874$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4113$
 ℓ' $E= 2.4002$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.4002$
 P' $E= 2.3174$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.1191$
 q_γ $E= 0.2811$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2811$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.653\text{e}+22$; $I_h=-1.038\text{e-}13$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_{L/T}=1.245\text{e-}26$; $I_{TL}=+7.596\text{e-}19$

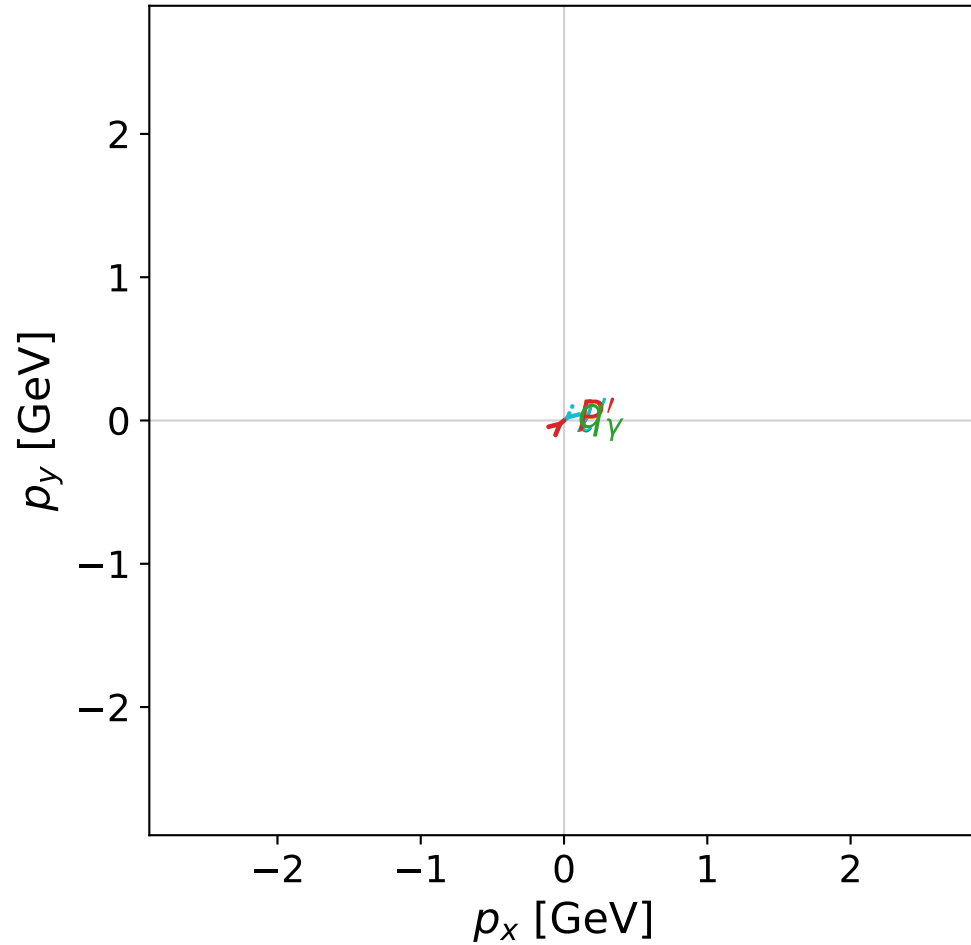


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

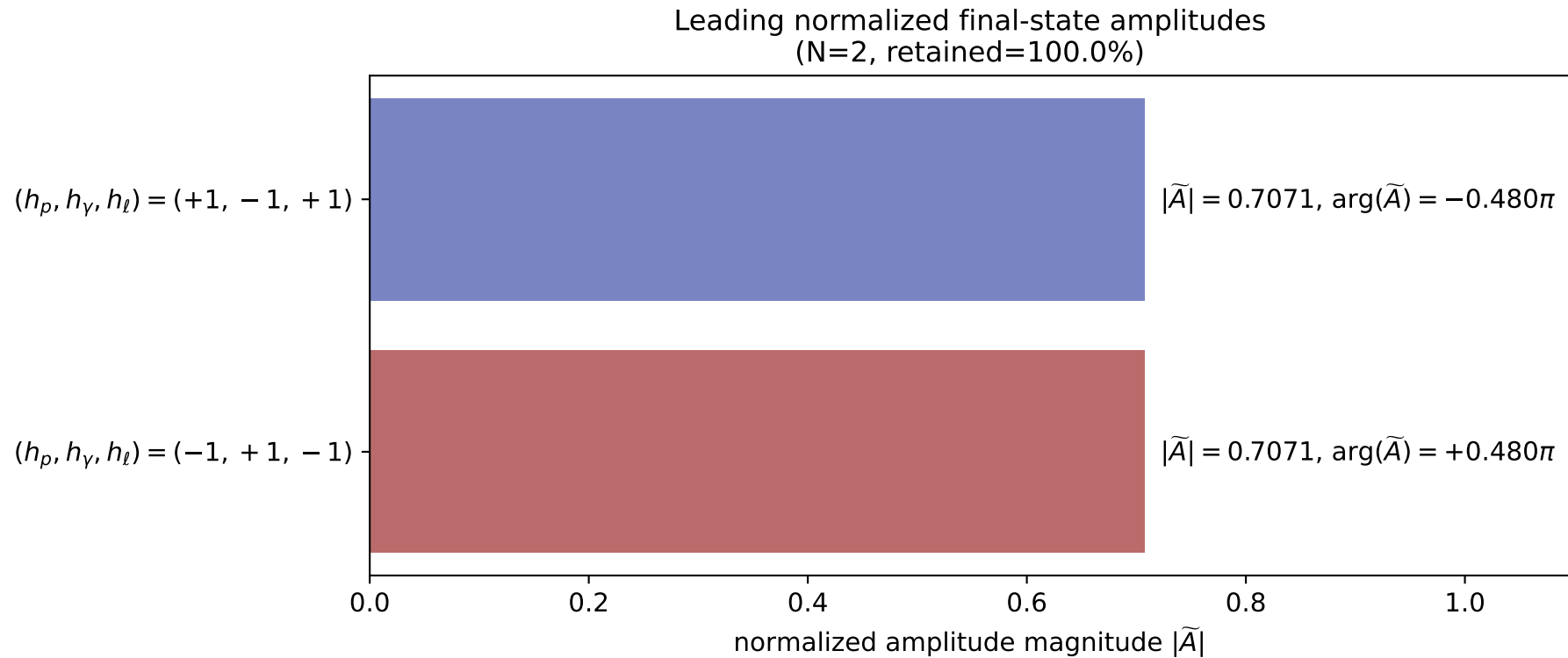


dghz_mixing_angles_polarization_03_local_minimum_13
 $d_{\text{GHZ}}=2.46222\text{e-}08$
 region: polarization_cluster_03_local_minimum_13
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0013
 lepton mixing angle: $\alpha_e=2.1691893$ rad
 incoming lepton: $-0.563315|+\rangle +0.826242|-\rangle$
 proton mixing angle: $\alpha_p=0.97239039$ rad
 incoming proton: $0.563326|+\rangle +0.826235|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=1.46356$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=0.711355$, $\phi_\gamma=2.34502$
 $E_\gamma=0.899044$, $\theta_{\ell'}=4.02607\text{e-}07$, $\phi_{\ell'}=3.85295$
 $Q^2=22.7946$, $x_B=0.978784$, $t=-14.2982$, $W^2=1.37394$, $y=0.965527$

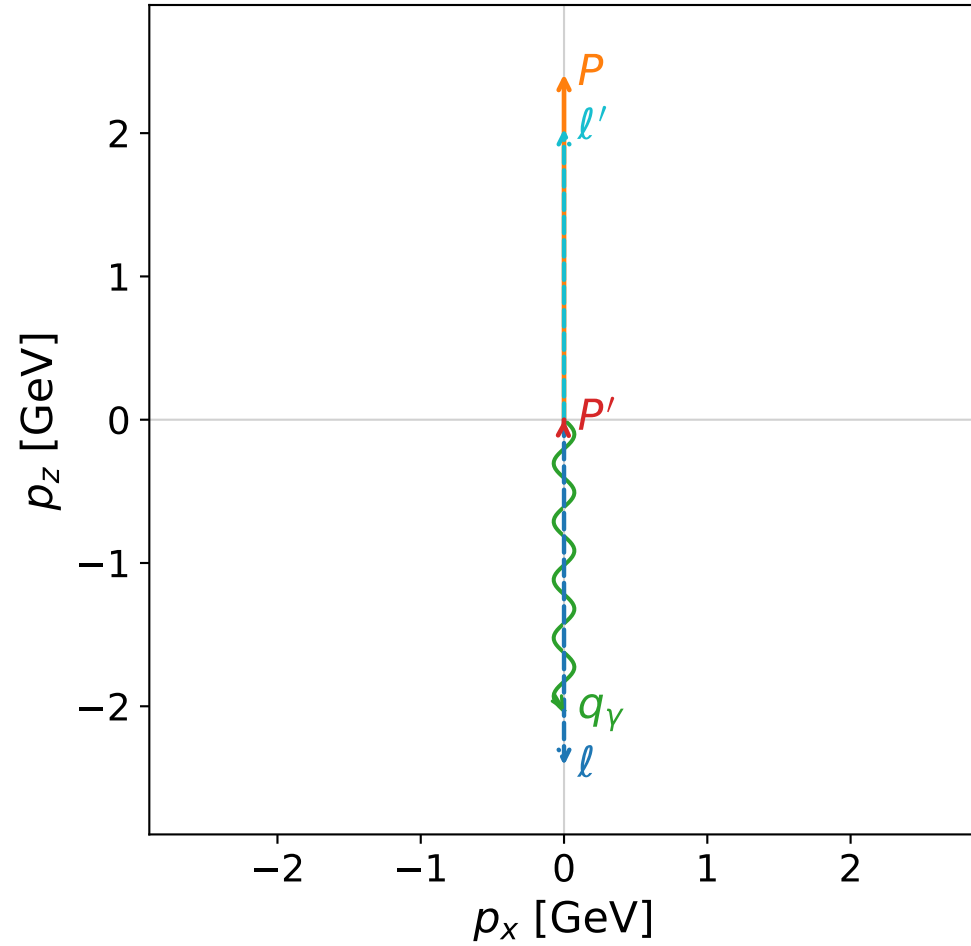
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.3626$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.3626$
 P' $E= 1.7383$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.4636$
 q_γ $E= 0.8990$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.8990$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.104\text{e+}20$; $I_h=+3.454\text{e-}11$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=9.917\text{e-}18$; $I_{TL}=-1.232\text{e-}09$

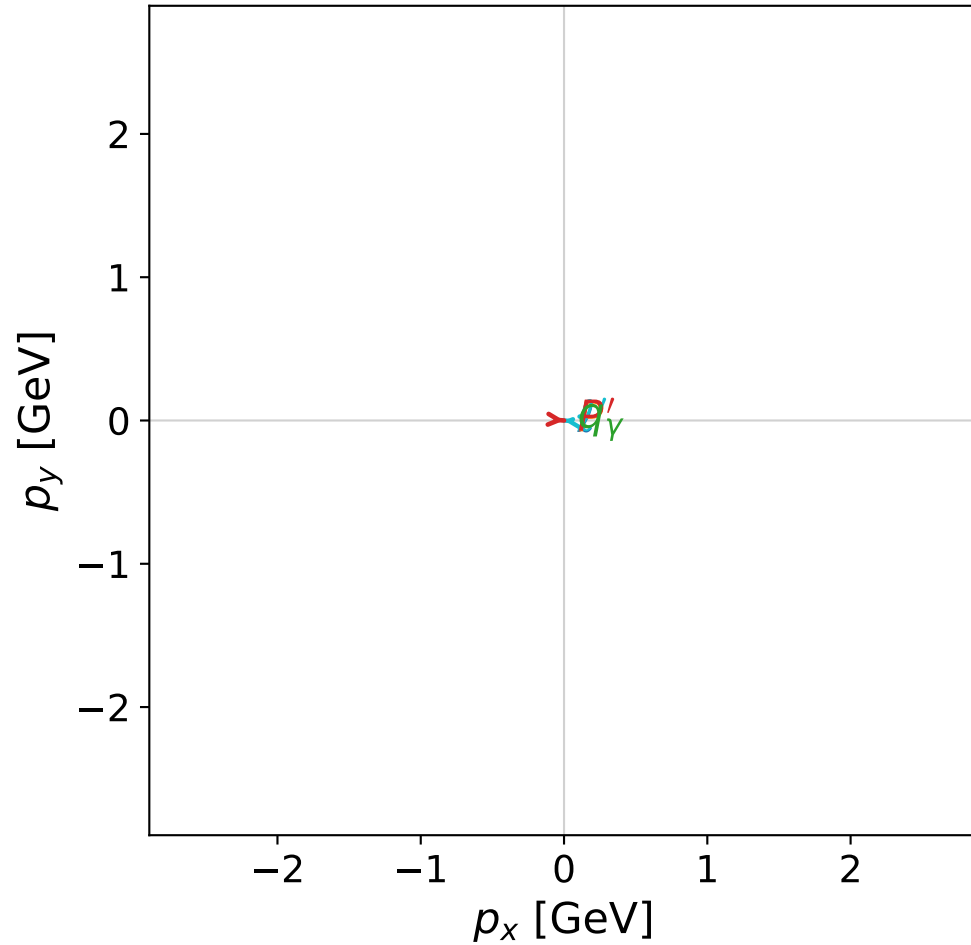


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

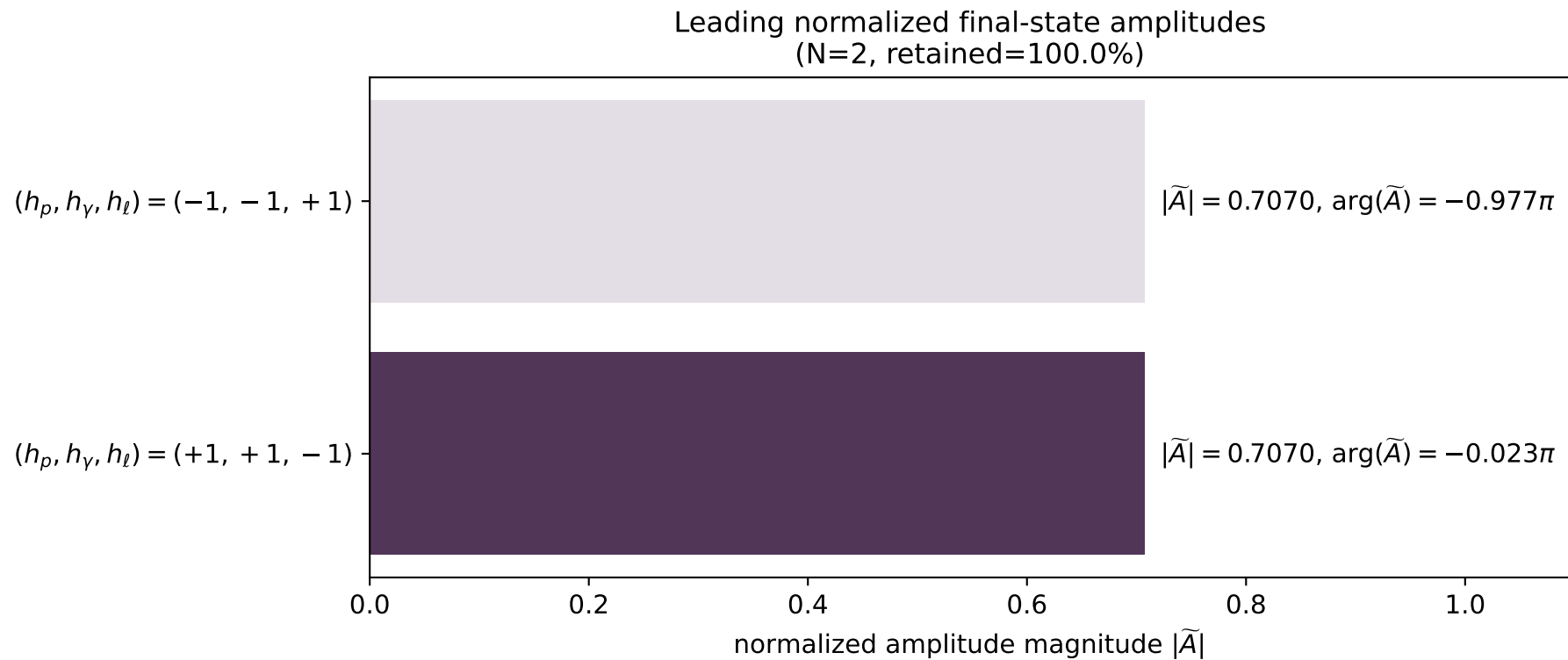


dghz_mixing_angles_polarization_03_local_minimum_15
 $d_{\text{GHZ}}=2.47594\text{e-}08$
 region: polarization_cluster_03_local_minimum_15
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0015
 lepton mixing angle: $\alpha_e=2.4020754$ rad
 incoming lepton: $-0.738794|+\rangle +0.673931|-\rangle$
 proton mixing angle: $\alpha_p=0.73950355$ rad
 incoming proton: $0.738803|+\rangle +0.673921|-\rangle$

$s=24.9985$, $\sqrt{s}=4.99985$, $|\vec{P}|=2.41194$, $|\vec{P}'|=0.00056453$
 $\theta_{P'}=0.0393096$, $\theta_\gamma=3.14159$, $\phi_{P'}=6.20497$, $\phi_\gamma=0.0735511$
 $E_\gamma=2.03121$, $\theta_{\ell'}=1.09255\text{e-}05$, $\phi_{\ell'}=3.06338$
 $Q^2=19.5912$, $x_B=0.837086$, $t=-3.09252$, $W^2=4.69268$, $y=0.970368$

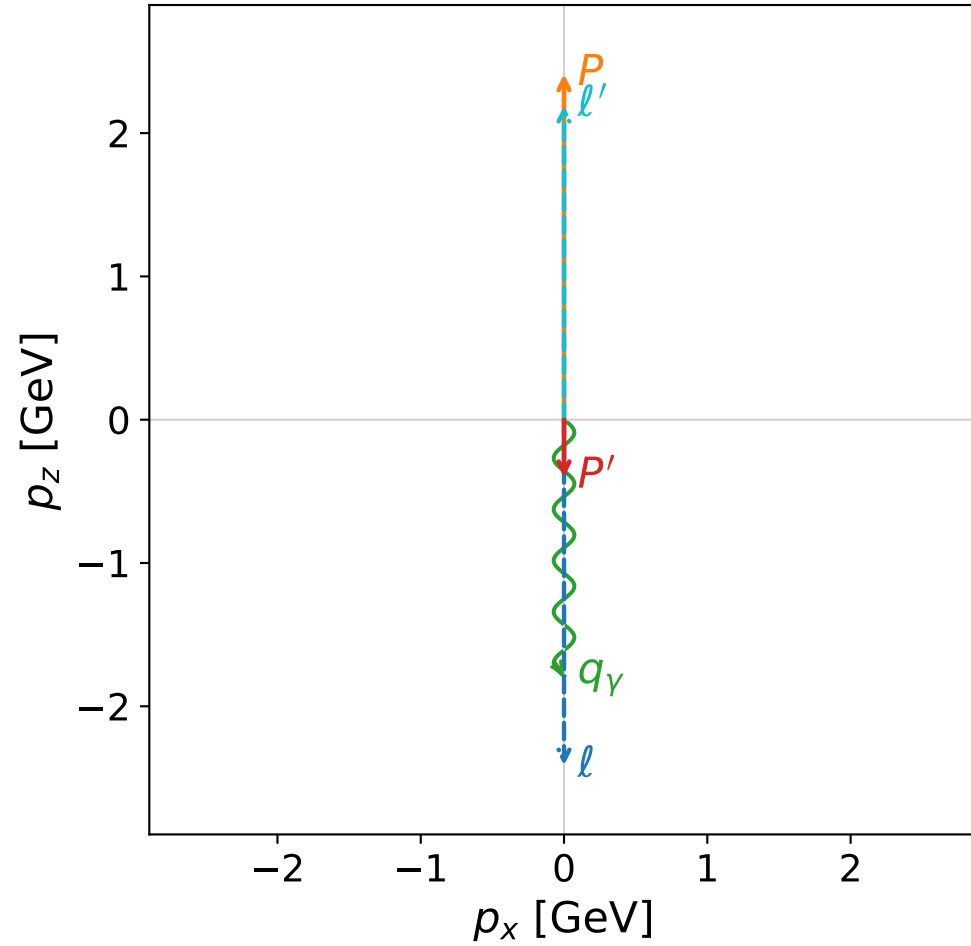
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4119$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4119$
 P $E= 2.5879$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4119$
 ℓ' $E= 2.0306$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0306$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0006$
 q_γ $E= 2.0312$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0312$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=6.985\text{e+}17$; $I_h=-1.261\text{e-}10$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=4.499\text{e-}13$; $I_{TL}=-8.659\text{e-}08$

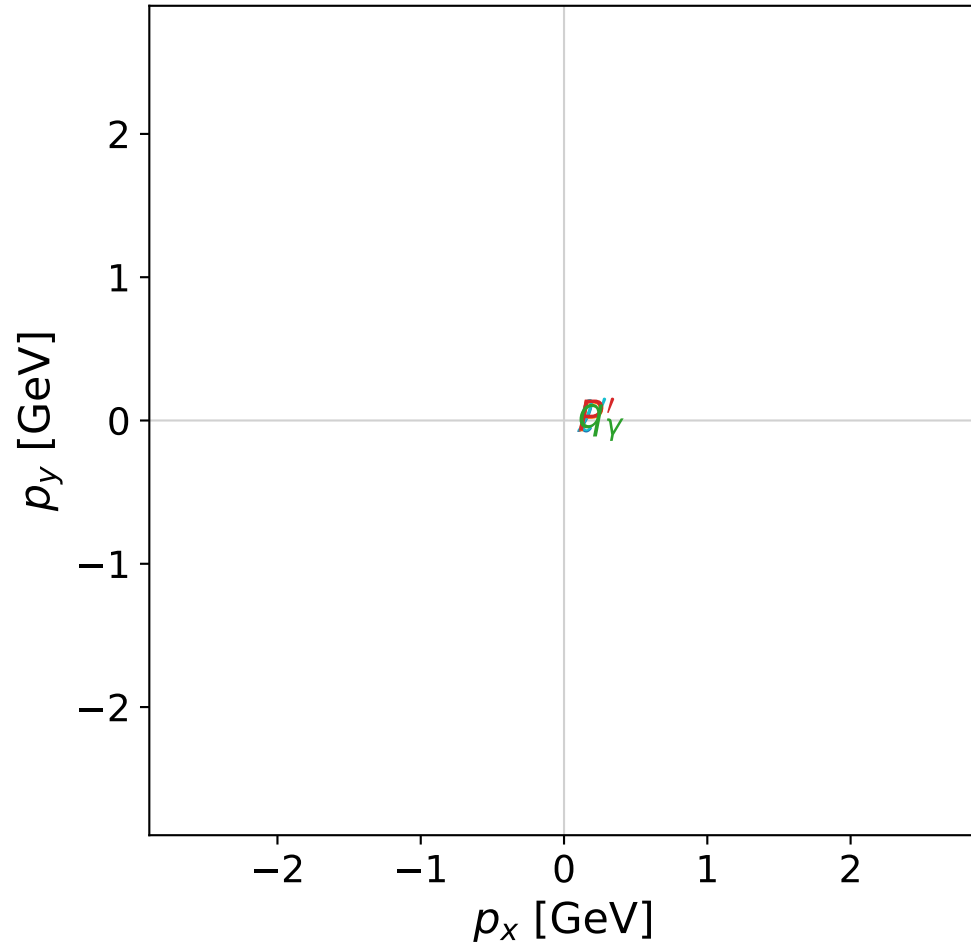


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

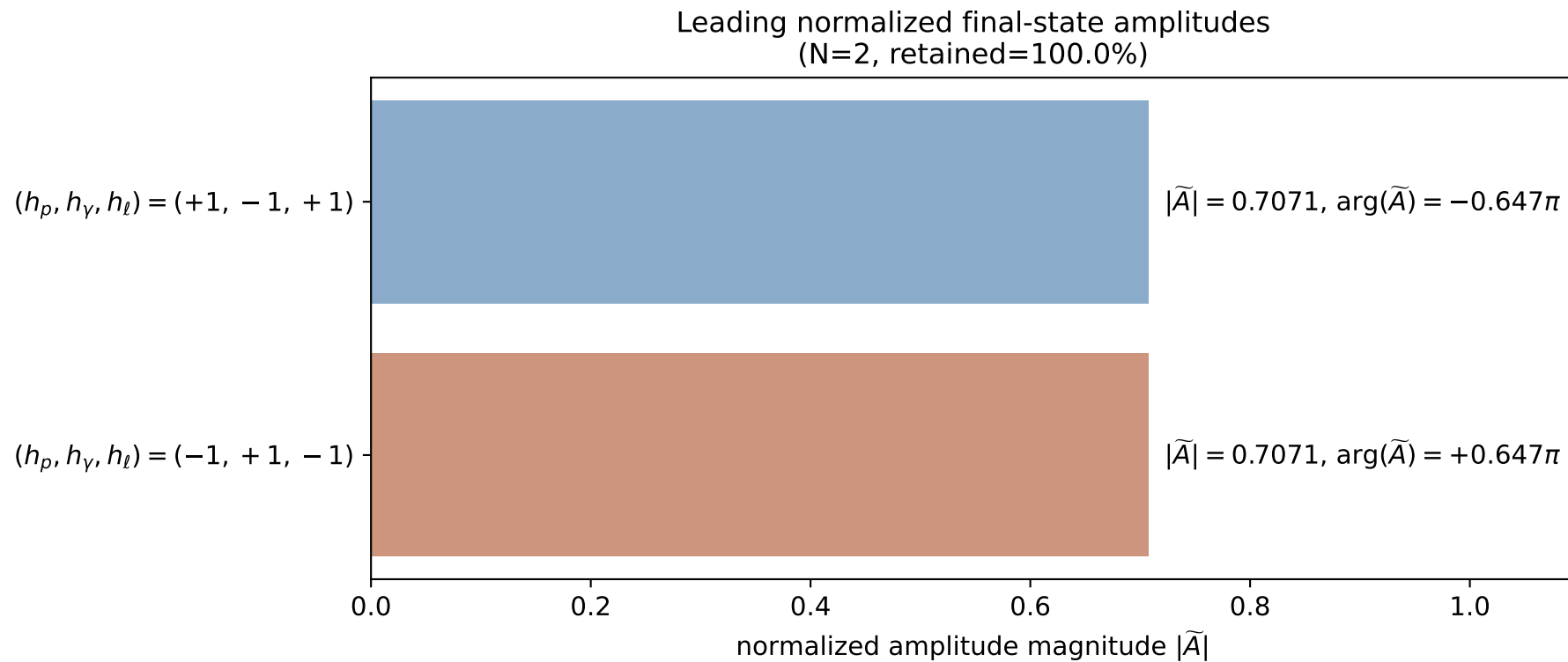


dghz_mixing_angles_polarization_03_local_minimum_17
 $d_{\text{GHZ}}=2.4853\text{e-}08$
 region: polarization_cluster_03_local_minimum_17
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0017
 lepton mixing angle: $\alpha_e=2.4106629$ rad
 incoming lepton: $-0.744554|+\rangle +0.667562|-\rangle$
 proton mixing angle: $\alpha_p=0.73092649$ rad
 incoming proton: $0.744556|+\rangle +0.66756|-\rangle$

$s=24.9975$, $\sqrt{s}=4.99975$, $|\vec{P}|=2.41189$, $|\vec{P}'|=0.403477$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.09629$, $\phi_\gamma=1.10925$
 $E_\gamma=1.78759$, $\theta_{\ell'}=0$, $\phi_{\ell'}=4.47422$
 $Q^2=21.1384$, $x_B=0.905421$, $t=-5.47151$, $W^2=3.08794$, $y=0.968025$

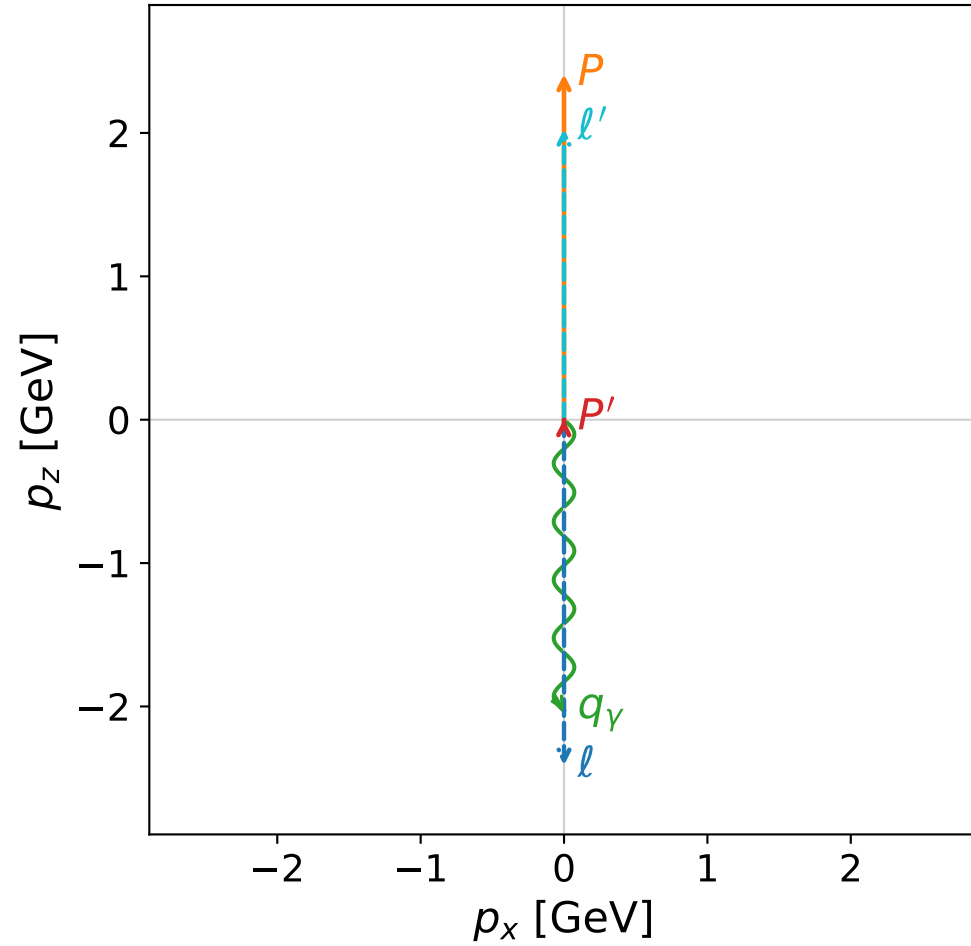
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4119$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4119$
 P $E= 2.5879$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4119$
 ℓ' $E= 2.1911$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.1911$
 P' $E= 1.0211$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.4035$
 q_γ $E= 1.7876$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.7876$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.506\text{e}+24$; $I_h=+1.967\text{e-}14$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=2.527\text{e-}28$; $I_{TL}=+9.512\text{e-}19$

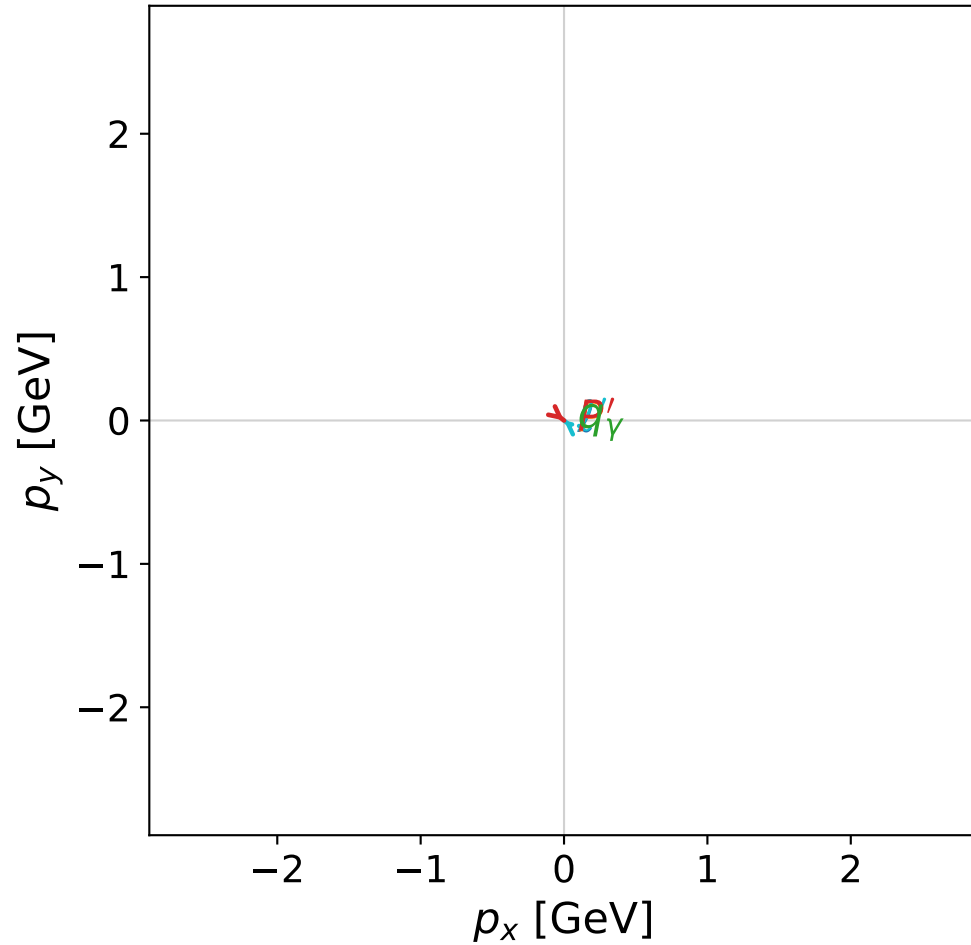


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

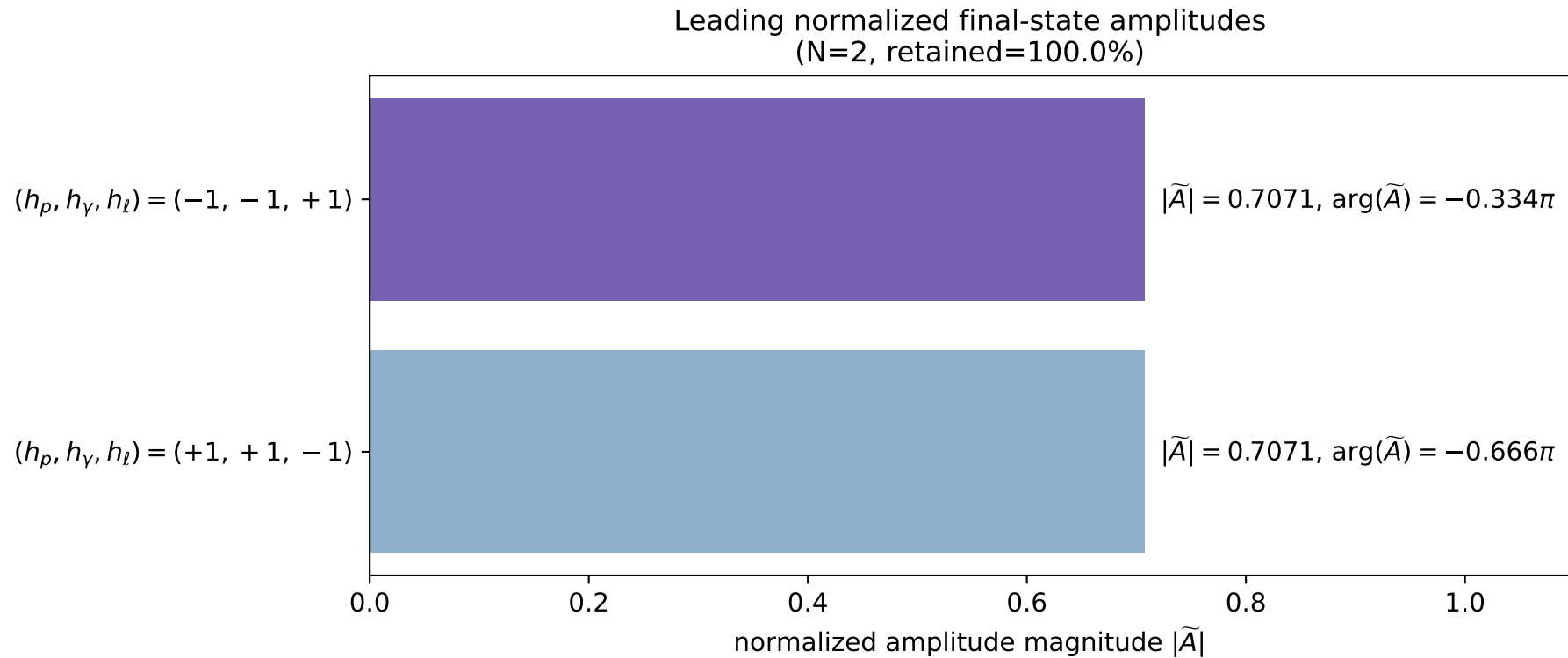


dghz_mixing_angles_polarization_03_local_minimum_20
 $d_{\text{GHZ}}=2.54044\text{e-}08$
 region: polarization_cluster_03_local_minimum_20
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0020
 lepton mixing angle: $\alpha_e=2.3449091$ rad
 incoming lepton: $-0.699082|+\rangle +0.715042|-\rangle$
 proton mixing angle: $\alpha_p=0.79667666$ rad
 incoming proton: $0.699087|+\rangle +0.715037|-\rangle$

$s=24.9686$, $\sqrt{s}=4.99686$, $|\vec{P}|=2.41039$, $|\vec{P}'|=0.0036789$
 $\theta_{P'}=0.019635$, $\theta_\gamma=3.14159$, $\phi_{P'}=5.61047$, $\phi_\gamma=2.09314$
 $E_\gamma=2.03127$, $\theta_{\ell'}=3.56238\text{e-}05$, $\phi_{\ell'}=2.46887$
 $Q^2=19.5491$, $x_B=0.836335$, $t=-3.07484$, $W^2=4.70547$, $y=0.970358$

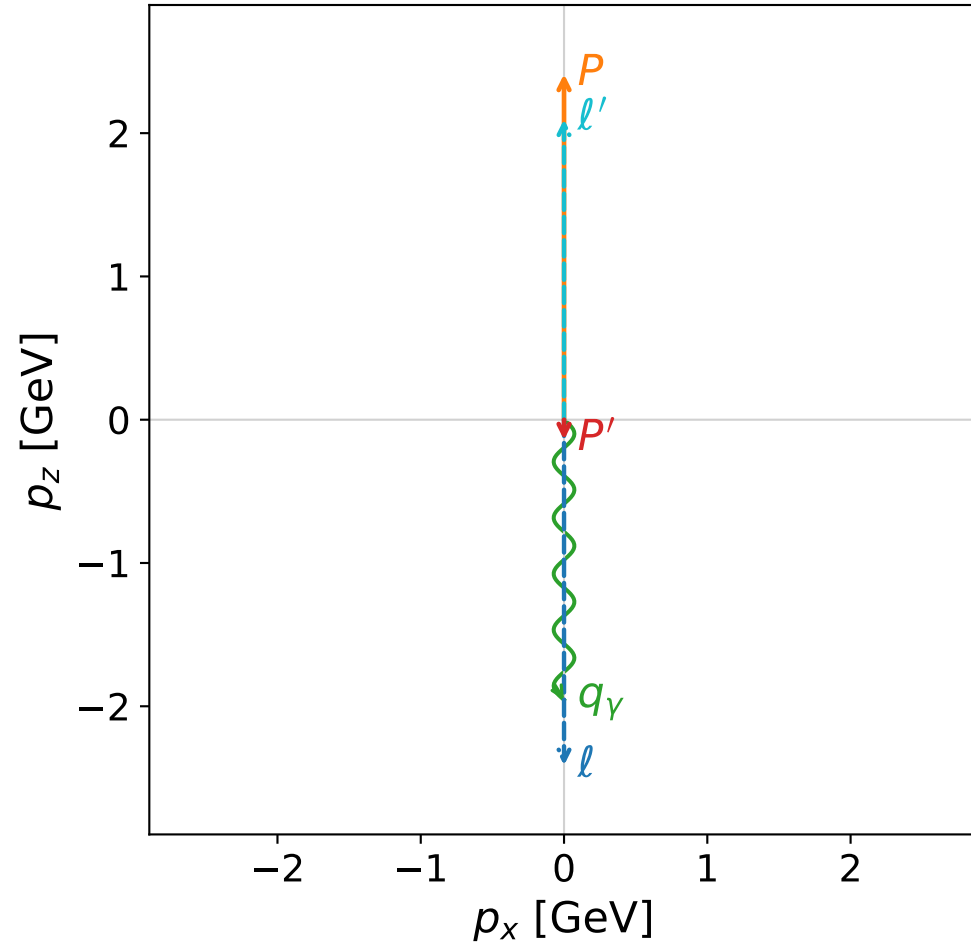
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4104$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4104$
 P $E= 2.5865$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4104$
 ℓ' $E= 2.0276$ $p_x= -0.0001$ $p_y= 0.0000$ $p_z= 2.0276$
 P' $E= 0.9380$ $p_x= 0.0001$ $p_y= -0.0000$ $p_z= 0.0037$
 q_γ $E= 2.0313$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0313$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=6.627\text{e+}16$; $I_h=+7.912\text{e-}11$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=4.794\text{e-}12$; $I_{TL}=+5.543\text{e-}08$

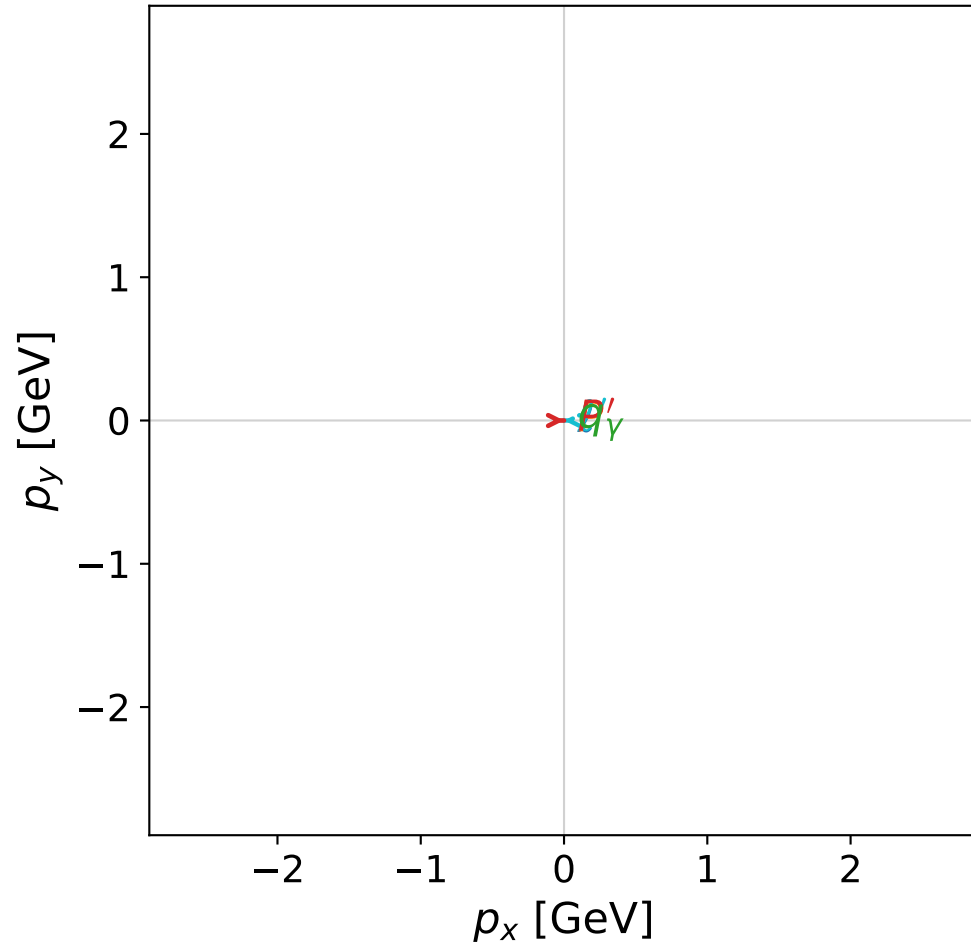


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

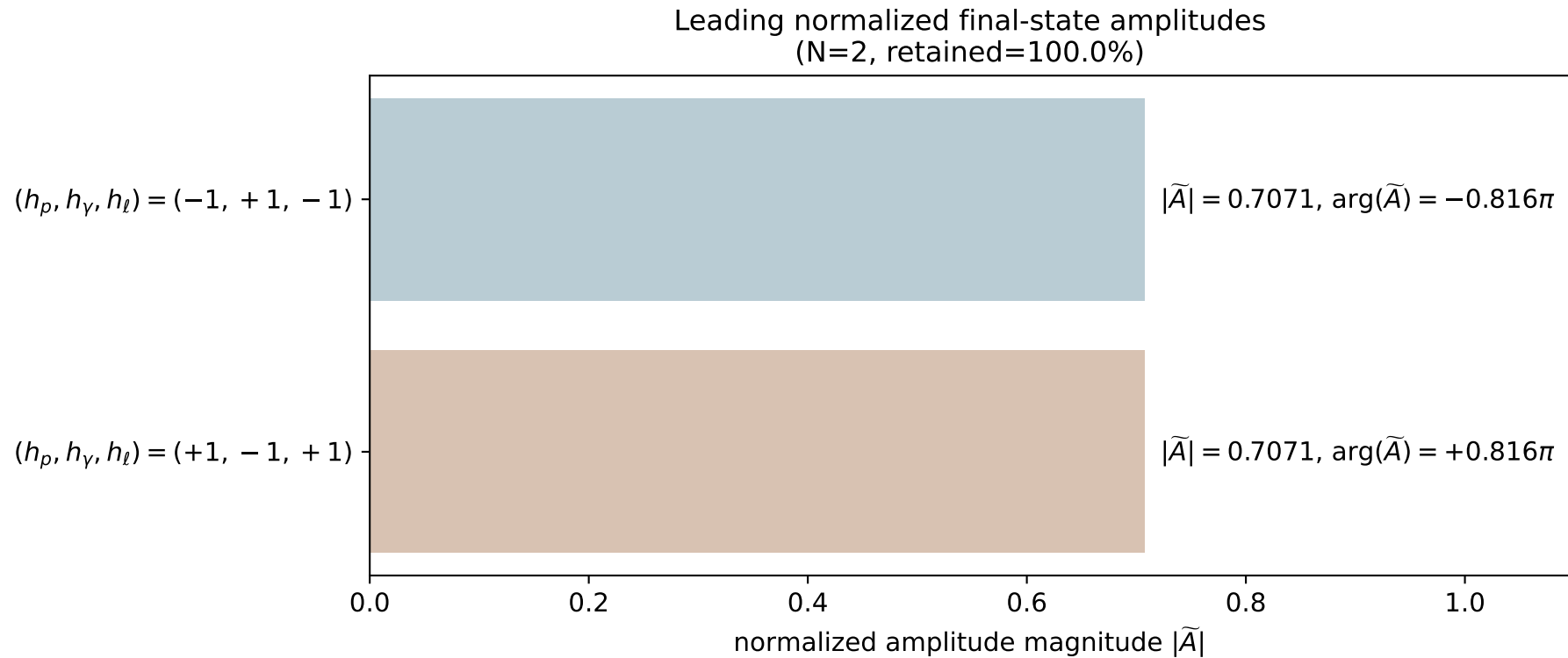


dghz_mixing_angles_polarization_03_local_minimum_38
 $d_{\text{GHZ}}=2.64248\text{e-}08$
 region: polarization_cluster_03_local_minimum_38
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0038
 lepton mixing angle: $\alpha_e=2.4362483$ rad
 incoming lepton: $-0.761388|+\rangle +0.648296|-\rangle$
 proton mixing angle: $\alpha_p=0.70535267$ rad
 incoming proton: $0.761383|+\rangle +0.648302|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.140649$
 $\theta_{p'}=3.14094$, $\theta_\gamma=3.14159$, $\phi_{p'}=0.000316874$, $\phi_\gamma=5.70698$
 $E_\gamma=1.95543$, $\theta_{\ell'}=4.39858\text{e-}05$, $\phi_{\ell'}=3.14191$
 $Q^2=20.2231$, $x_B=0.864884$, $t=-3.82814$, $W^2=4.03919$, $y=0.969416$

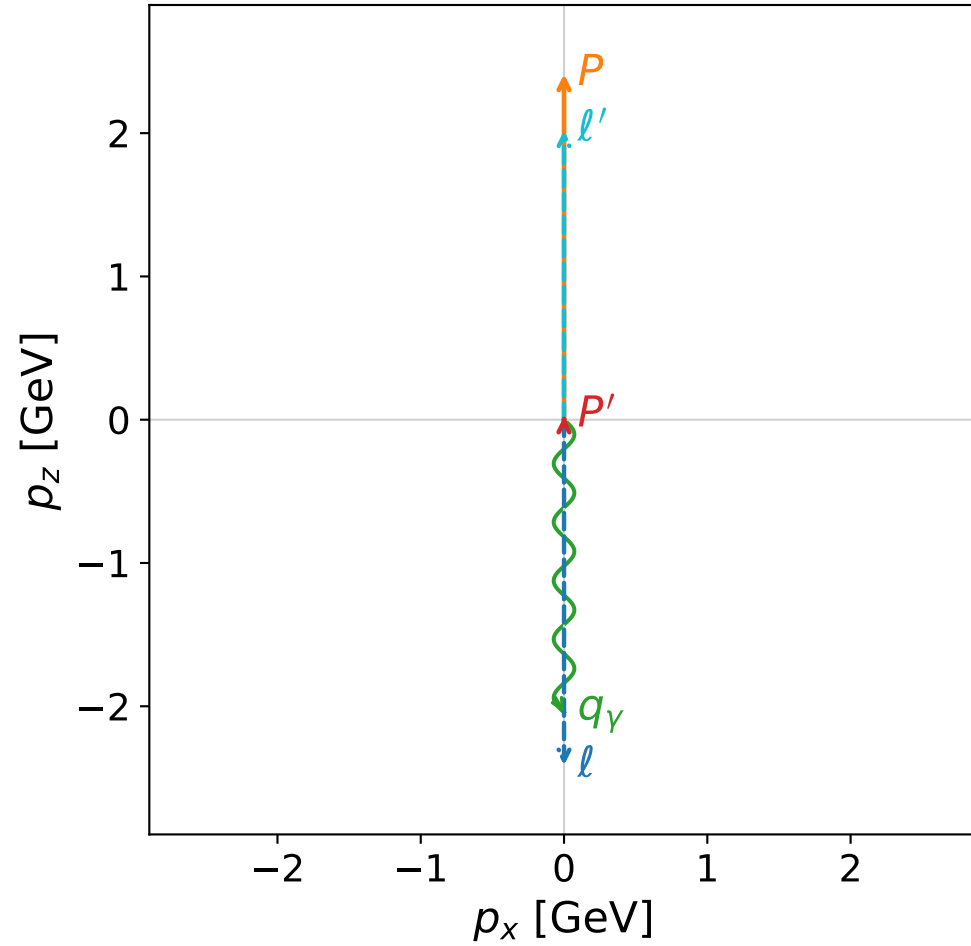
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0961$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= 2.0961$
 P' $E= 0.9485$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= -0.1406$
 q_γ $E= 1.9554$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.9554$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=4.072\text{e+}16$; $I_h=-9.279\text{e-}10$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=5.075\text{e-}12$; $I_{TL}=-5.068\text{e-}07$

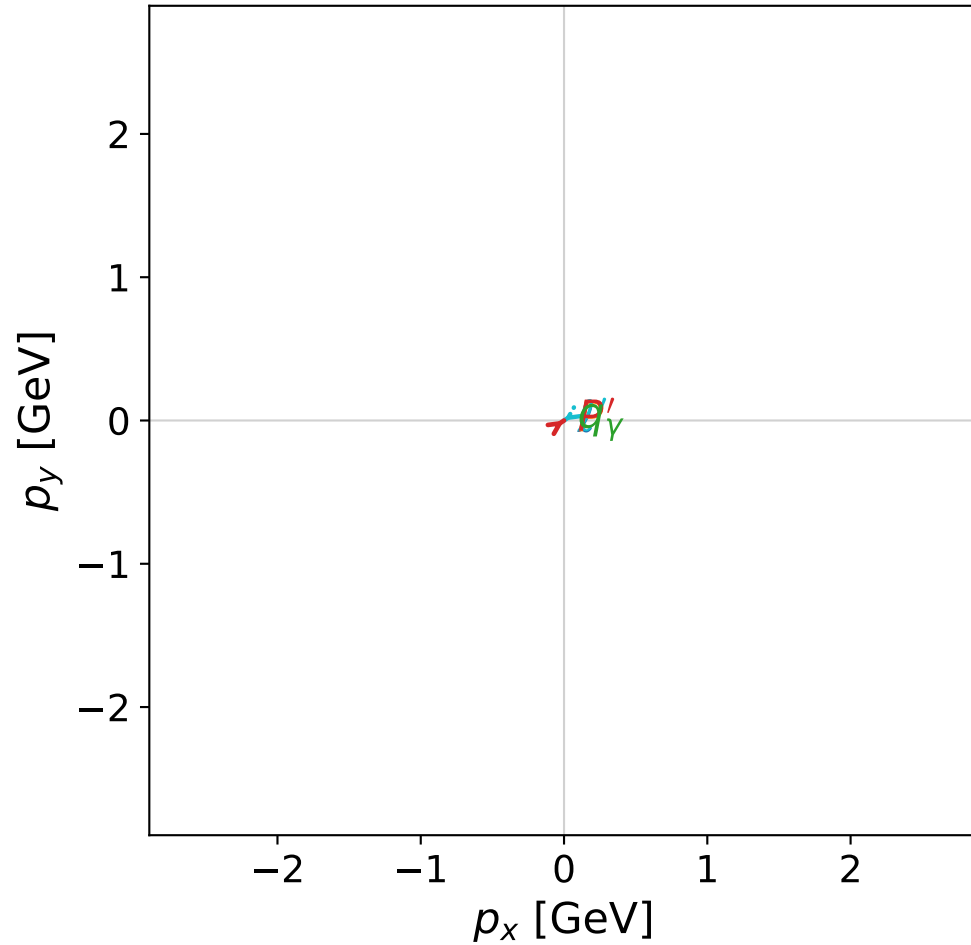


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

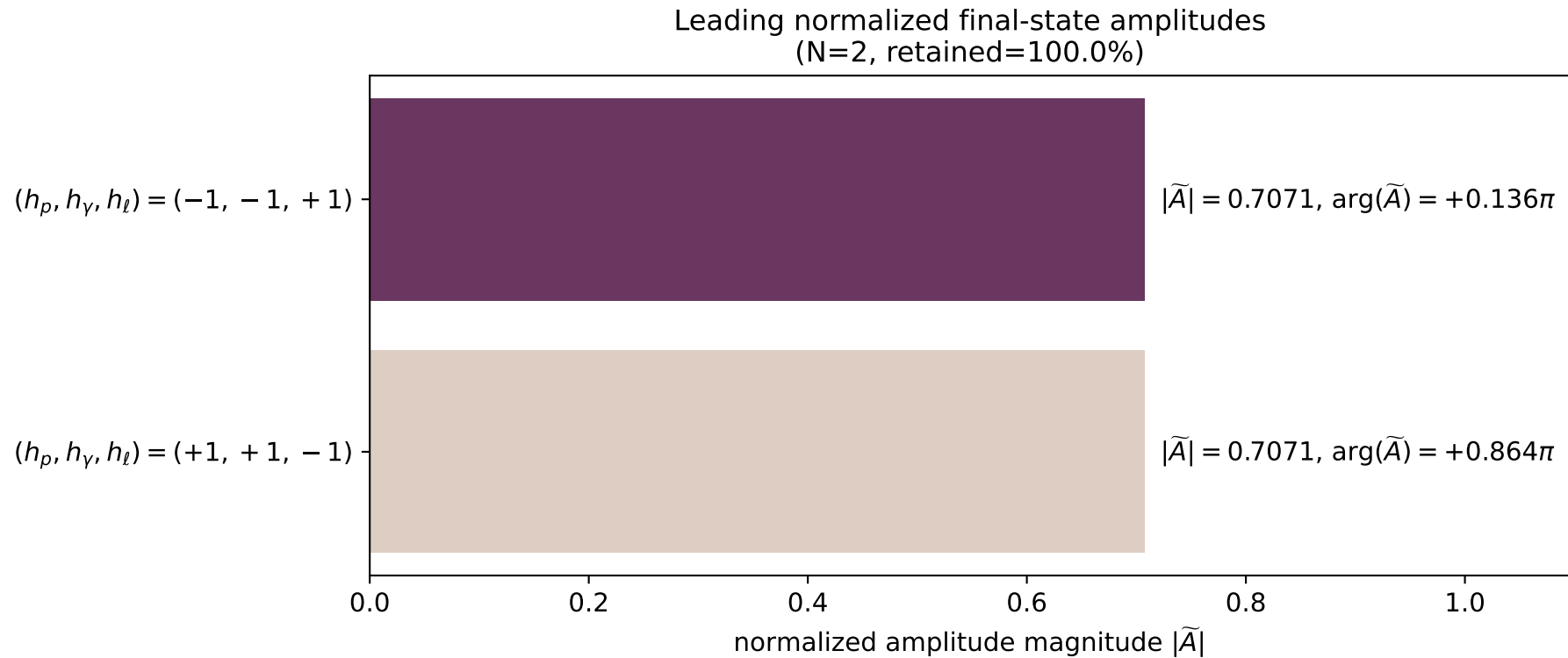


dghz_mixing_angles_polarization_03_local_minimum_43
 $d_{\text{GHZ}}=2.6533\text{e-}08$
 region: polarization_cluster_03_local_minimum_43
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0043
 lepton mixing angle: $\alpha_e=2.3049352$ rad
 incoming lepton: $-0.669948|+\rangle +0.742408|-\rangle$
 proton mixing angle: $\alpha_p=0.83665018$ rad
 incoming proton: $0.669953|+\rangle +0.742403|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.023742$
 $\theta_{p'}=3.55227\text{e-}05$, $\theta_\gamma=3.14159$, $\phi_{p'}=0.592674$, $\phi_\gamma=3.5688$
 $E_\gamma=2.04272$, $\theta_{\ell'}=4.1803\text{e-}07$, $\phi_{\ell'}=3.73427$
 $Q^2=19.4792$, $x_B=0.832104$, $t=-2.98239$, $W^2=4.81021$, $y=0.970541$

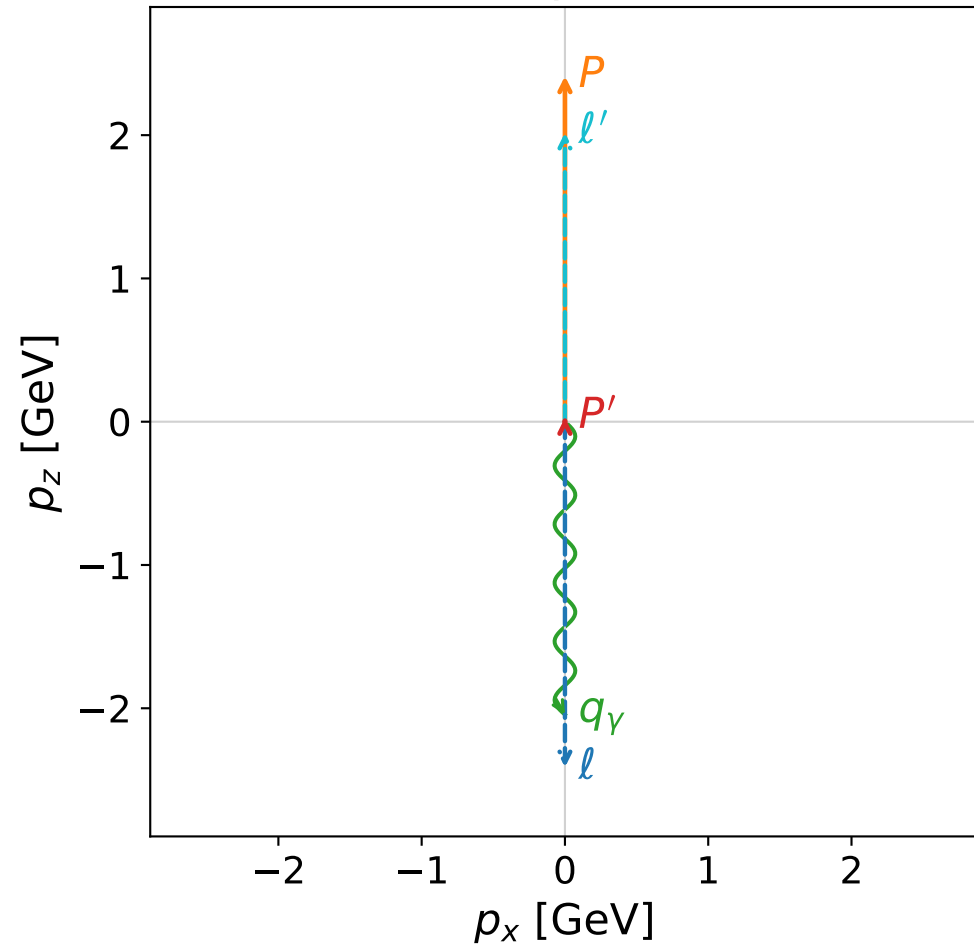
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0190$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.0190$
 P' $E= 0.9383$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0237$
 q_γ $E= 2.0427$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.0427$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=4.685\text{e+}20$; $I_h=+4.477\text{e-}12$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=6.982\text{e-}16$; $I_{TL}=+3.172\text{e-}09$

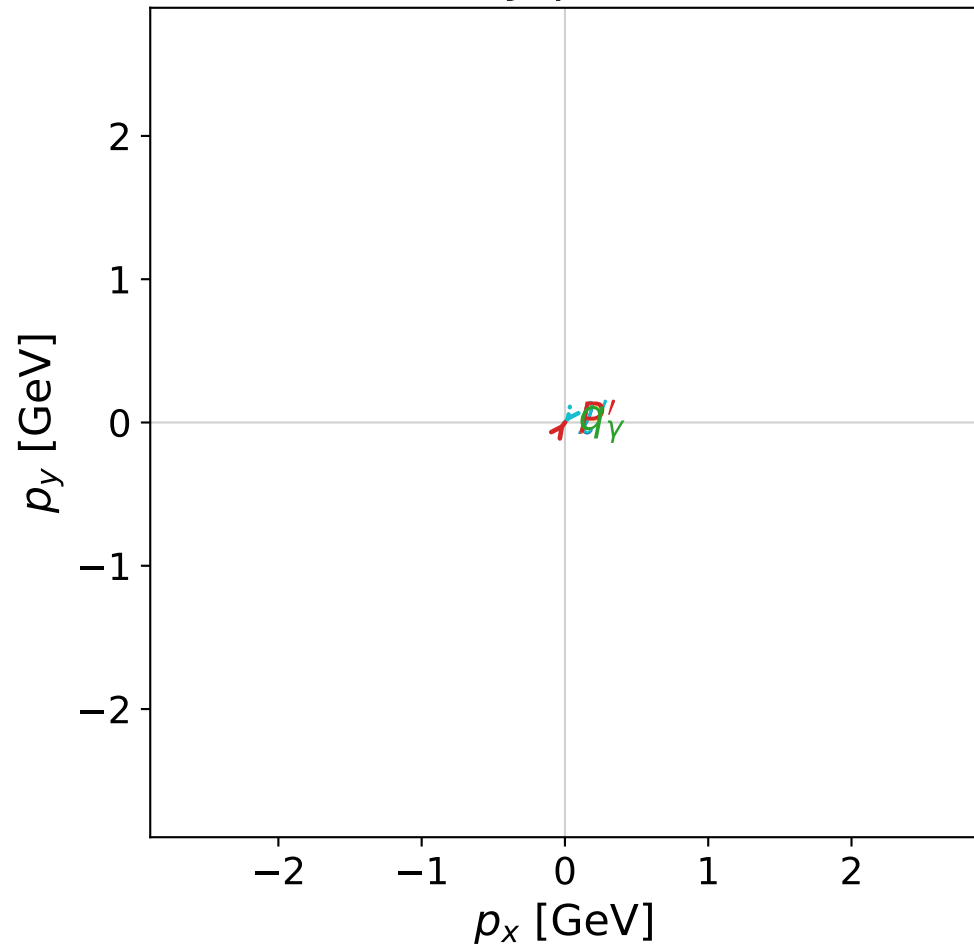


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



dghz_mixing_angles_polarization_03_local_minimum_45
 $d_{\text{GHZ}}=2.6597\text{e-}08$
 region: polarization_cluster_03_local_minimum_45
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0045
 lepton mixing angle: $\alpha_e=2.3367051$ rad
 incoming lepton: $-0.693192|+\rangle +0.720753|-\rangle$
 proton mixing angle: $\alpha_p=0.80488895$ rad
 incoming proton: $0.693191|+\rangle +0.720754|-\rangle$

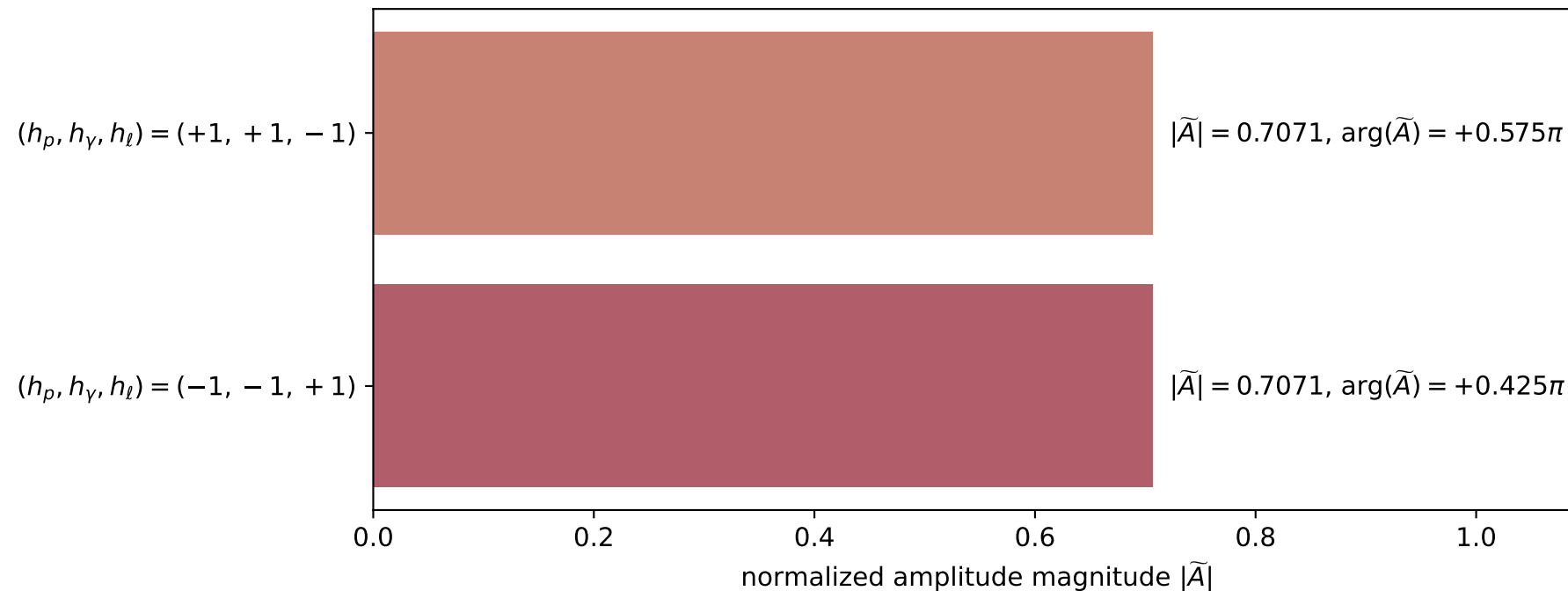
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0277887$
 $\theta_{P'}=0.00245437$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.937225$, $\phi_\gamma=4.47598$
 $E_\gamma=2.04469$, $\theta_{\ell'}=3.38161\text{e-}05$, $\phi_{\ell'}=4.07882$
 $Q^2=19.4592$, $x_B=0.831222$, $t=-2.96345$, $W^2=4.831$, $y=0.970571$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

Particle	E	p_x	p_y	p_z
ℓ	2.4120	0.0000	0.0000	-2.4120
P	2.5880	0.0000	0.0000	2.4120
ℓ'	2.0169	-0.0000	-0.0001	2.0169
P'	0.9384	0.0000	0.0001	0.0278
q_γ	2.0447	-0.0000	-0.0000	-2.0447

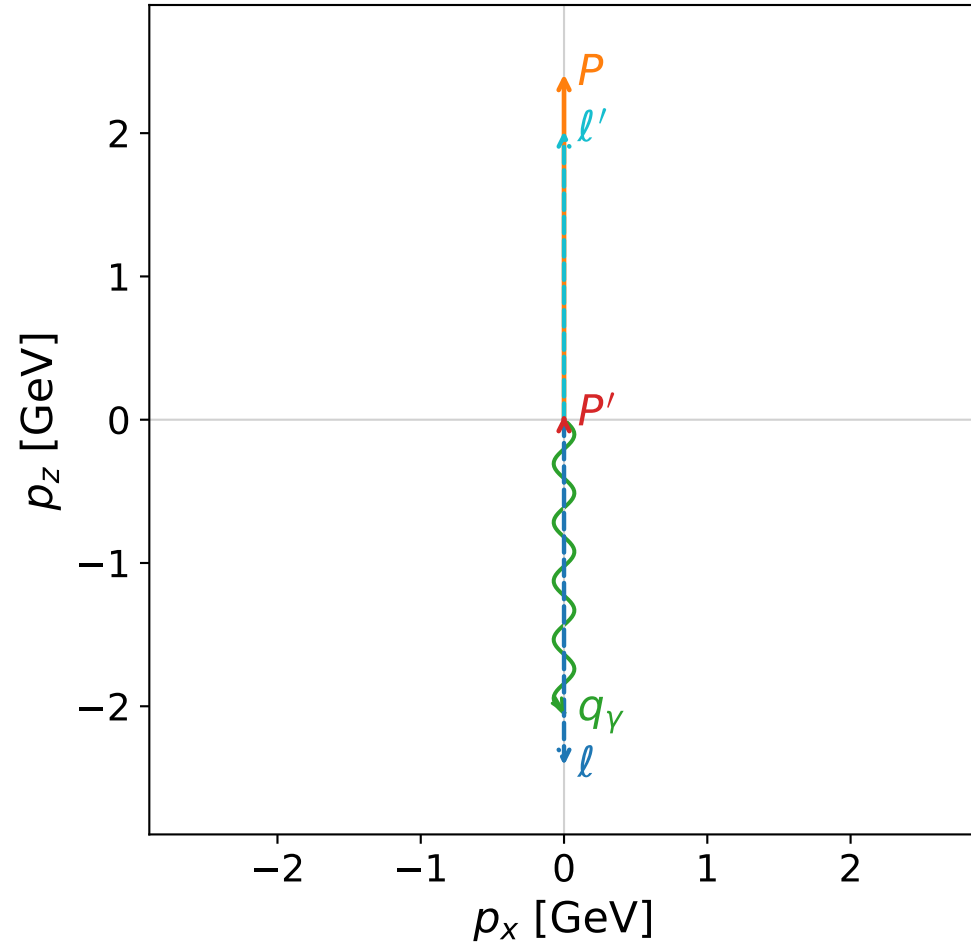
GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=7.382\text{e+}16$; $I_h=+9.759\text{e-}11$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=4.580\text{e-}12$; $I_{TL}=+7.052\text{e-}08$

Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

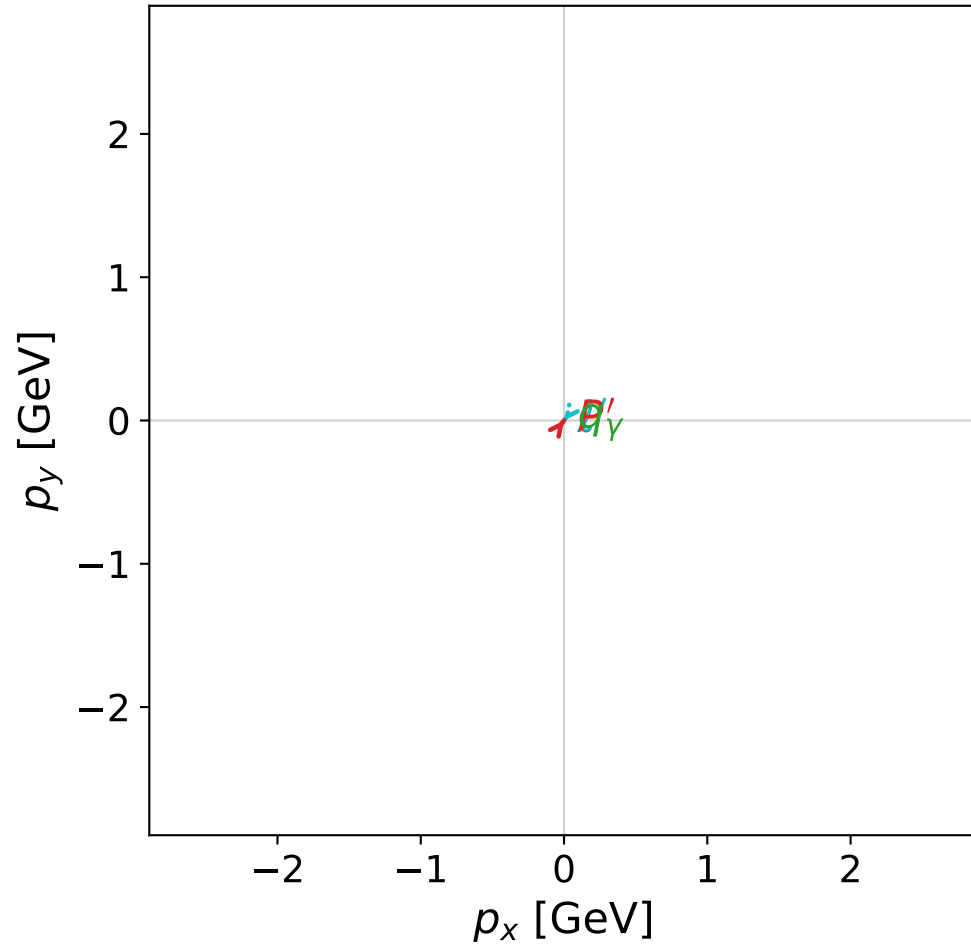


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

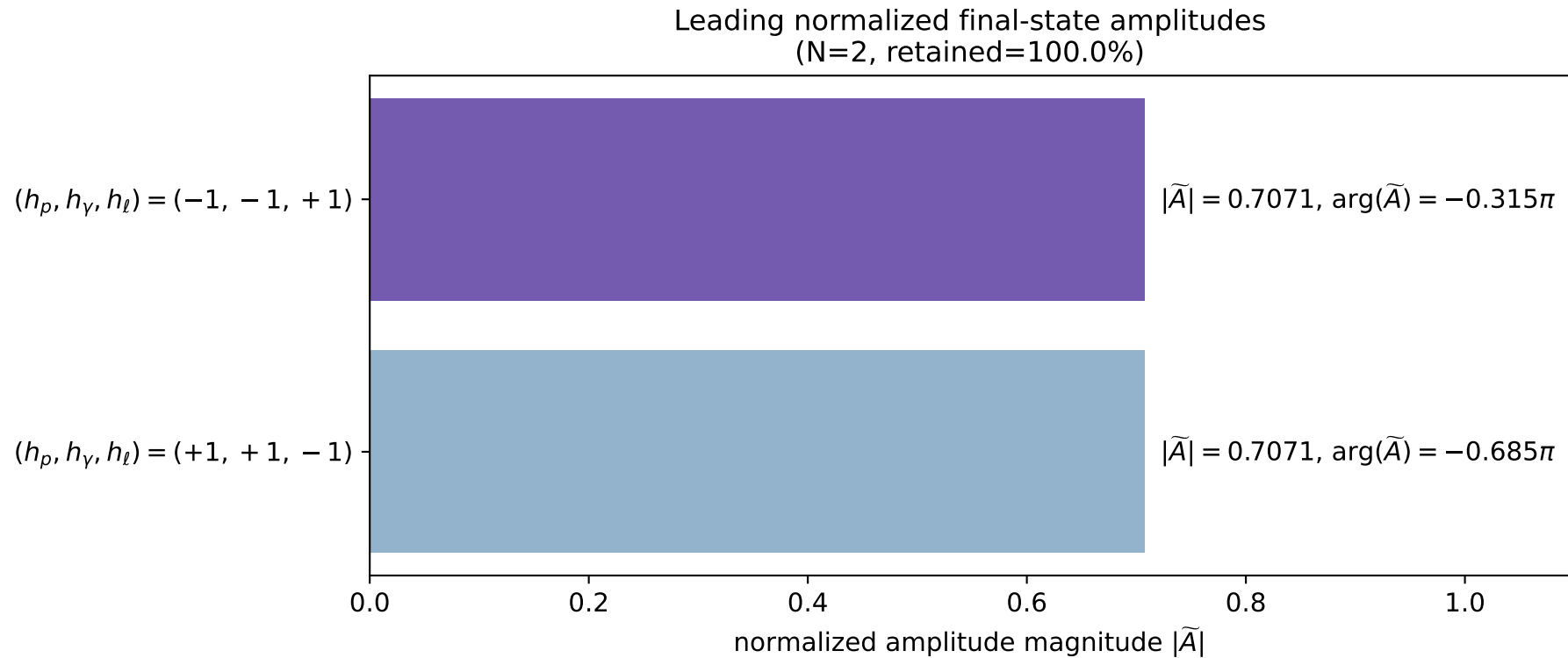


dghz_mixing_angles_polarization_03_local_minimum_48
 $d_{\text{GHZ}}=2.67266\text{e-}08$
 region: polarization_cluster_03_local_minimum_48
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0048
 lepton mixing angle: $\alpha_e=2.3692897$ rad
 incoming lepton: $-0.716306|+\rangle +0.697787|-\rangle$
 proton mixing angle: $\alpha_p=0.772294$ rad
 incoming proton: $0.716312|+\rangle +0.69778|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0314813$
 $\theta_{P'}=0.000613592$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.92167$, $\phi_\gamma=2.15105$
 $E_\gamma=2.04648$, $\theta_{\ell'}=9.58645\text{e-}06$, $\phi_{\ell'}=4.06326$
 $Q^2=19.4408$, $x_B=0.830413$, $t=-2.94624$, $W^2=4.85005$, $y=0.970599$

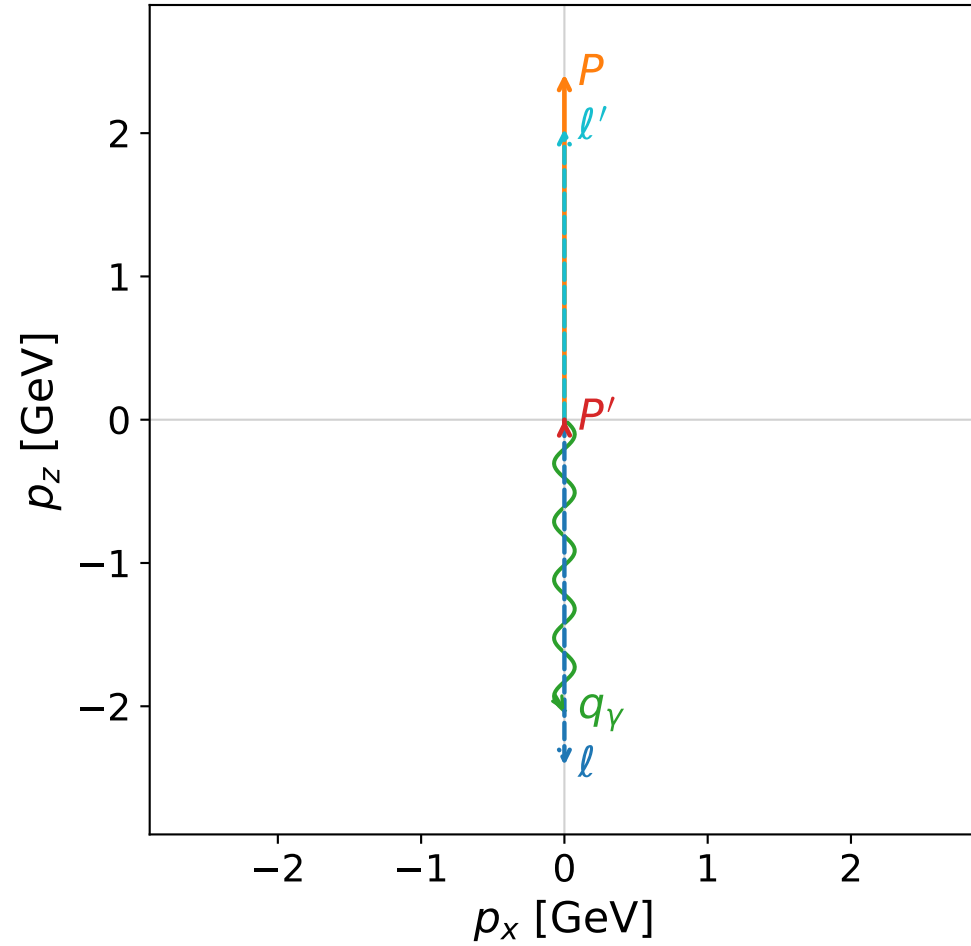
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0150$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.0150$
 P' $E= 0.9385$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0315$
 q_γ $E= 2.0465$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0465$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=9.206\text{e+}17$; $I_h=-1.897\text{e-}11$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=3.711\text{e-}13$; $I_{TL}=-1.358\text{e-}08$

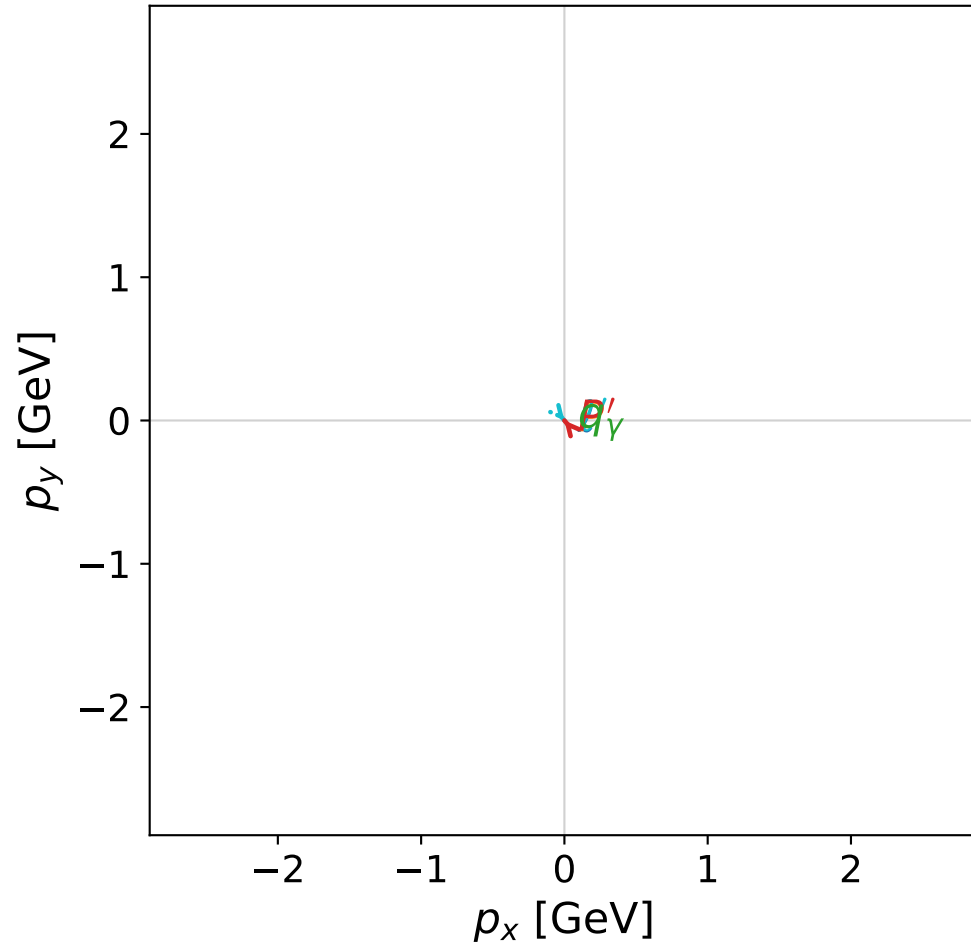


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



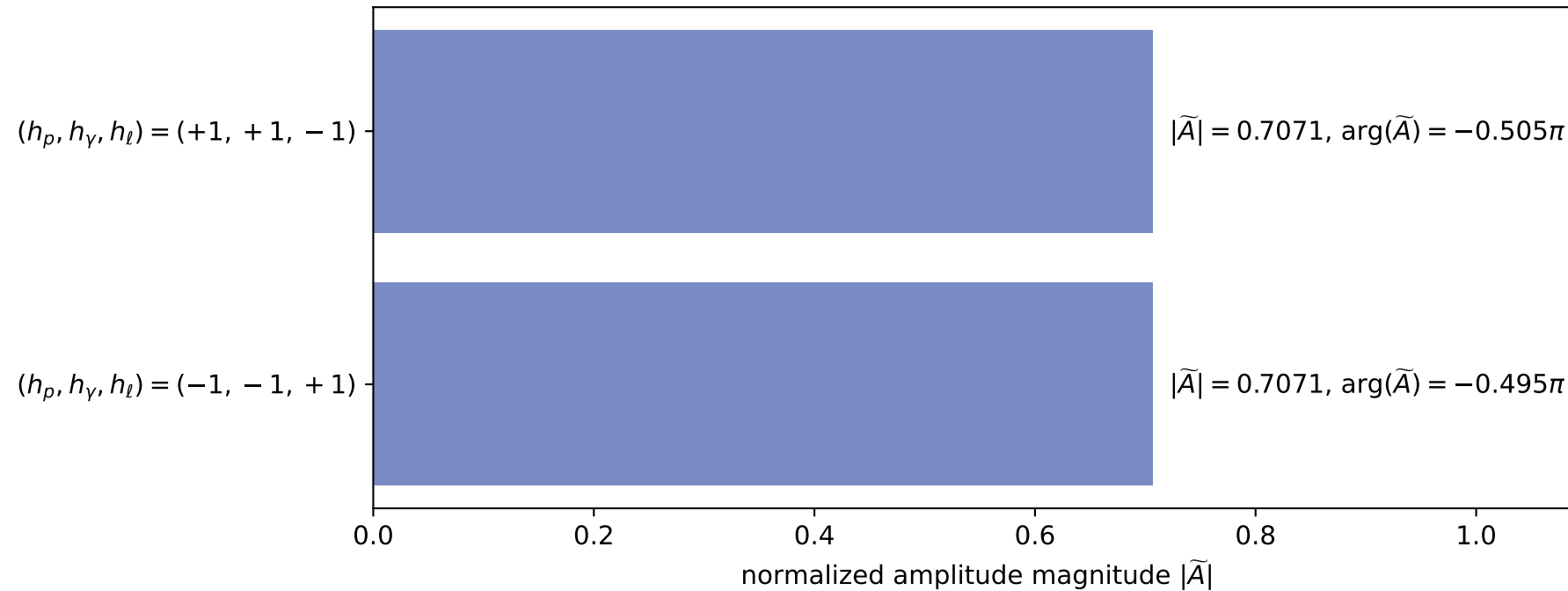
dghz_mixing_angles_polarization_03_local_minimum_51
 $d_{\text{GHZ}}=2.68417\text{e-}08$
 region: polarization_cluster_03_local_minimum_51
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0051
 lepton mixing angle: $\alpha_e=2.2220181$ rad
 incoming lepton: $-0.606159|+\rangle +0.795344|-\rangle$
 proton mixing angle: $\alpha_p=0.91958237$ rad
 incoming proton: $0.606152|+\rangle +0.795349|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.001359$
 $\theta_{P'}=0.00981748$, $\theta_Y=3.14159$, $\phi_{P'}=2.27173$, $\phi_Y=1.5862$
 $E_Y=2.03168$, $\theta_{l'}=6.57126\text{e-}06$, $\phi_{l'}=5.41332$
 $Q^2=19.5887$, $x_B=0.836921$, $t=-3.08882$, $W^2=4.6968$, $y=0.970376$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0303$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.0303$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0014$
 q_Y $E= 2.0317$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0317$

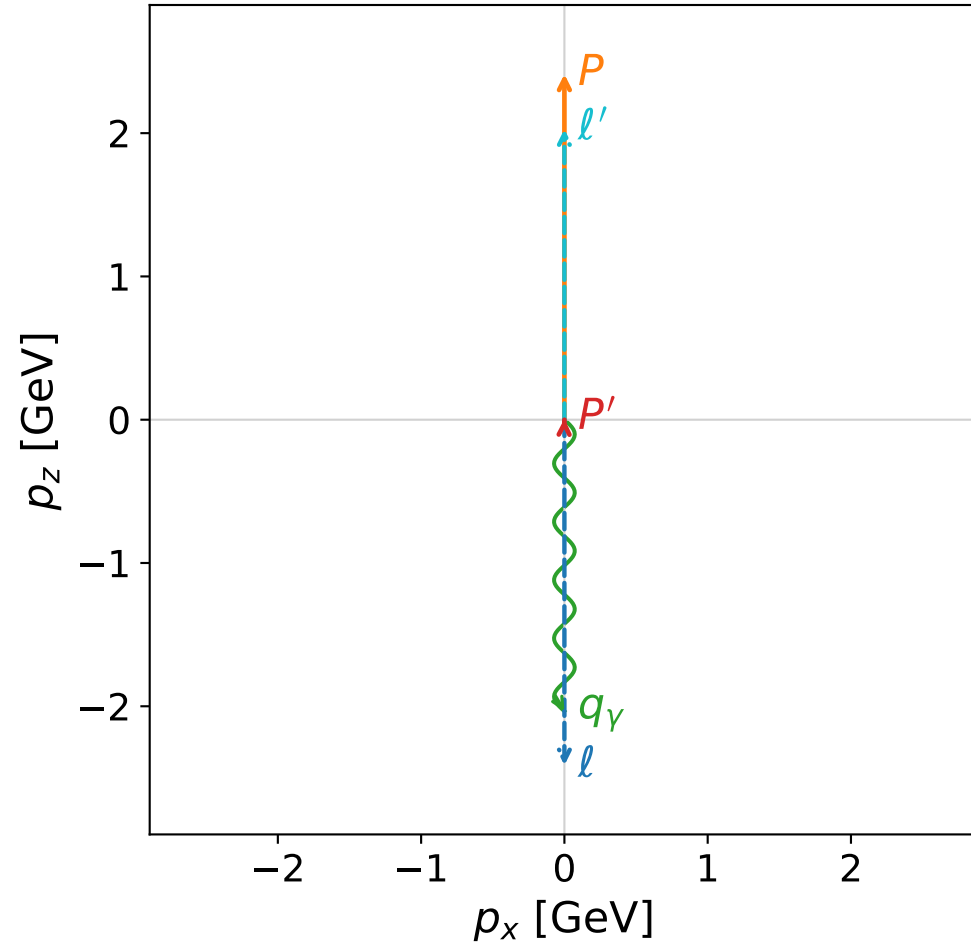
GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.775\text{e+}18$; $I_h=-1.466\text{e-}10$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.739\text{e-}13$; $I_{TL}=-1.008\text{e-}07$

Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

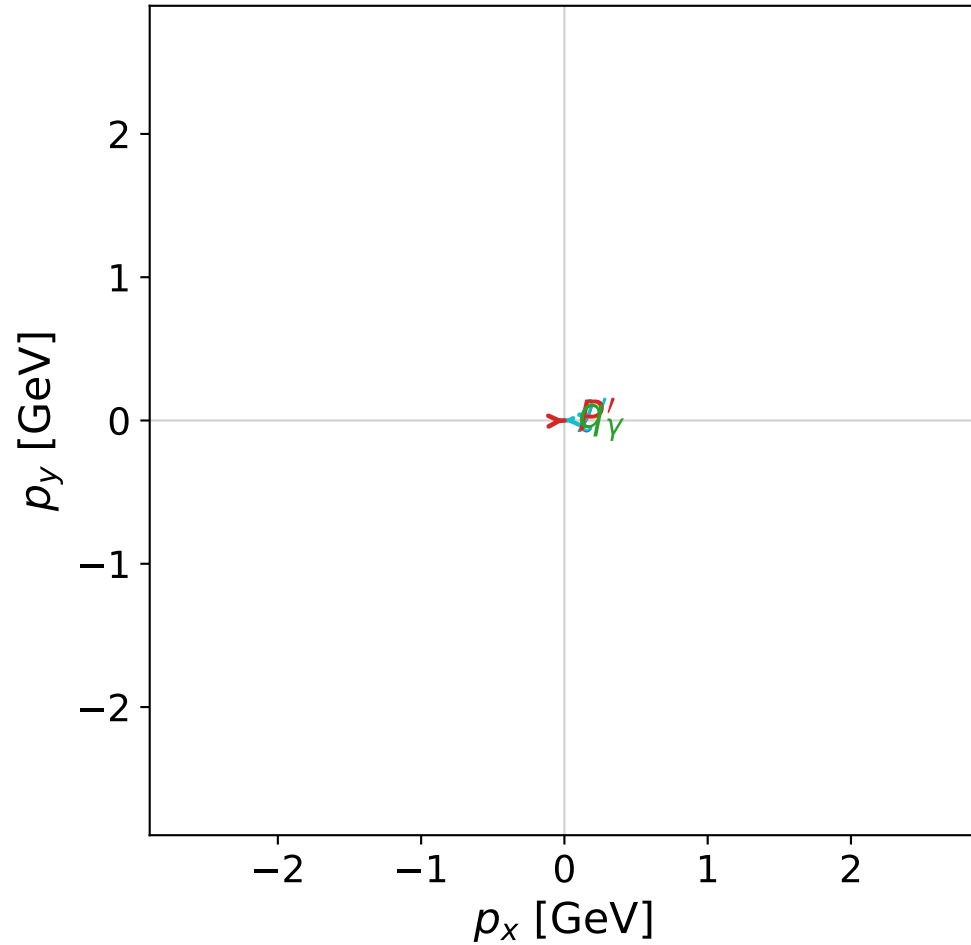


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



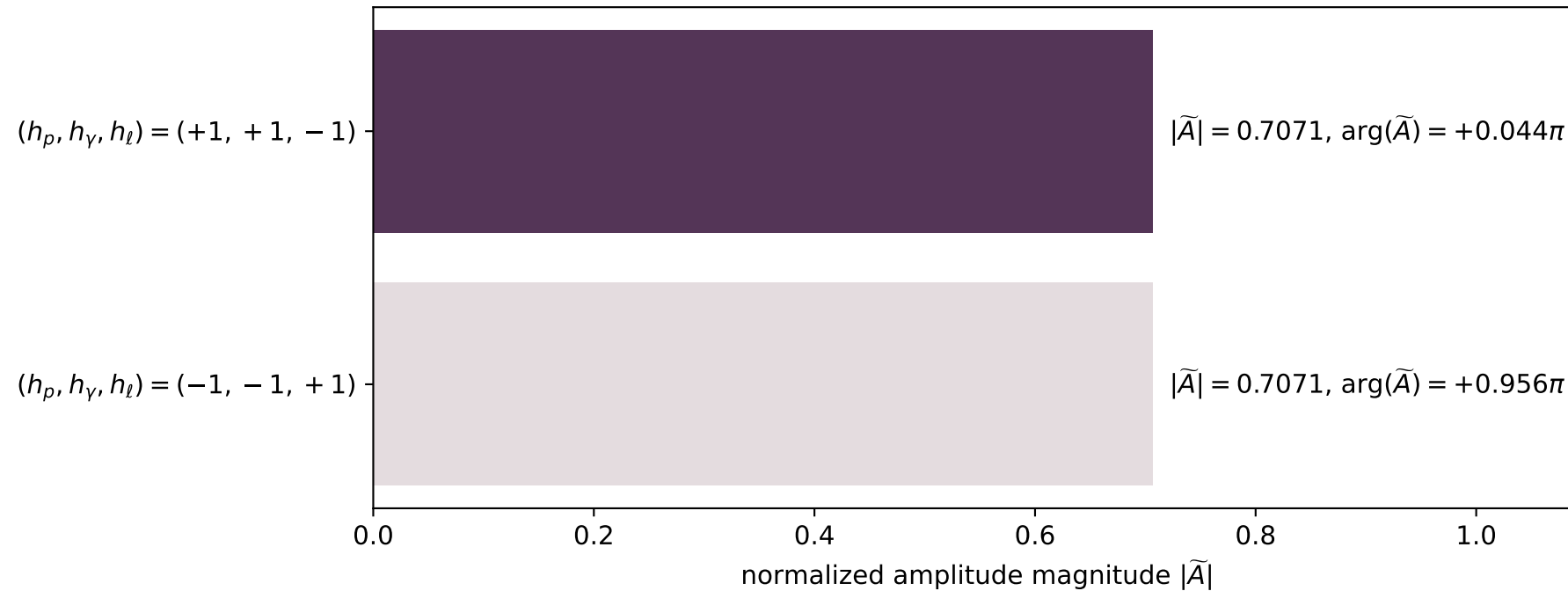
dghz_mixing_angles_polarization_03_local_minimum_52
 $d_{\text{GHZ}}=2.69014\text{e-}08$
 region: polarization_cluster_03_local_minimum_52
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0052
 lepton mixing angle: $\alpha_e=2.4223145$ rad
 incoming lepton: $-0.752281|+\rangle +0.658842|-\rangle$
 proton mixing angle: $\alpha_p=0.71927442$ rad
 incoming proton: $0.752284|+\rangle +0.658839|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.00684209$
 $\theta_{P'}=0.0196328$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.0386213$, $\phi_\gamma=6.14533$
 $E_\gamma=2.03441$, $\theta_{\ell'}=6.62473\text{e-}05$, $\phi_{\ell'}=3.18021$
 $Q^2=19.5621$, $x_B=0.835752$, $t=-3.0625$, $W^2=4.72433$, $y=0.970416$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0276$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= 2.0276$
 P' $E= 0.9380$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= 0.0068$
 q_γ $E= 2.0344$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0344$

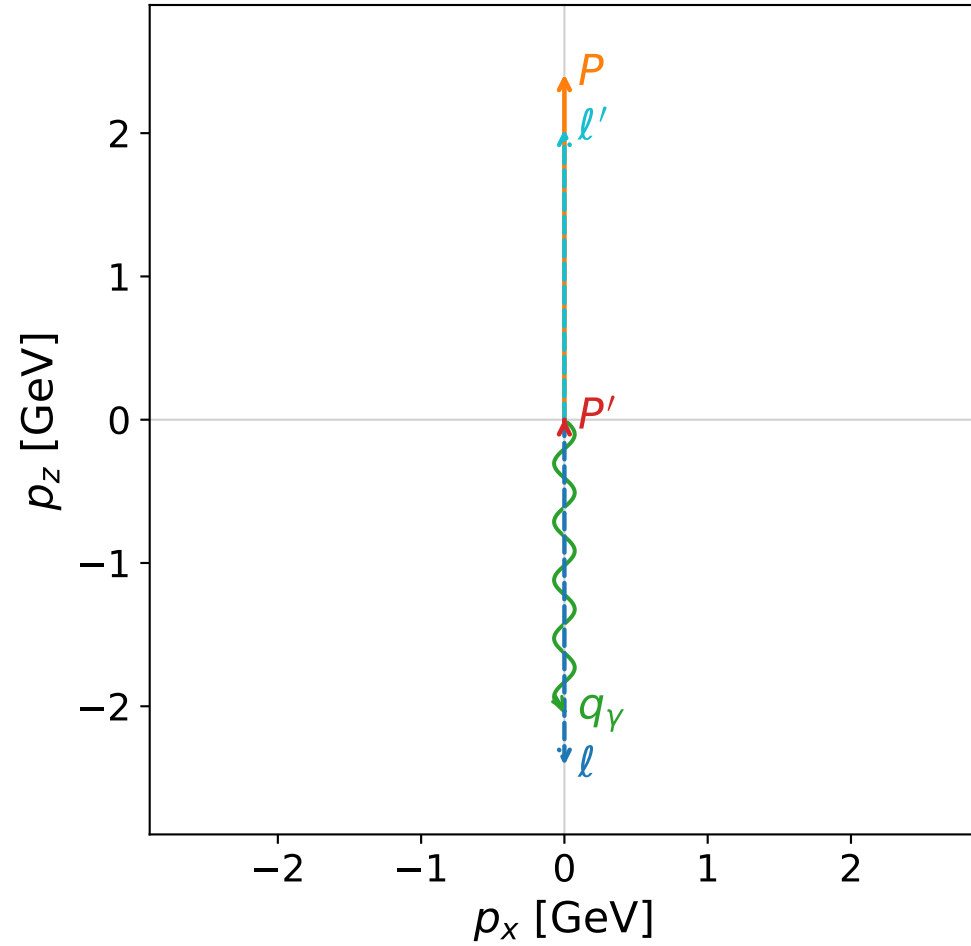
GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.878\text{e+}16$; $I_h=-1.106\text{e-}09$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.696\text{e-}11$; $I_{TL}=-7.646\text{e-}07$

Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

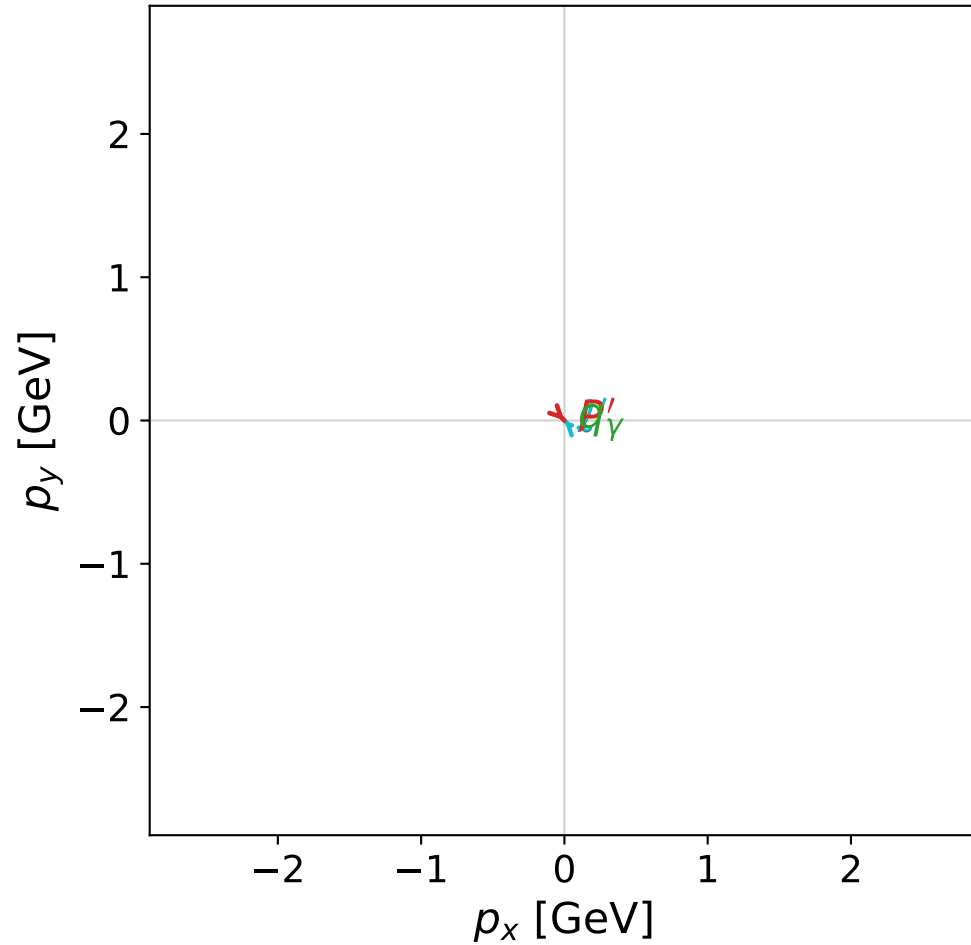


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

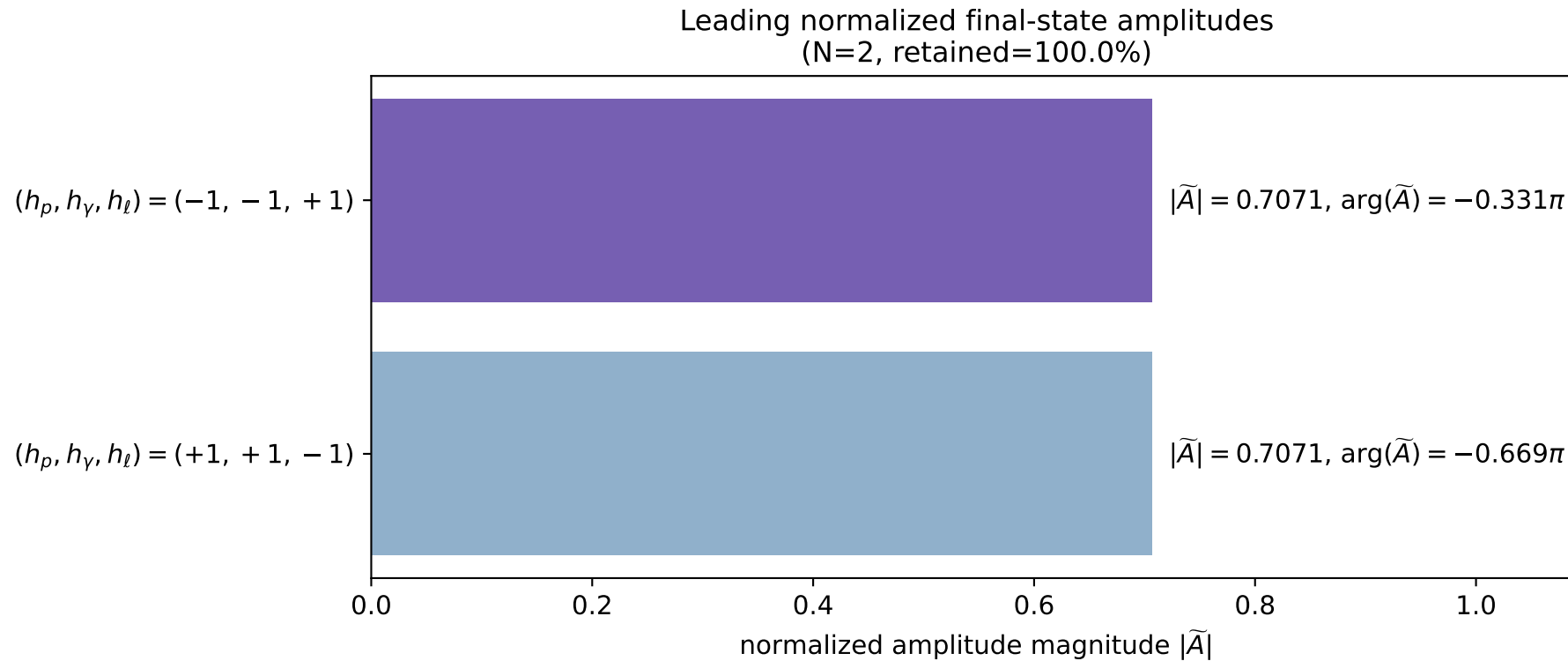


dghz_mixing_angles_polarization_03_local_minimum_53
 $d_{\text{GHZ}}=2.69649\text{e-}08$
 region: polarization_cluster_03_local_minimum_53
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0053
 lepton mixing angle: $\alpha_e=2.2615246$ rad
 incoming lepton: $-0.637099|+\rangle +0.770782|-\rangle$
 proton mixing angle: $\alpha_p=0.8800608$ rad
 incoming proton: $0.637104|+\rangle +0.770778|-\rangle$

$s=24.9999$, $\sqrt{s}=4.99999$, $|\vec{P}|=2.41201$, $|\vec{P}'|=0.00908808$
 $\theta_{P'}=0.000472367$, $\theta_\gamma=3.14159$, $\phi_{P'}=5.49656$, $\phi_\gamma=2.10193$
 $E_\gamma=2.03552$, $\theta_{\ell'}=2.11843\text{e-}06$, $\phi_{\ell'}=2.35497$
 $Q^2=19.5511$, $x_B=0.83527$, $t=-3.05175$, $W^2=4.73565$, $y=0.970432$

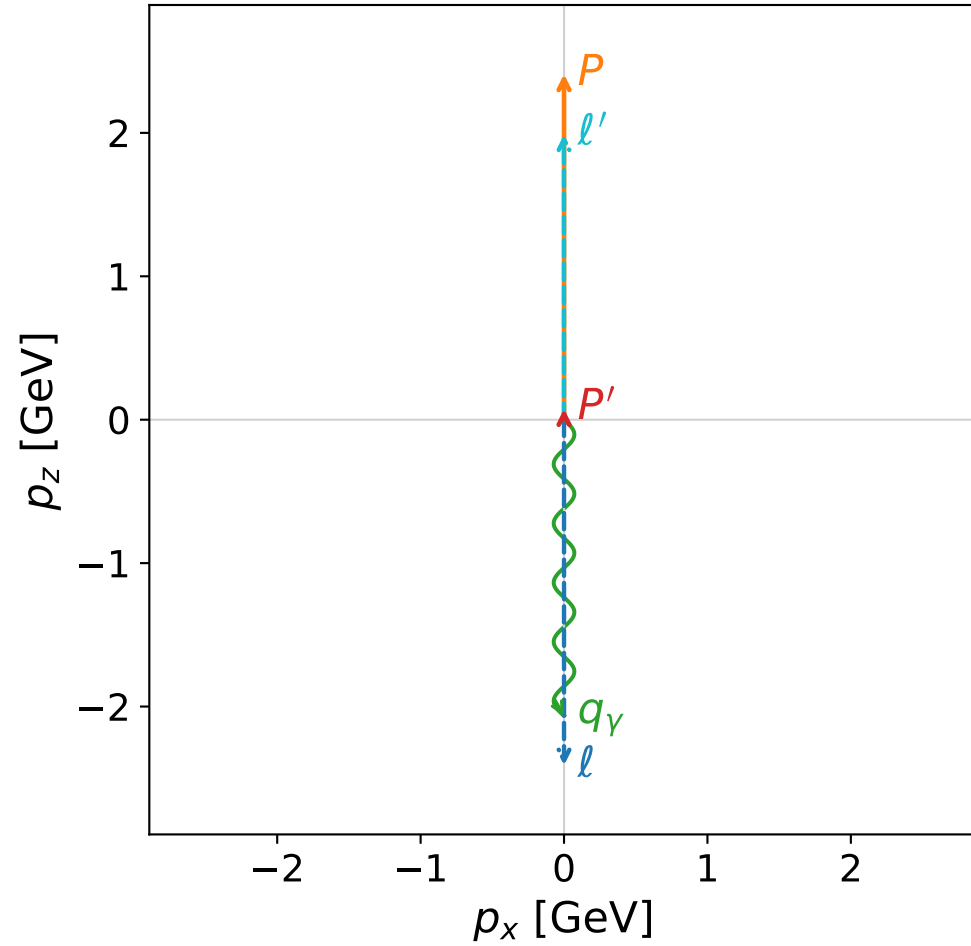
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0264$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0264$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0091$
 q_γ $E= 2.0355$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0355$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.786\text{e+}19$; $I_h=+3.601\text{e-}11$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.776\text{e-}14$; $I_{TL}=+2.506\text{e-}08$

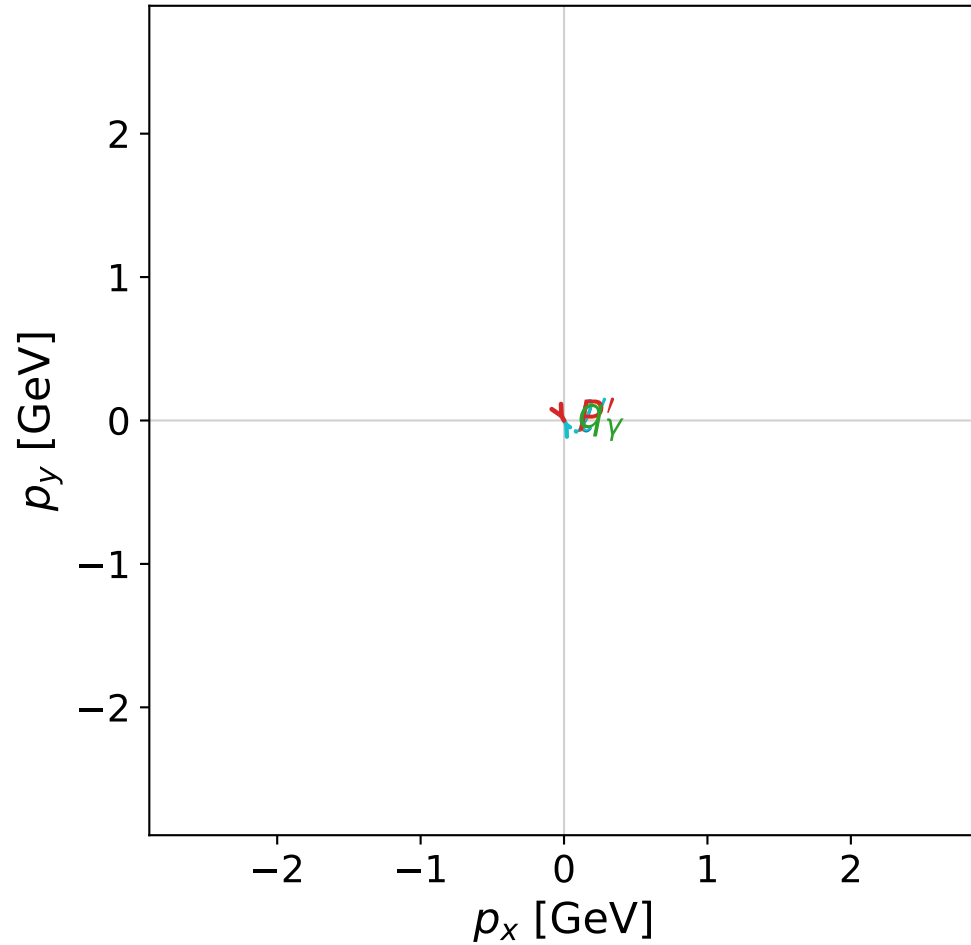


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

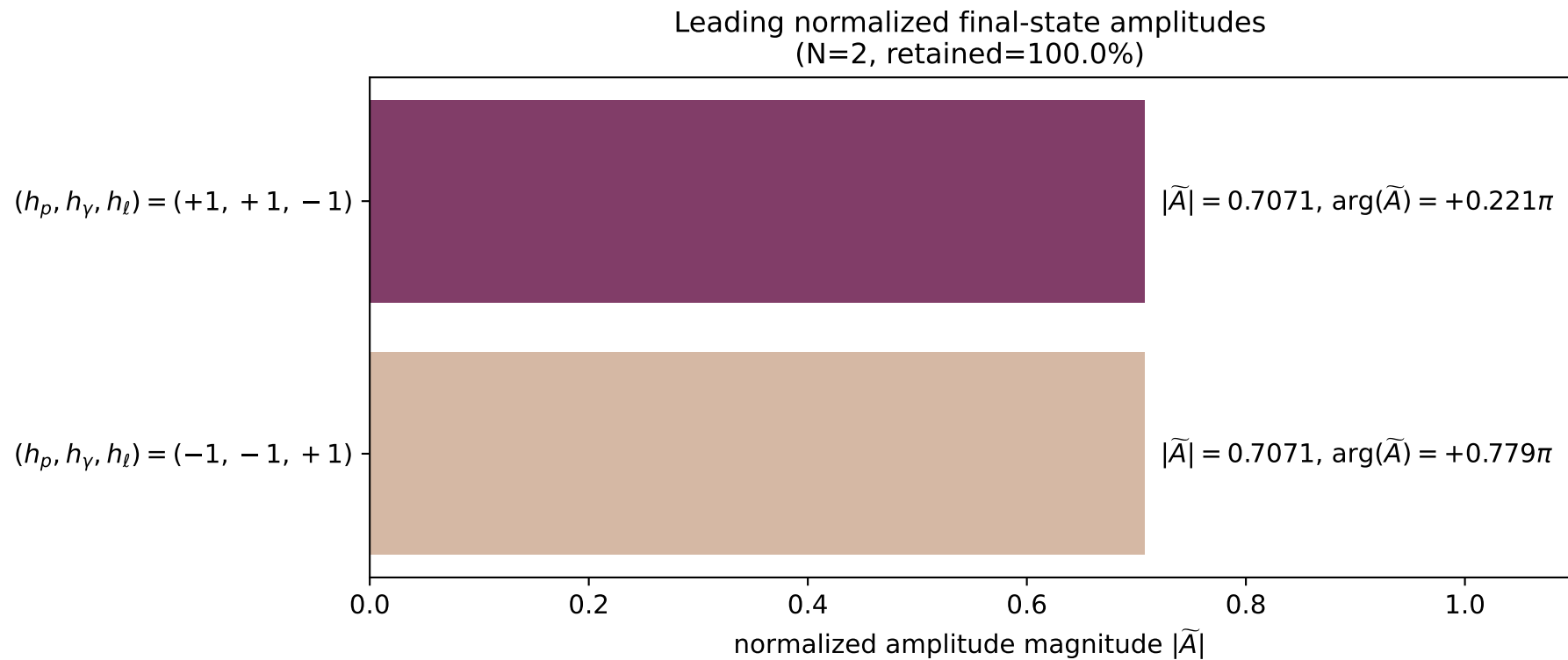


dghz_mixing_angles_polarization_03_local_minimum_55
 $d_{\text{GHZ}}=2.70614\text{e-}08$
 region: polarization_cluster_03_local_minimum_55
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0055
 lepton mixing angle: $\alpha_e=2.360551$ rad
 incoming lepton: $-0.710181|+\rangle +0.70402|-\rangle$
 proton mixing angle: $\alpha_p=0.78104501$ rad
 incoming proton: $0.710178|+\rangle +0.704022|-\rangle$

$s=24.9555$, $\sqrt{s}=4.99555$, $|\vec{P}|=2.40971$, $|\vec{P}'|=0.0757091$
 $\theta_{P'}=7.31897\text{e-}05$, $\theta_\gamma=3.14159$, $\phi_{P'}=5.21638$, $\phi_\gamma=5.58978$
 $E_\gamma=2.0651$, $\theta_{\ell'}=2.78532\text{e-}06$, $\phi_{\ell'}=2.07479$
 $Q^2=19.1755$, $x_B=0.820345$, $t=-2.74225$, $W^2=5.07927$, $y=0.970893$

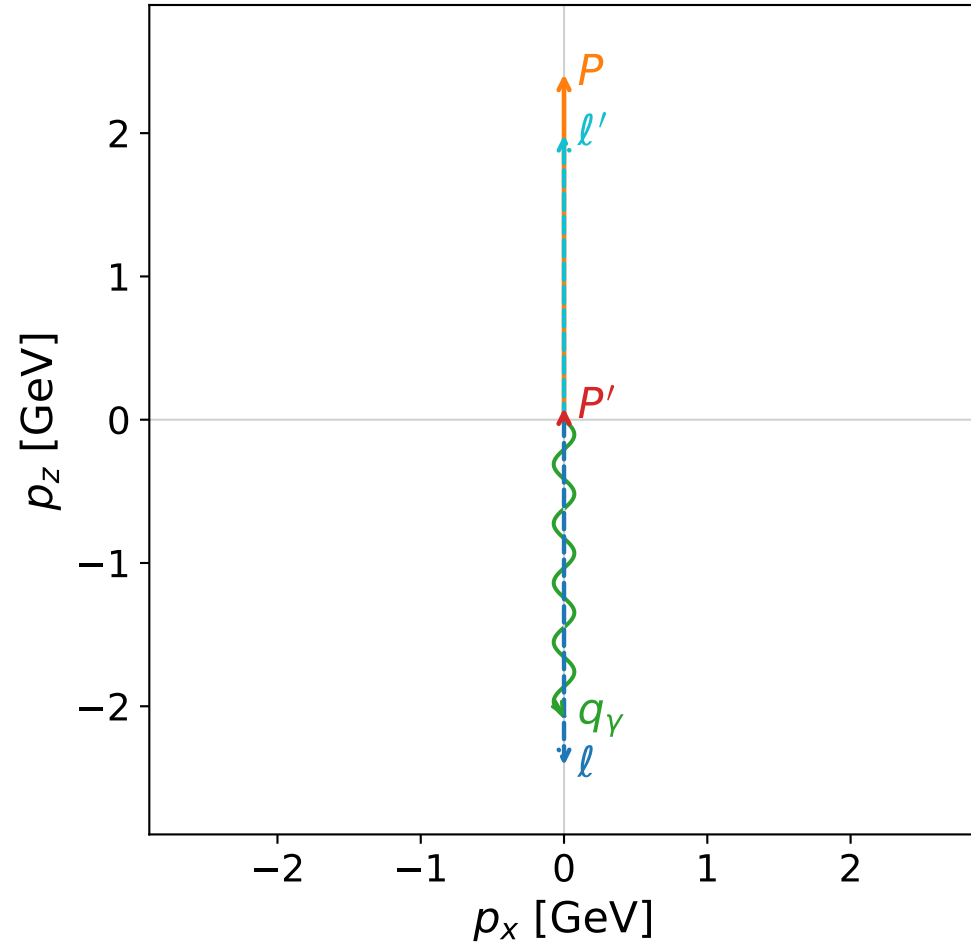
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4097$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4097$
 P $E= 2.5858$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4097$
 ℓ' $E= 1.9894$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.9894$
 P' $E= 0.9411$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0757$
 q_γ $E= 2.0651$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0651$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.097\text{e+}19$; $I_h=-1.448\text{e-}12$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=3.491\text{e-}14$; $I_{TL}=-1.107\text{e-}09$

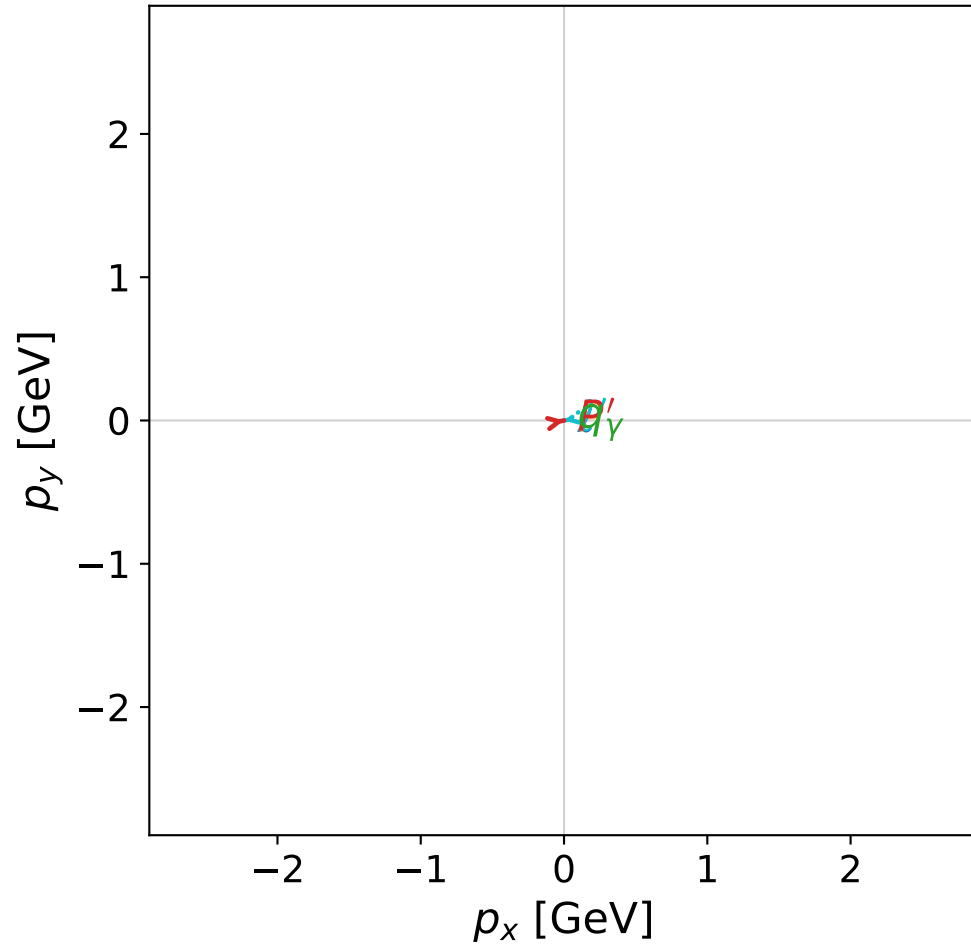


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

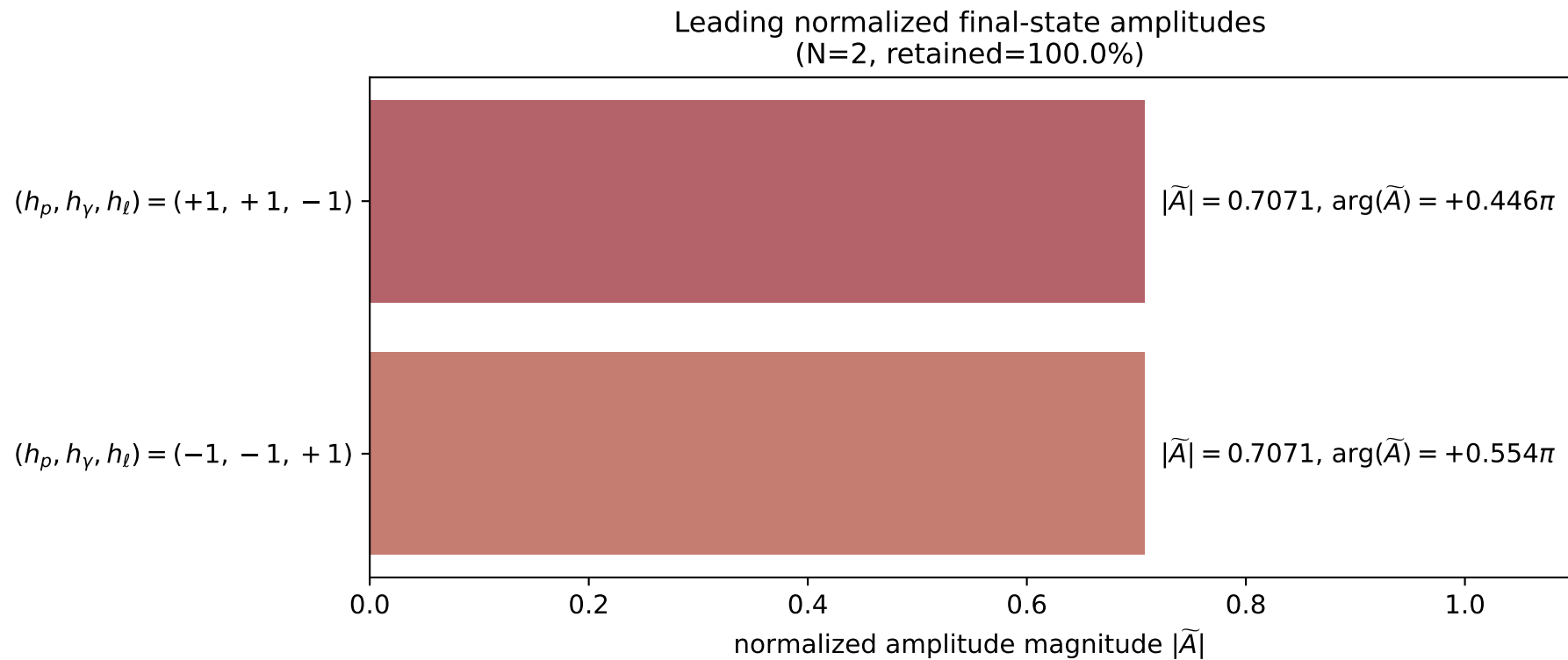


dghz_mixing_angles_polarization_03_local_minimum_56
 $d_{\text{GHZ}}=2.70735\text{e-}08$
 region: polarization_cluster_03_local_minimum_56
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0056
 lepton mixing angle: $\alpha_e=2.3601707$ rad
 incoming lepton: $-0.709913|+\rangle +0.70429|-\rangle$
 proton mixing angle: $\alpha_p=0.78143038$ rad
 incoming proton: $0.709907|+\rangle +0.704296|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0826338$
 $\theta_{P'}=7.6699\text{e-}05$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.192921$, $\phi_\gamma=4.88356$
 $E_\gamma=2.0705$, $\theta_{\ell'}=3.18833\text{e-}06$, $\phi_{\ell'}=3.33451$
 $Q^2=19.1791$, $x_B=0.818899$, $t=-2.71555$, $W^2=5.12133$, $y=0.970995$

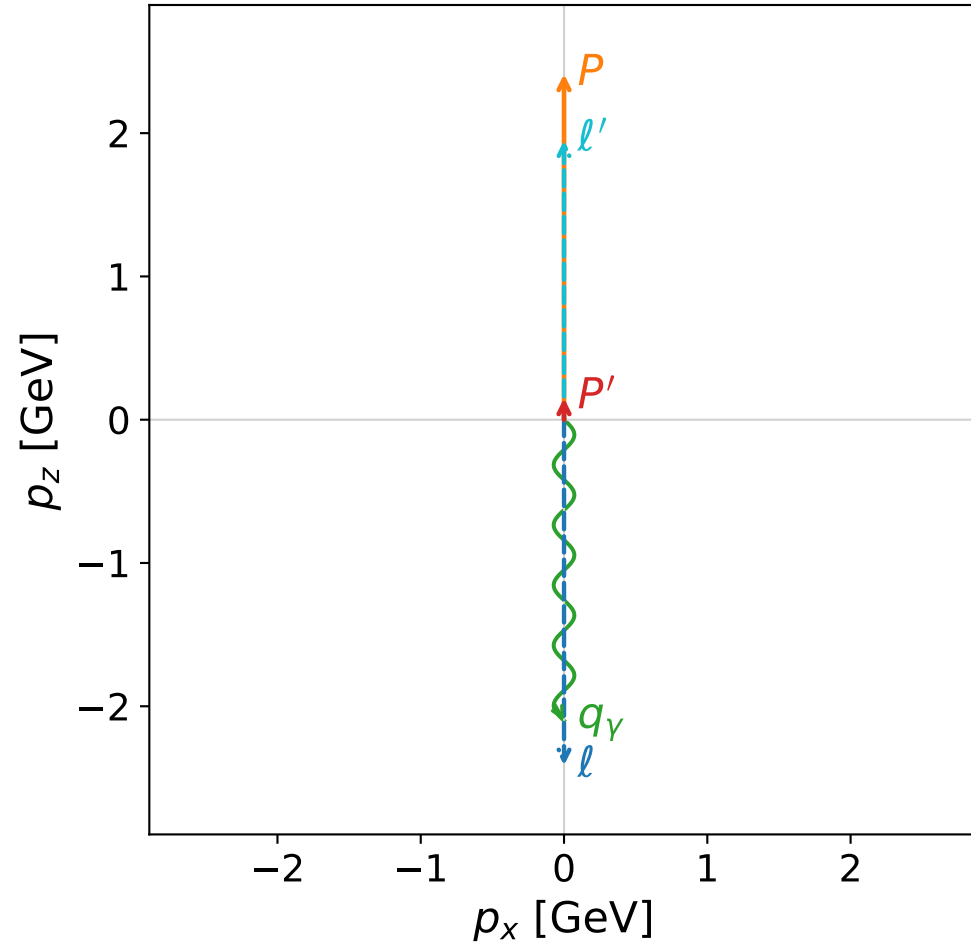
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 1.9879$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.9879$
 P' $E= 0.9416$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0826$
 q_γ $E= 2.0705$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0705$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=8.376\text{e+}18$; $I_h=-3.063\text{e-}12$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=4.634\text{e-}14$; $I_{TL}=-2.370\text{e-}09$

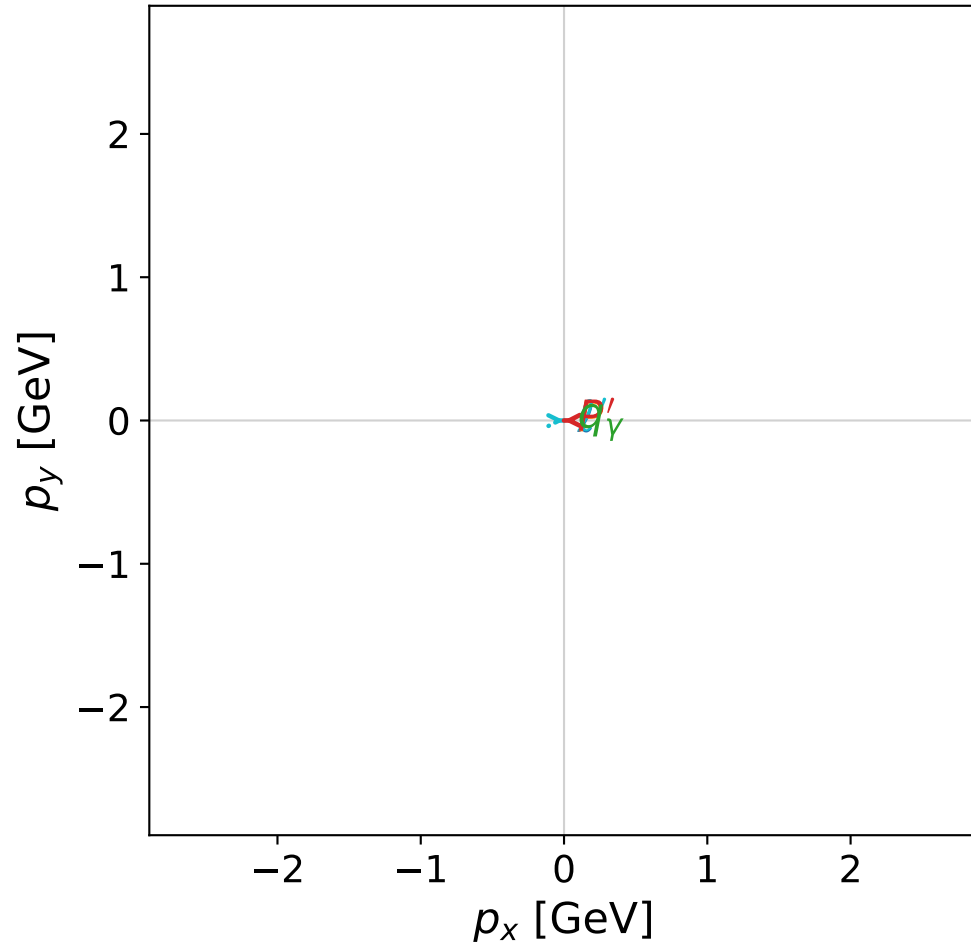


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

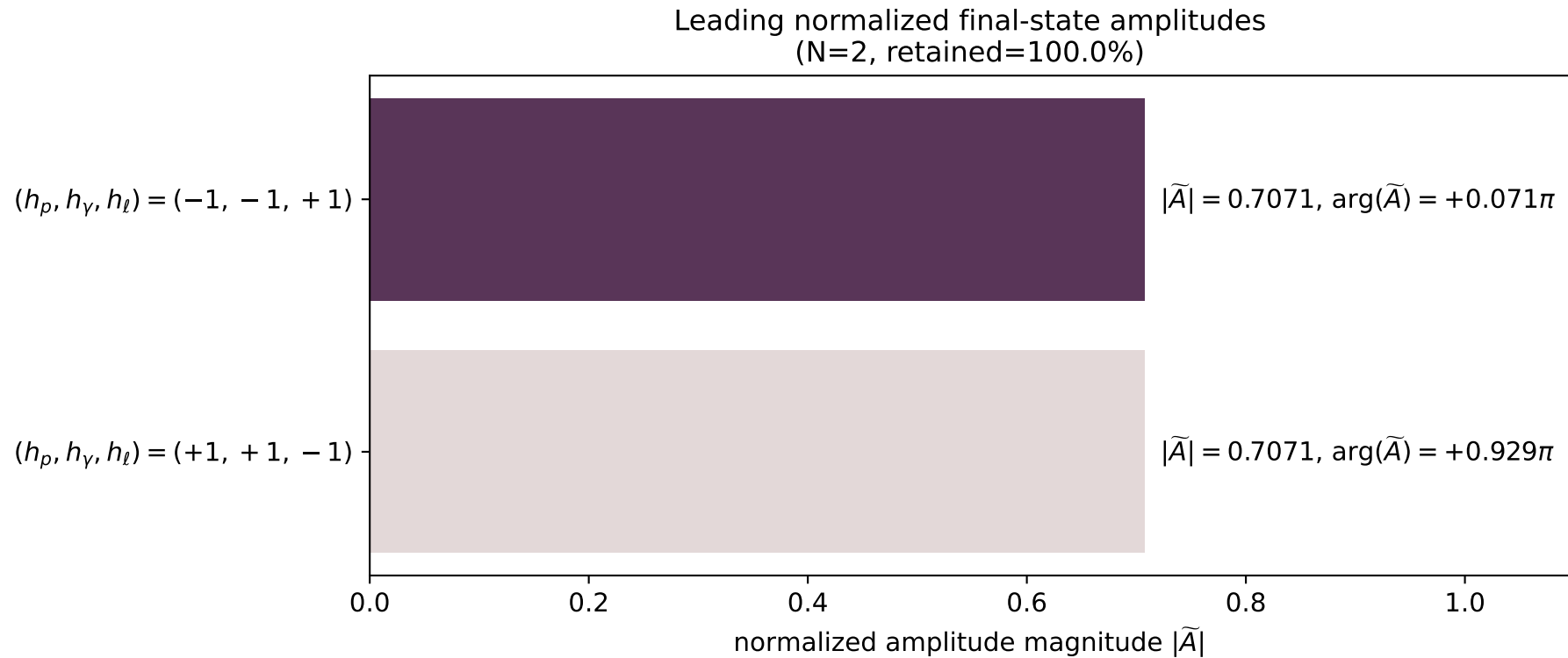


dghz_mixing_angles_polarization_03_local_minimum_60
 $d_{\text{GHZ}}=2.74796\text{e-}08$
 region: polarization_cluster_03_local_minimum_60
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0060
 lepton mixing angle: $\alpha_e=2.3243577$ rad
 incoming lepton: $-0.68424|+\rangle +0.729257|-\rangle$
 proton mixing angle: $\alpha_p=0.8172232$ rad
 incoming proton: $0.684249|+\rangle +0.729249|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.150619$
 $\theta_{P'}=5.43472\text{e-}05$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.14933$, $\phi_\gamma=3.36397$
 $E_\gamma=2.1003$, $\theta_{\ell'}=4.19848\text{e-}06$, $\phi_{\ell'}=0.00773789$
 $Q^2=18.8107$, $x_B=0.802708$, $t=-2.43097$, $W^2=5.50317$, $y=0.971552$

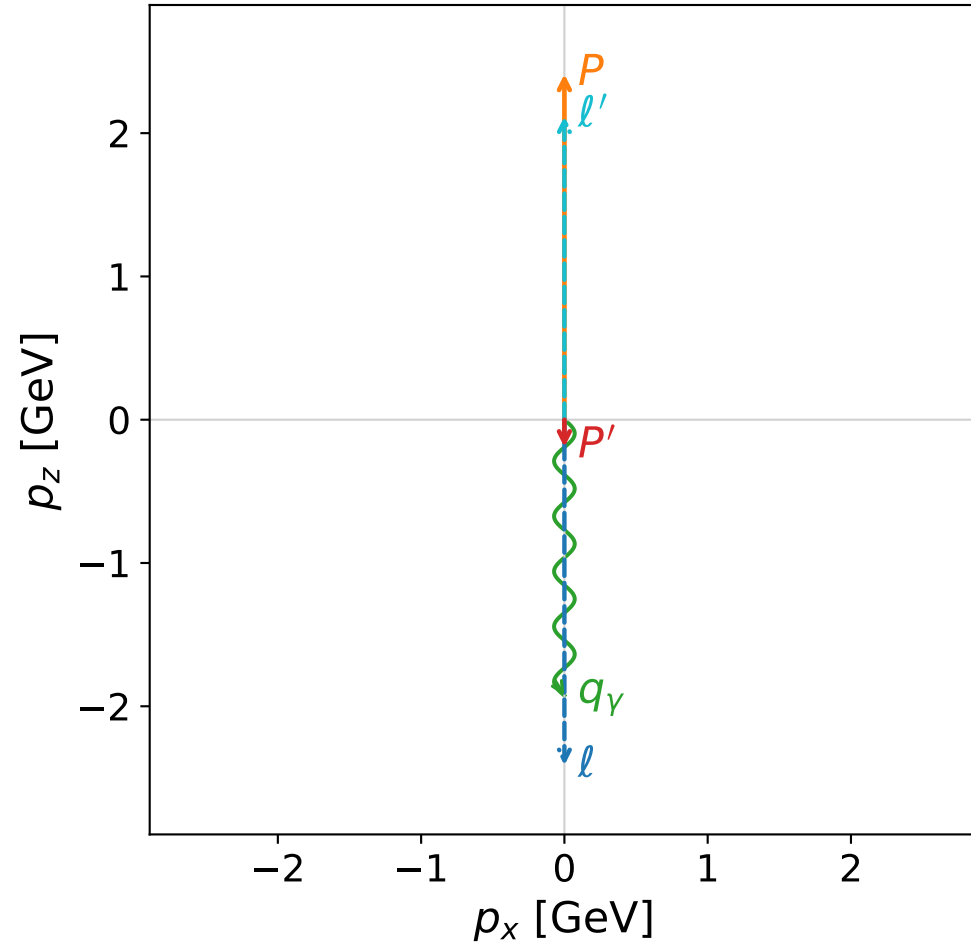
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 1.9497$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9497$
 P' $E= 0.9500$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.1506$
 q_γ $E= 2.1003$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.1003$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=4.836\text{e+}18$; $I_h=-3.254\text{e-}11$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=9.409\text{e-}14$; $I_{TL}=-2.759\text{e-}08$

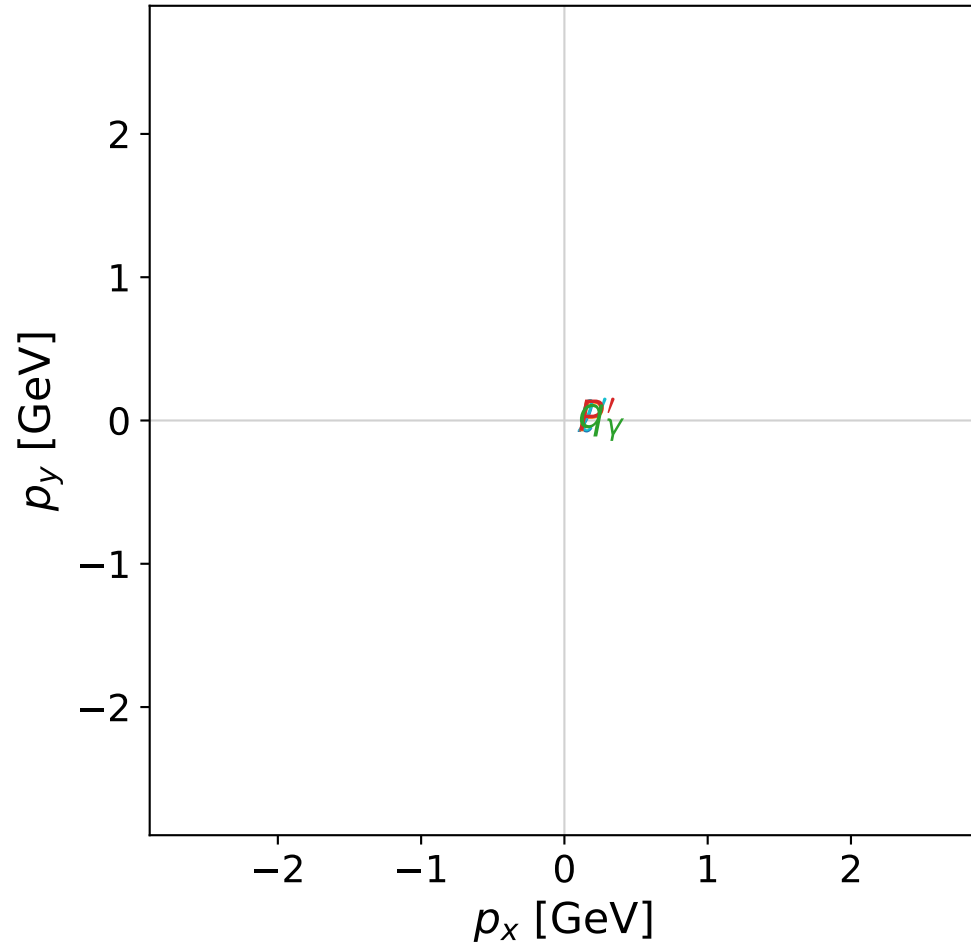


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



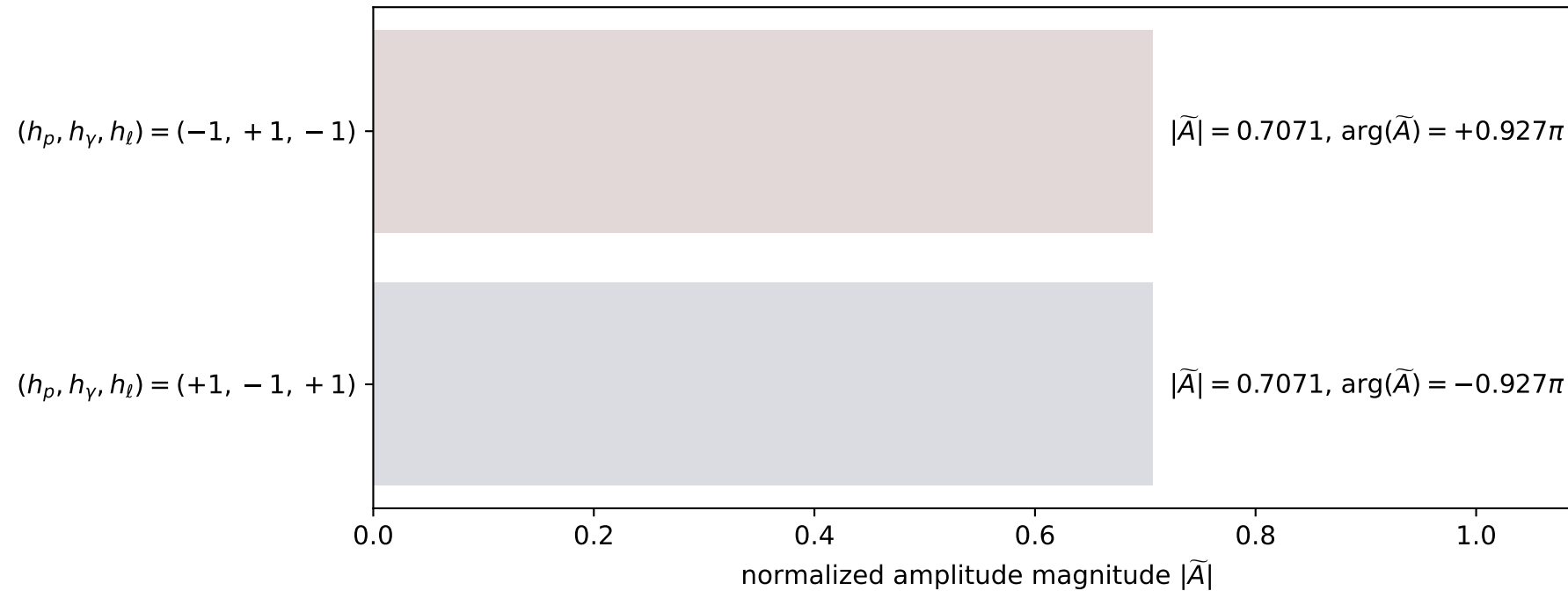
dghz_mixing_angles_polarization_03_local_minimum_66
 $d_{\text{GHZ}}=2.78662\text{e-}08$
 region: polarization_cluster_03_local_minimum_66
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0066
 lepton mixing angle: $\alpha_e=2.5600686$ rad
 incoming lepton: $-0.835626|+\rangle +0.549298|-\rangle$
 proton mixing angle: $\alpha_p=0.58153866$ rad
 incoming proton: $0.835618|+\rangle +0.54931|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.192212$
 $\theta_{P'}=3.14159$, $\theta_Y=3.14159$, $\phi_{P'}=3.14281$, $\phi_Y=0.229158$
 $E_Y=1.92515$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.39574$
 $Q^2=20.4284$, $x_B=0.873944$, $t=-4.12349$, $W^2=3.8264$, $y=0.969106$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.1174$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.1174$
 P' $E= 0.9575$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.1922$
 q_Y $E= 1.9251$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9251$

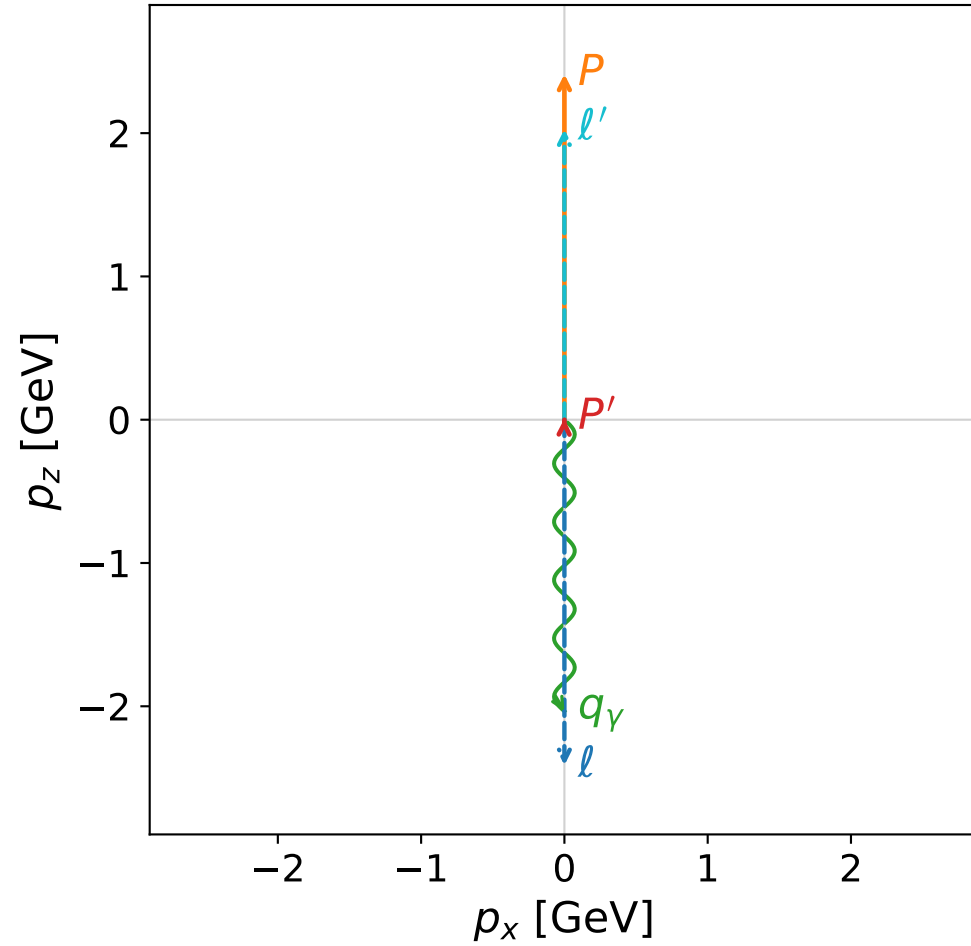
GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.362\text{e}+24$; $I_h=+1.230\text{e-}13$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.874\text{e-}27$; $I_{TL}=+9.779\text{e-}18$

Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

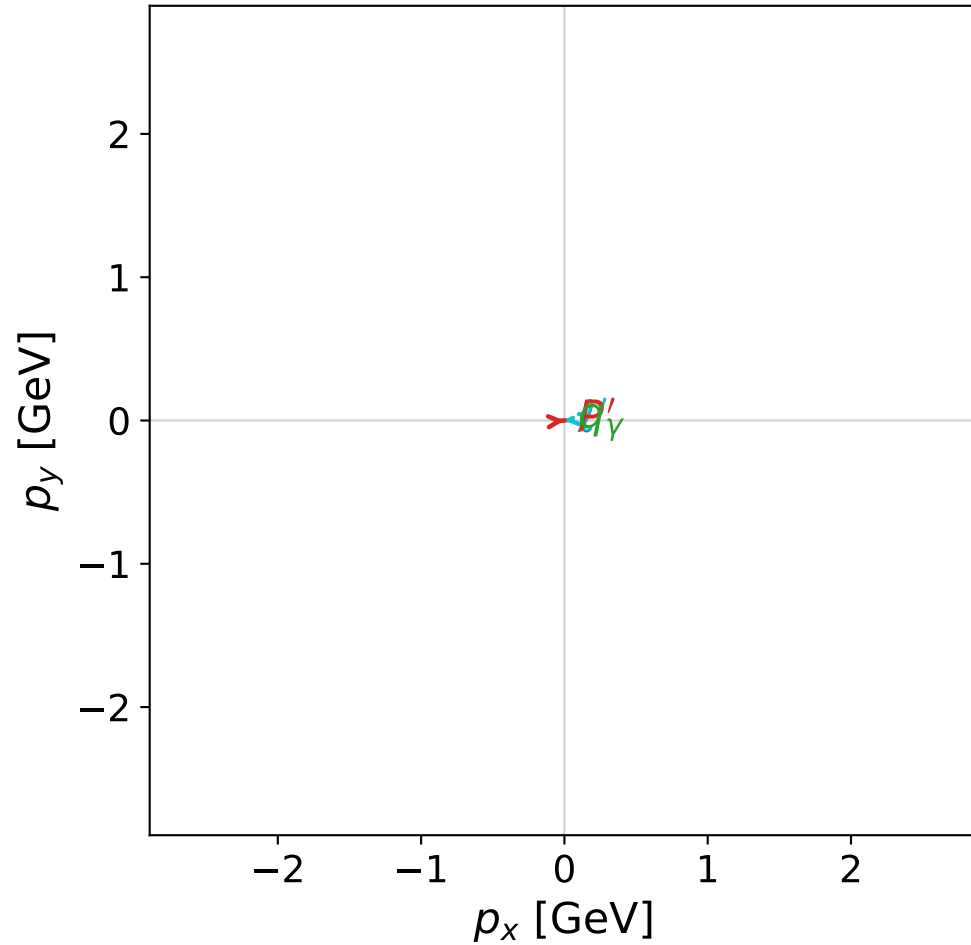


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



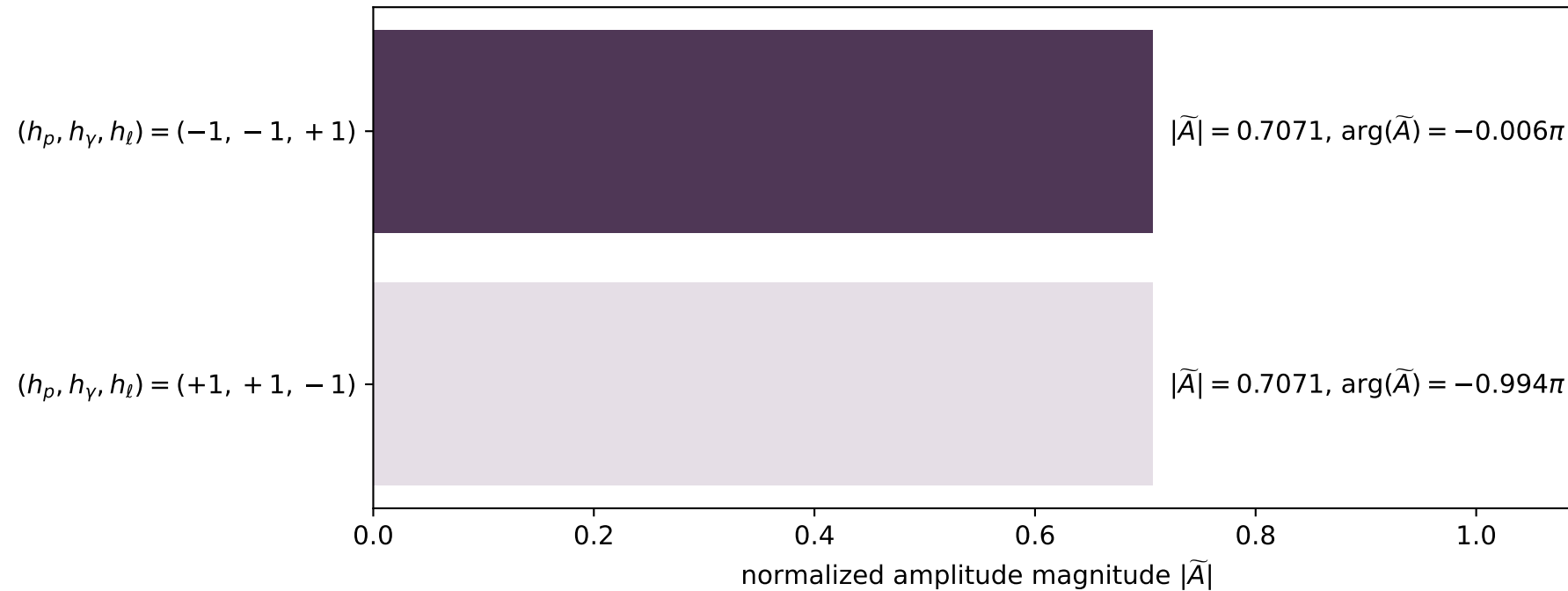
dghz_mixing_angles_polarization_03_local_minimum_67
 $d_{\text{GHZ}}=2.80689\text{e-}08$
 region: polarization_cluster_03_local_minimum_67
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0067
 lepton mixing angle: $\alpha_e=2.235145$ rad
 incoming lepton: $-0.616546|+\rangle +0.787319|-\rangle$
 proton mixing angle: $\alpha_p=0.90642644$ rad
 incoming proton: $0.616563|+\rangle +0.787305|-\rangle$

$s=24.9985$, $\sqrt{s}=4.99985$, $|\vec{P}|=2.41194$, $|\vec{P}'|=0.0078299$
 $\theta_{P'}=0.0184851$, $\theta_Y=3.14159$, $\phi_{P'}=0.0794118$, $\phi_Y=3.1225$
 $E_Y=2.03482$, $\theta_{\ell'}=7.14003\text{e-}05$, $\phi_{\ell'}=3.221$
 $Q^2=19.556$, $x_B=0.835536$, $t=-3.05764$, $W^2=4.72917$, $y=0.970421$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4119$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4119$
 P $E= 2.5879$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4119$
 ℓ' $E= 2.0270$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= 2.0270$
 P' $E= 0.9380$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= 0.0078$
 q_Y $E= 2.0348$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0348$

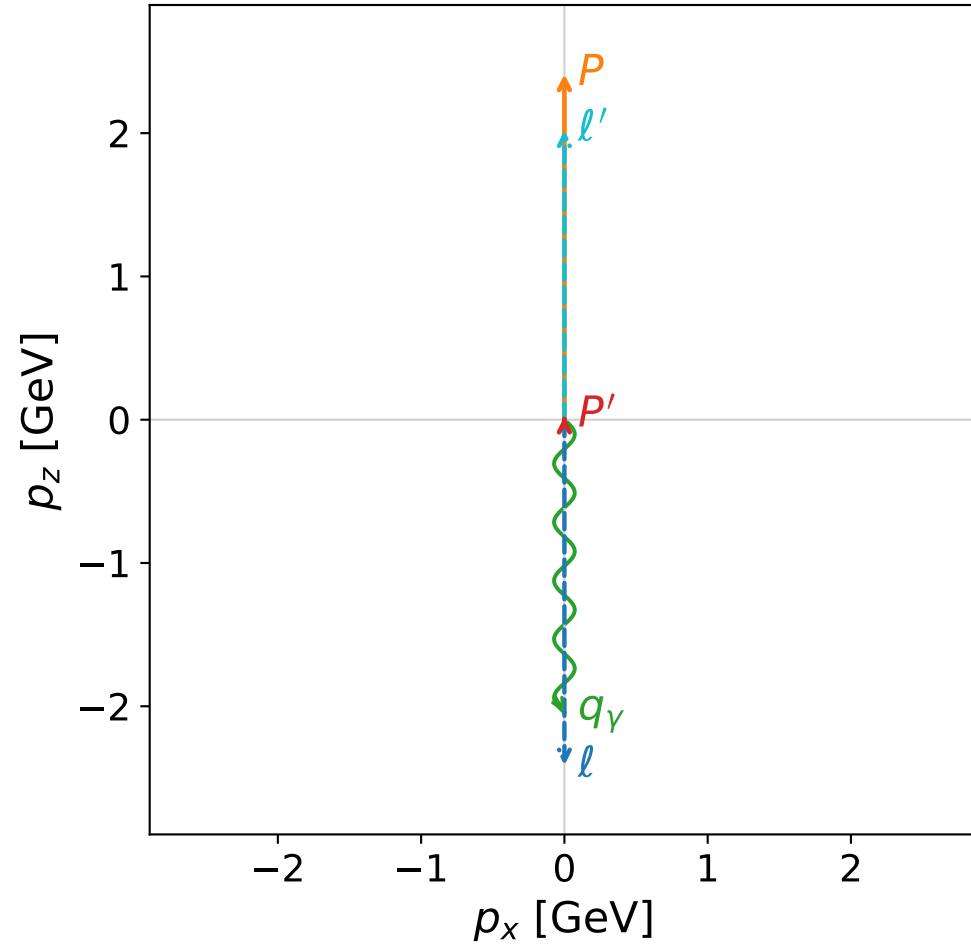
GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.530\text{e+}16$; $I_h=+2.207\text{e-}09$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=2.059\text{e-}11$; $I_{TL}=+1.537\text{e-}06$

Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

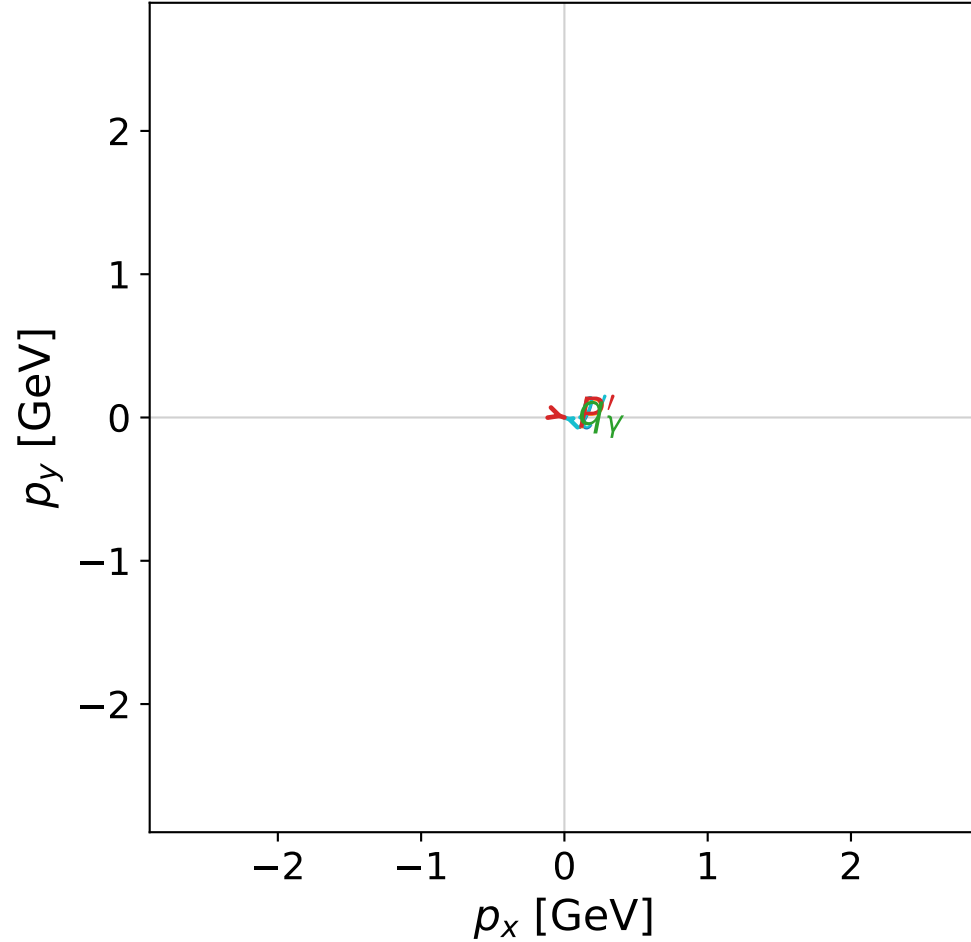


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

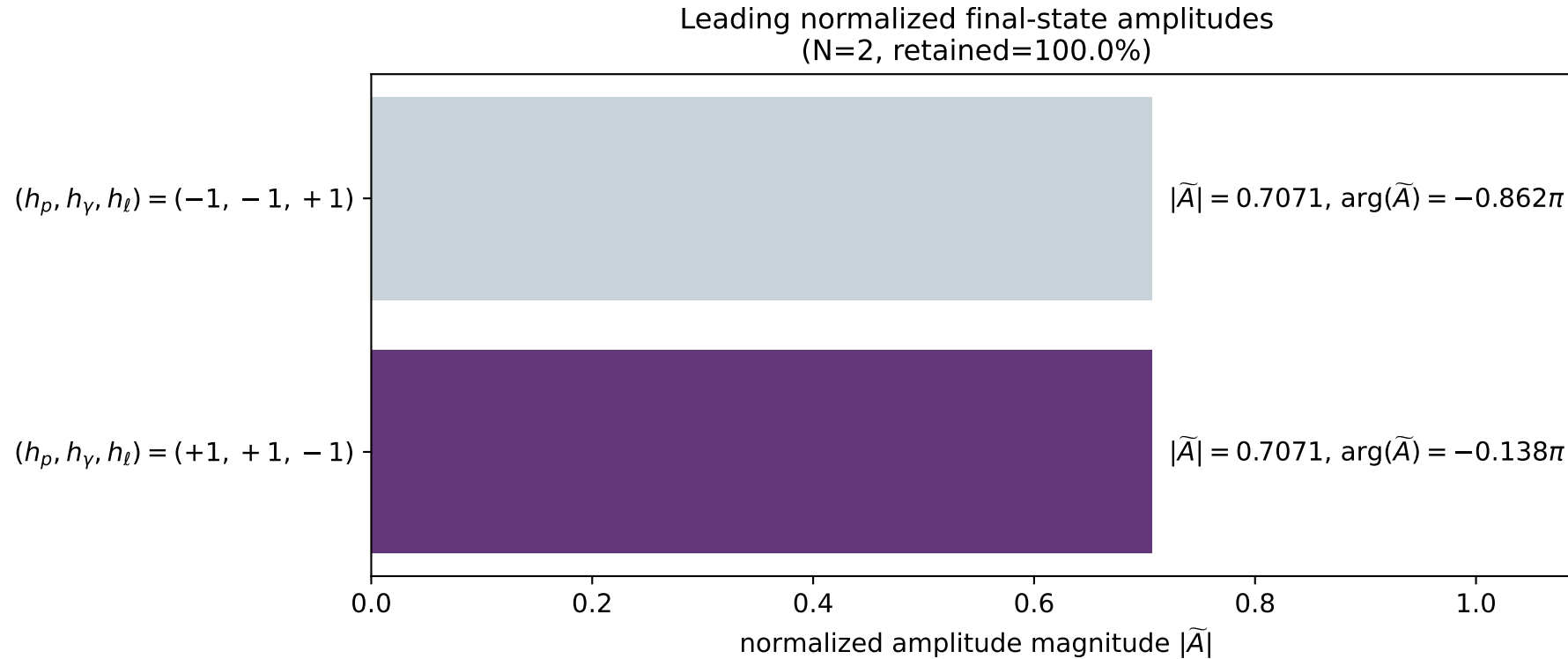


dghz_mixing_angles_polarization_03_local_minimum_73
 $d_{\text{GHZ}}=2.91539\text{e-}08$
 region: polarization_cluster_03_local_minimum_73
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0073
 lepton mixing angle: $\alpha_e=2.1495745$ rad
 incoming lepton: $-0.547002|+\rangle +0.837132|-\rangle$
 proton mixing angle: $\alpha_p=0.99200753$ rad
 incoming proton: $0.54701|+\rangle +0.837126|-\rangle$

$s=24.9997$, $\sqrt{s}=4.99997$, $|\vec{P}|=2.412$, $|\vec{P}'|=0.0232778$
 $\theta_{P'}=0.000120512$, $\theta_\gamma=3.14159$, $\phi_{P'}=5.96763$, $\phi_\gamma=0.433102$
 $E_\gamma=2.04248$, $\theta_{\ell'}=1.38933\text{e-}06$, $\phi_{\ell'}=2.82604$
 $Q^2=19.4813$, $x_B=0.832205$, $t=-2.98455$, $W^2=4.8078$, $y=0.970537$

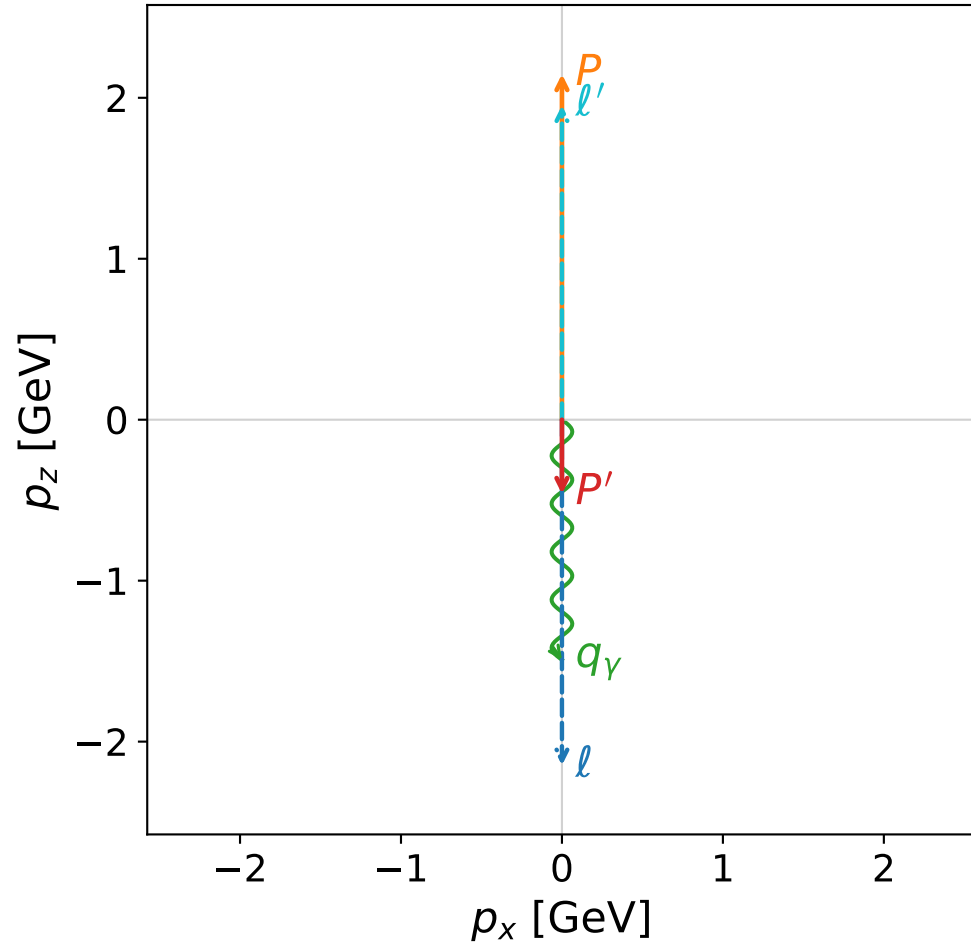
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0192$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0192$
 P' $E= 0.9383$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0233$
 q_γ $E= 2.0425$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0425$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=3.509\text{e+}19$; $I_h=+7.236\text{e-}11$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=9.101\text{e-}15$; $I_{TL}=+5.150\text{e-}08$

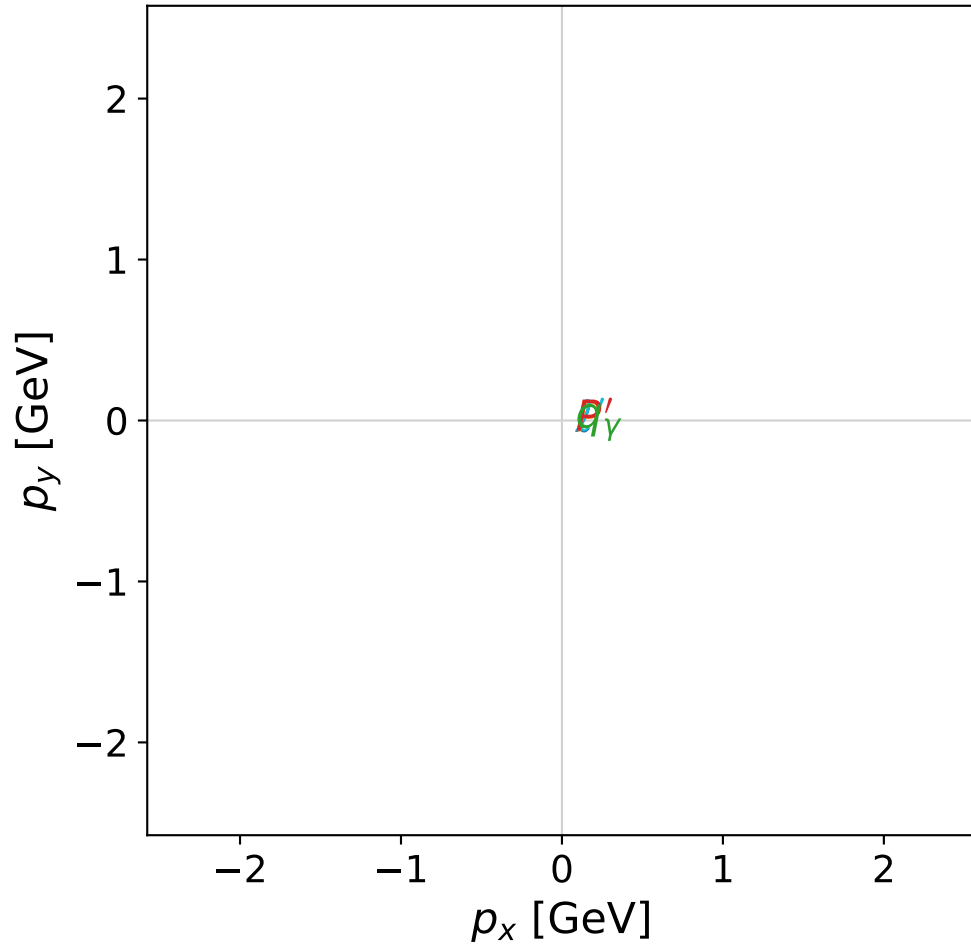


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



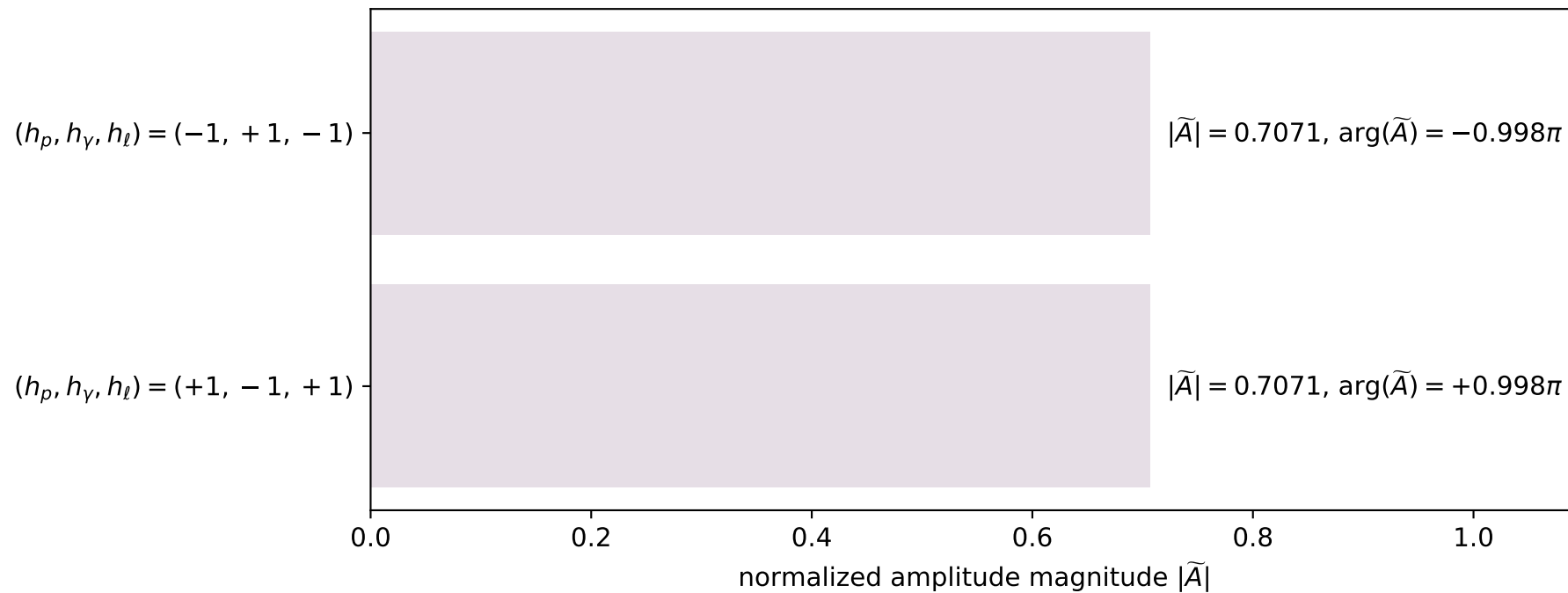
dghz_mixing_angles_polarization_03_local_minimum_87
 $d_{\text{GHZ}}=3.19716\text{e-}08$
 region: polarization_cluster_03_local_minimum_87
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0087
 lepton mixing angle: $\alpha_e=2.4687294$ rad
 incoming lepton: $-0.78204|+\rangle +0.623228|-\rangle$
 proton mixing angle: $\alpha_p=0.67288163$ rad
 incoming proton: $0.782029|+\rangle +0.623242|-\rangle$

$s=20.1618$, $\sqrt{s}=4.49019$, $|\vec{P}|=2.14712$, $|\vec{P}'|=0.461231$
 $\theta_{P'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.72593$, $\phi_\gamma=6.27676$
 $E_\gamma=1.49185$, $\theta_{l'}=0$, $\phi_{l'}=2.90749$
 $Q^2=16.774$, $x_B=0.905891$, $t=-5.1192$, $W^2=2.62242$, $y=0.960305$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.1471$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1471$
 P $E= 2.3431$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1471$
 l' $E= 1.9531$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.9531$
 P' $E= 1.0453$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.4612$
 q_γ $E= 1.4918$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.4918$

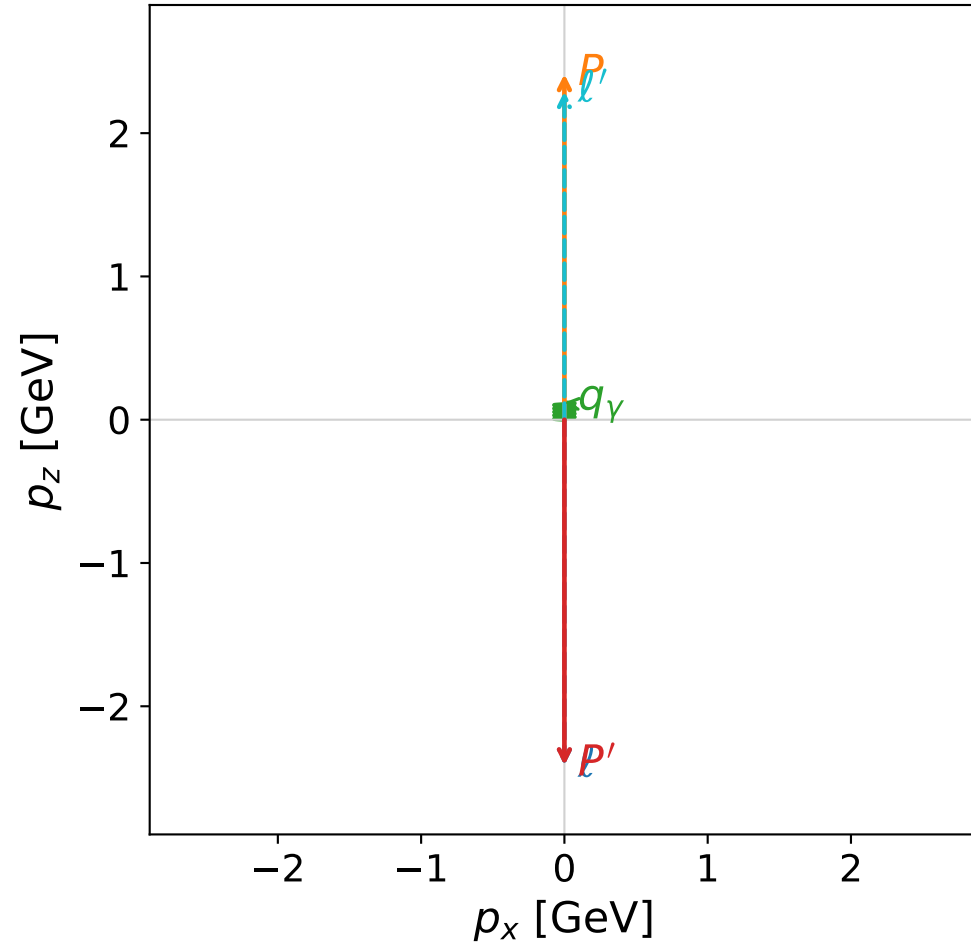
GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.510\text{e+}24$; $I_h=+1.035\text{e-}13$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.677\text{e-}31$; $I_{TL}=+5.242\text{e-}18$

Leading normalized final-state amplitudes
(N=2, retained=100.0%)

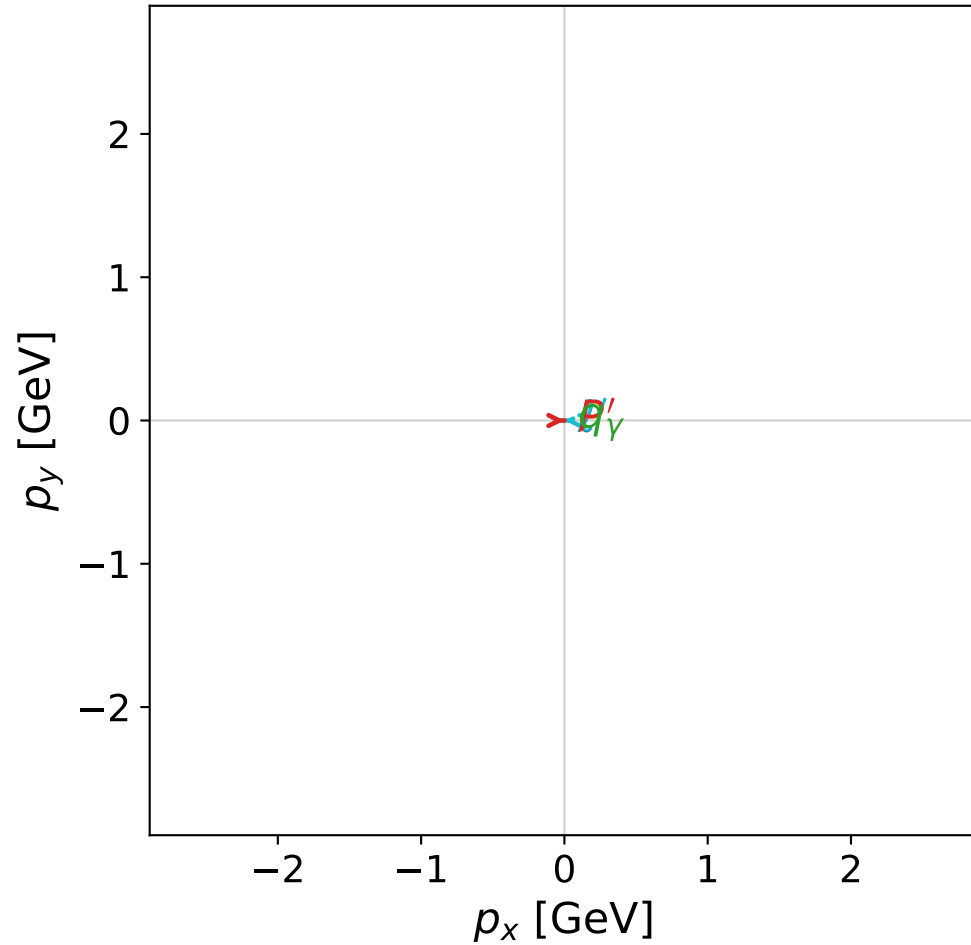


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



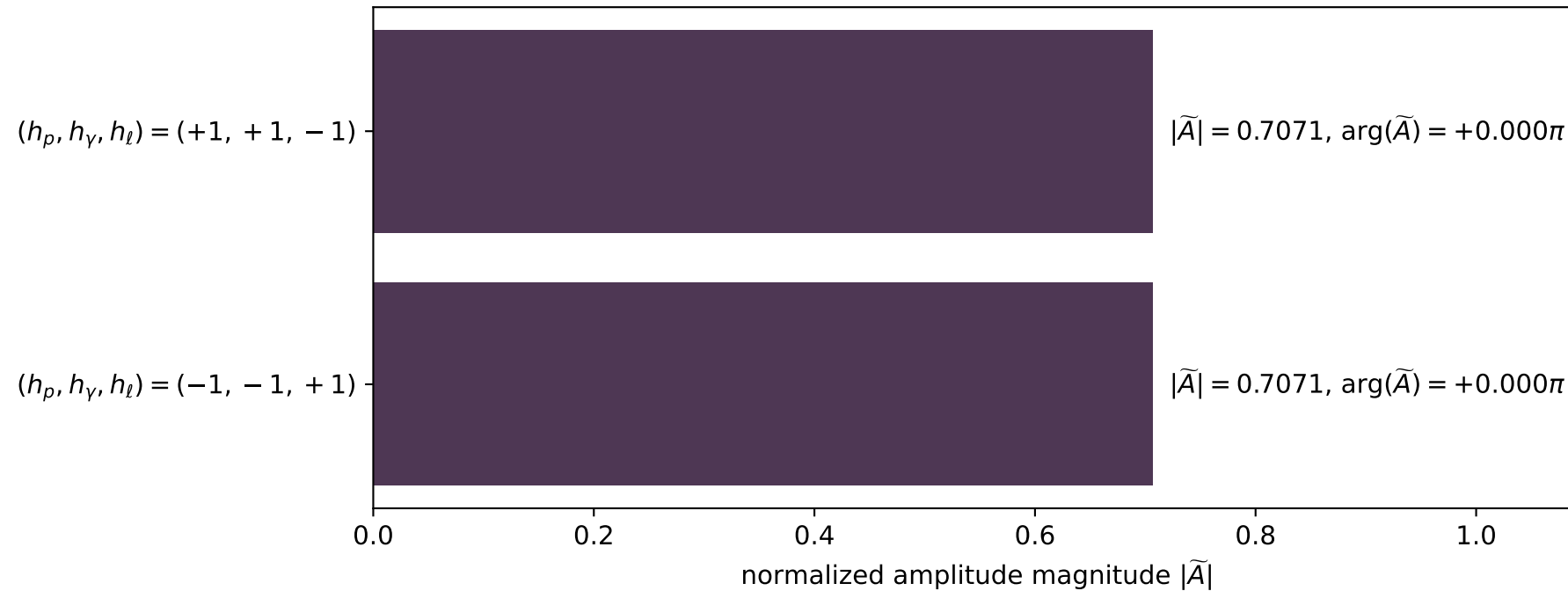
dghz_mixing_angles_polarization_03_local_minimum_89
 $d_{\text{GHZ}}=3.21673\text{e-}08$
 region: polarization_cluster_03_local_minimum_89
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0089
 lepton mixing angle: $\alpha_e=1.9834494$ rad
 incoming lepton: $-0.401041|+\rangle +0.91606|-\rangle$
 proton mixing angle: $\alpha_p=0.41264296$ rad
 incoming proton: $0.916064|+\rangle +0.401032|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.41202$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=0$, $\phi_\gamma=0$
 $E_\gamma=0.120596$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.14159$
 $Q^2=22.1078$, $x_B=0.948273$, $t=-23.2713$, $W^2=2.0858$, $y=0.966566$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.2914$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.2914$
 P' $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 q_γ $E= 0.1206$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1206$

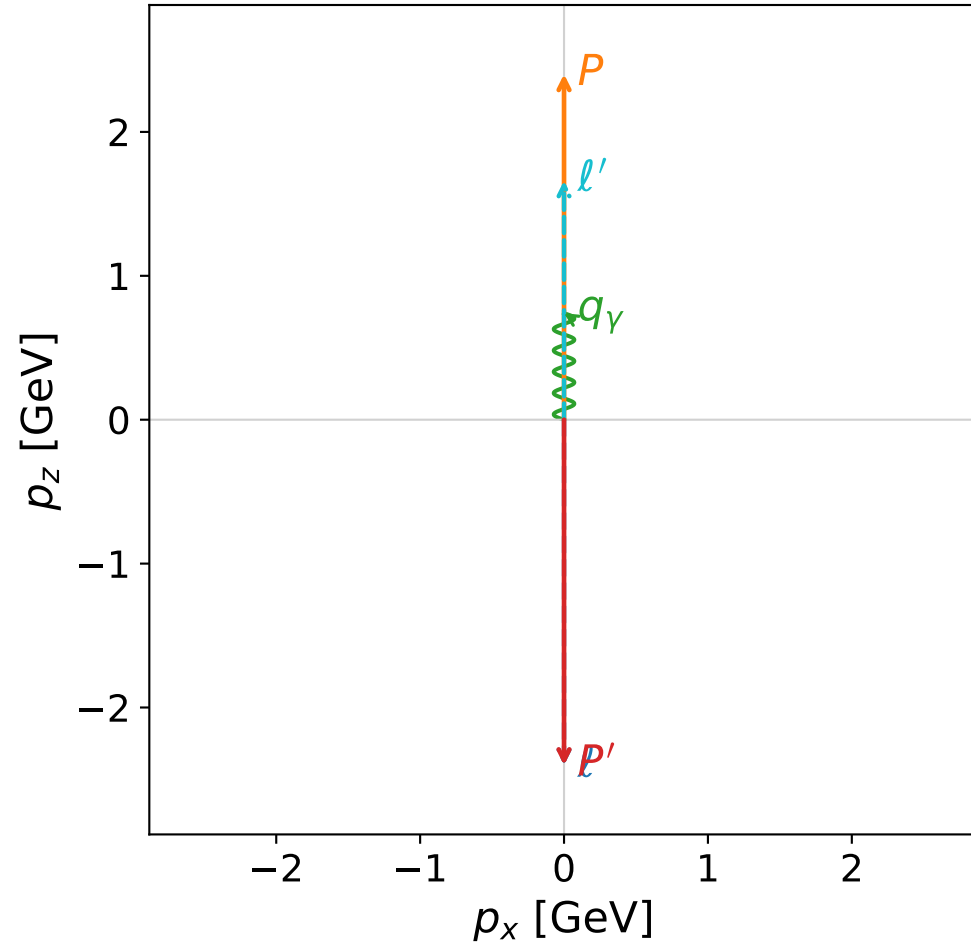
GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=3.928\text{e+}21$; $I_h=-2.607\text{e-}14$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=6.295\text{e-}28$; $I_{TL}=-2.595\text{e-}14$

Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

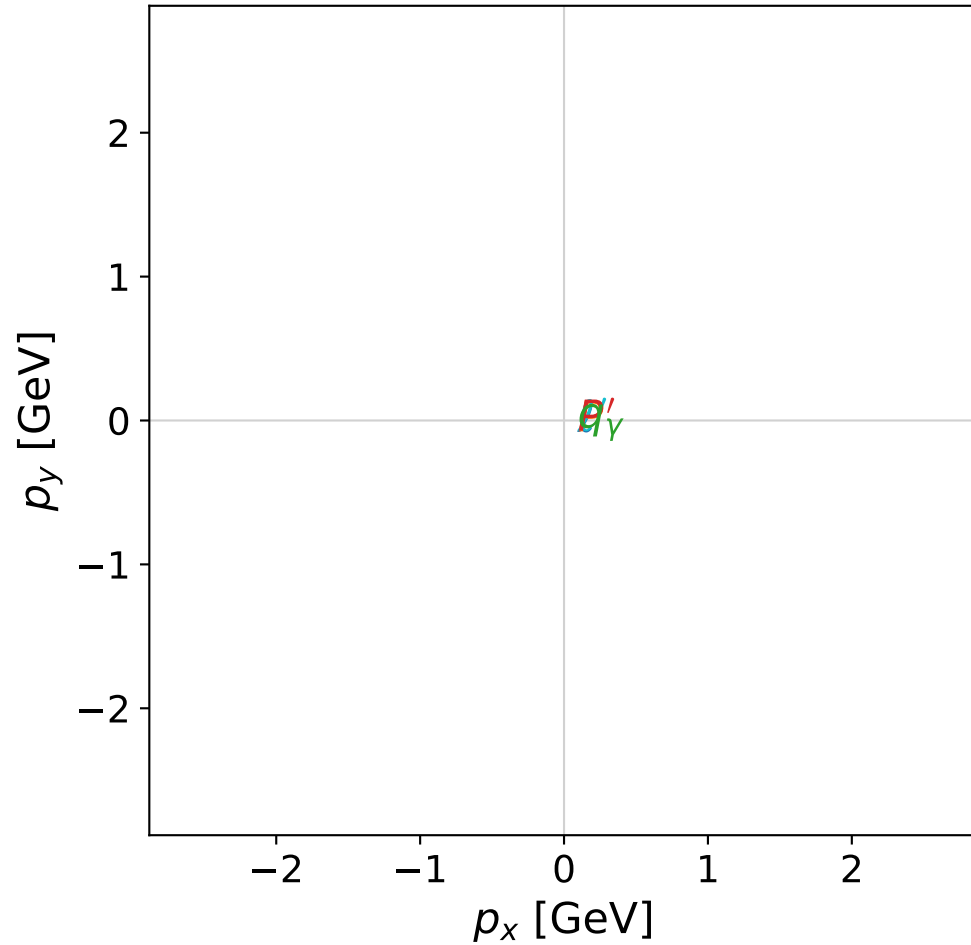


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

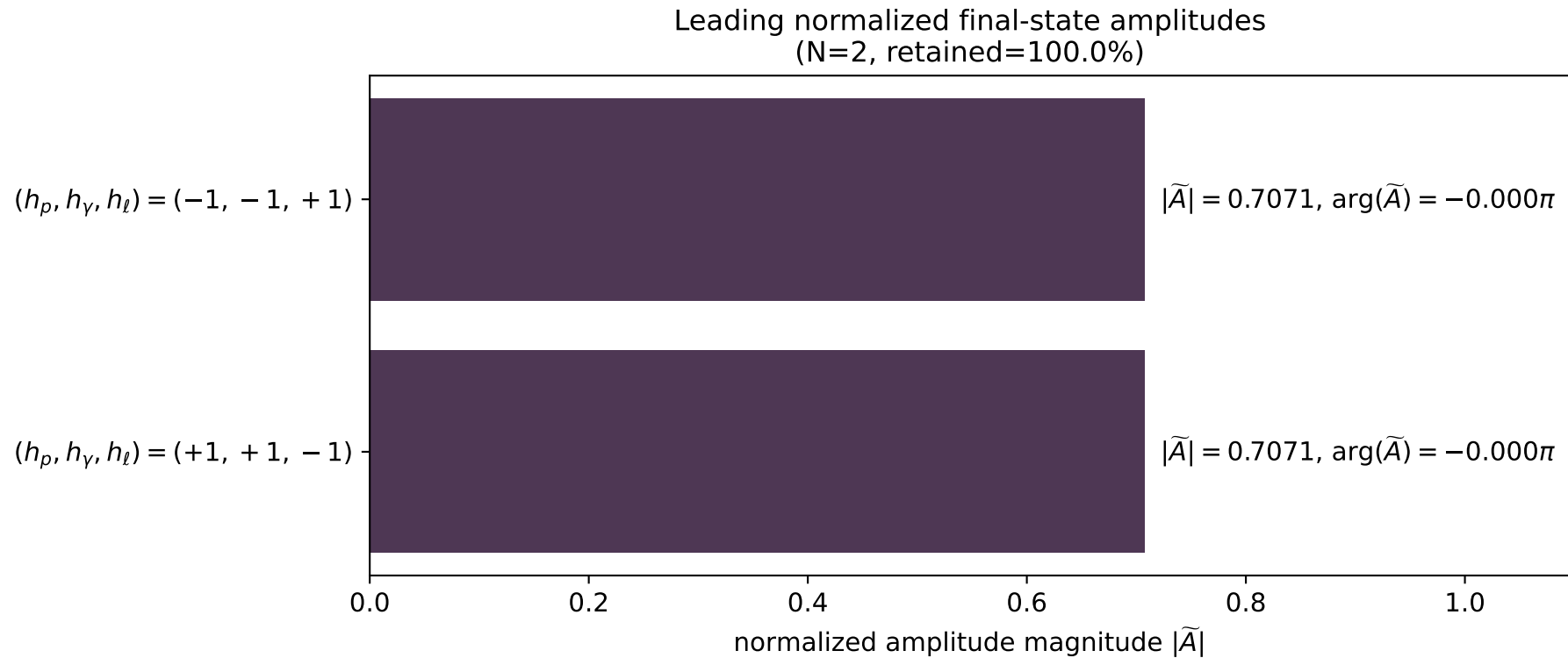


dghz_mixing_angles_polarization_03_local_minimum_92
 $d_{\text{GHZ}}=3.31027\text{e-}08$
 region: polarization_cluster_03_local_minimum_92
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0092
 lepton mixing angle: $\alpha_e=2.4281603$ rad
 incoming lepton: $-0.75612|+\rangle +0.654433|-\rangle$
 proton mixing angle: $\alpha_p=0.85738407$ rad
 incoming proton: $0.654418|+\rangle +0.756133|-\rangle$

$s=24.8013$, $\sqrt{s}=4.98009$, $|\vec{P}|=2.40171$, $|\vec{P}'|=2.40171$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=5.23216$, $\phi_\gamma=5.8557$
 $E_\gamma=0.740299$, $\theta_{l'}=0$, $\phi_{l'}=2.09057$
 $Q^2=15.9609$, $x_B=0.684007$, $t=-23.0728$, $W^2=8.25336$, $y=0.975459$

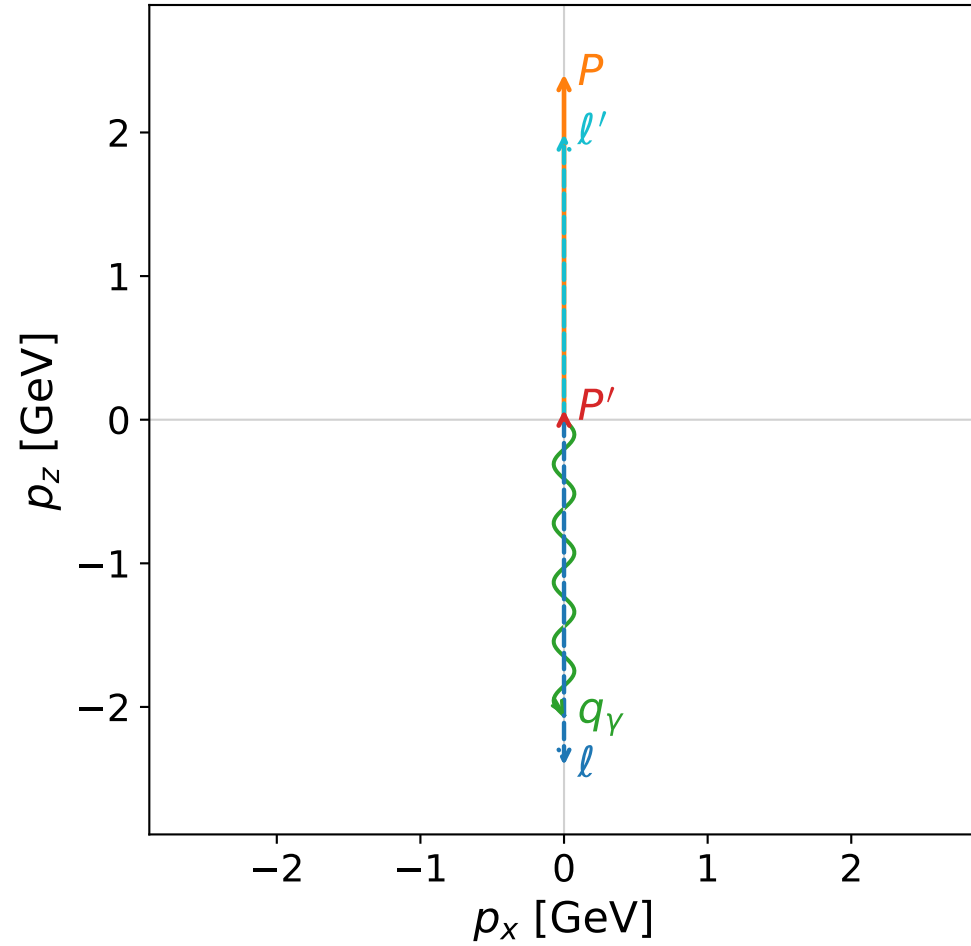
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4017$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4017$
 P $E= 2.5784$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4017$
 l' $E= 1.6614$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.6614$
 P' $E= 2.5784$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.4017$
 q_γ $E= 0.7403$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.7403$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.936\text{e}+23$; $I_h=+6.687\text{e-}14$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=2.283\text{e-}27$; $I_{TL}=+6.687\text{e-}14$

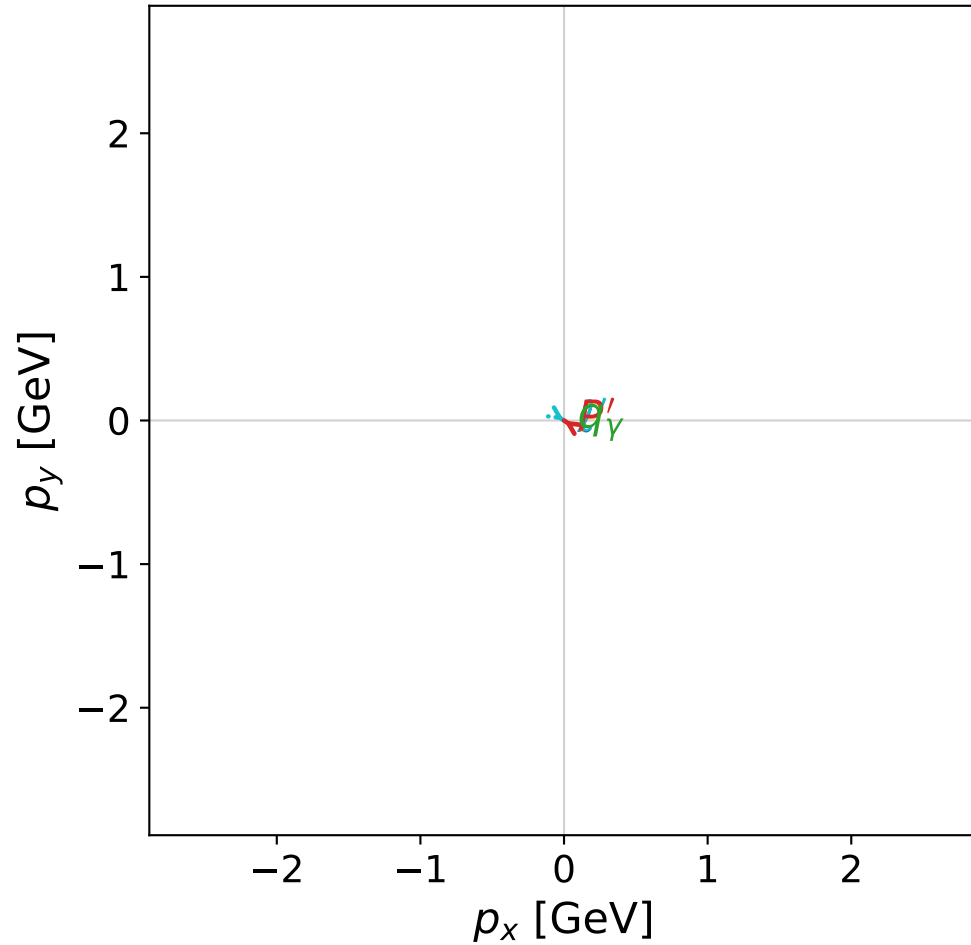


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

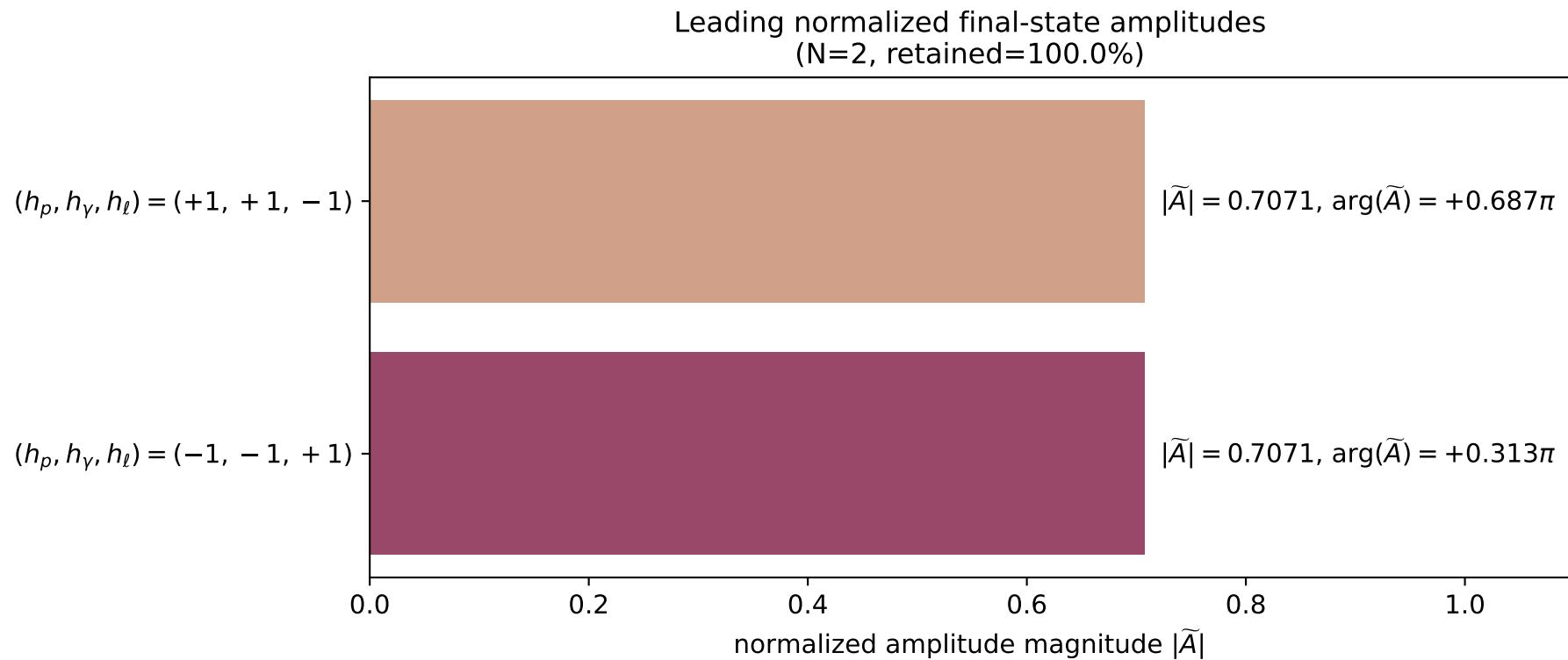


dghz_mixing_angles_polarization_03_local_minimum_93
 $d_{\text{GHZ}}=3.31851\text{e-}08$
 region: polarization_cluster_03_local_minimum_93
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0093
 lepton mixing angle: $\alpha_e=2.6709187$ rad
 incoming lepton: $-0.891263|+\rangle +0.453487|-\rangle$
 proton mixing angle: $\alpha_p=0.47067803$ rad
 incoming proton: $0.891261|+\rangle +0.453491|-\rangle$

$s=24.8907$, $\sqrt{s}=4.98906$, $|\vec{P}|=2.40635$, $|\vec{P}'|=0.0697271$
 $\theta_{P'}=1.43494\text{e-}05$, $\theta_\gamma=3.14159$, $\phi_{P'}=2.55848$, $\phi_\gamma=4.12637$
 $E_\gamma=2.0591$, $\theta_{\ell'}=5.029\text{e-}07$, $\phi_{\ell'}=5.70007$
 $Q^2=19.1485$, $x_B=0.821501$, $t=-2.76326$, $W^2=5.04052$, $y=0.970777$

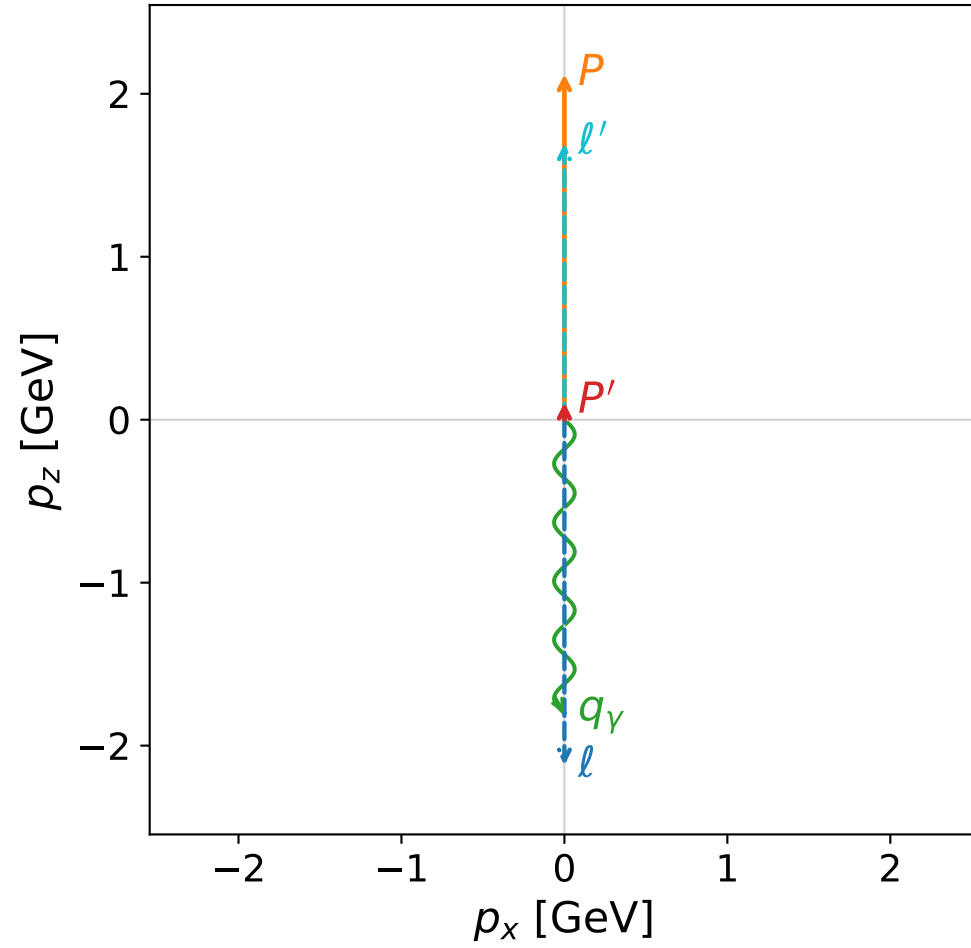
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4064$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4064$
 P $E= 2.5827$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4064$
 ℓ' $E= 1.9894$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.9894$
 P' $E= 0.9406$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0697$
 q_γ $E= 2.0591$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.0591$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.978\text{e+}20$; $I_h=+3.786\text{e-}11$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.726\text{e-}15$; $I_{TL}=+2.887\text{e-}08$

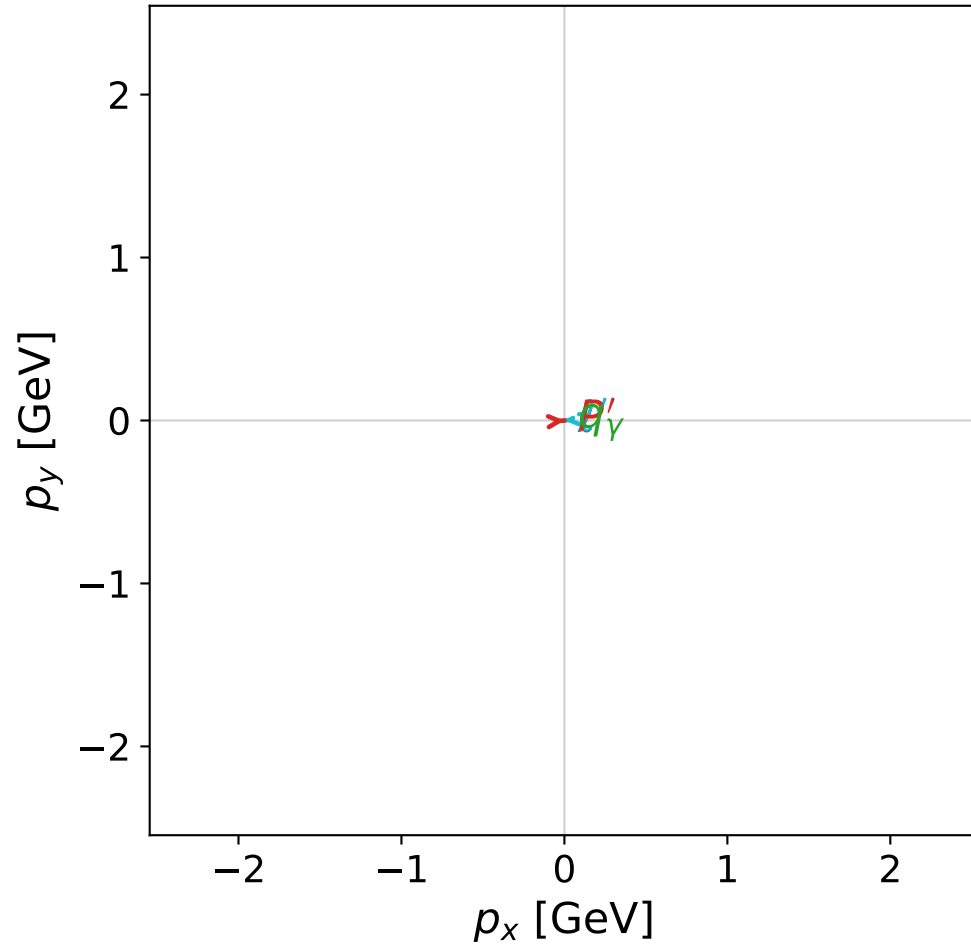


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

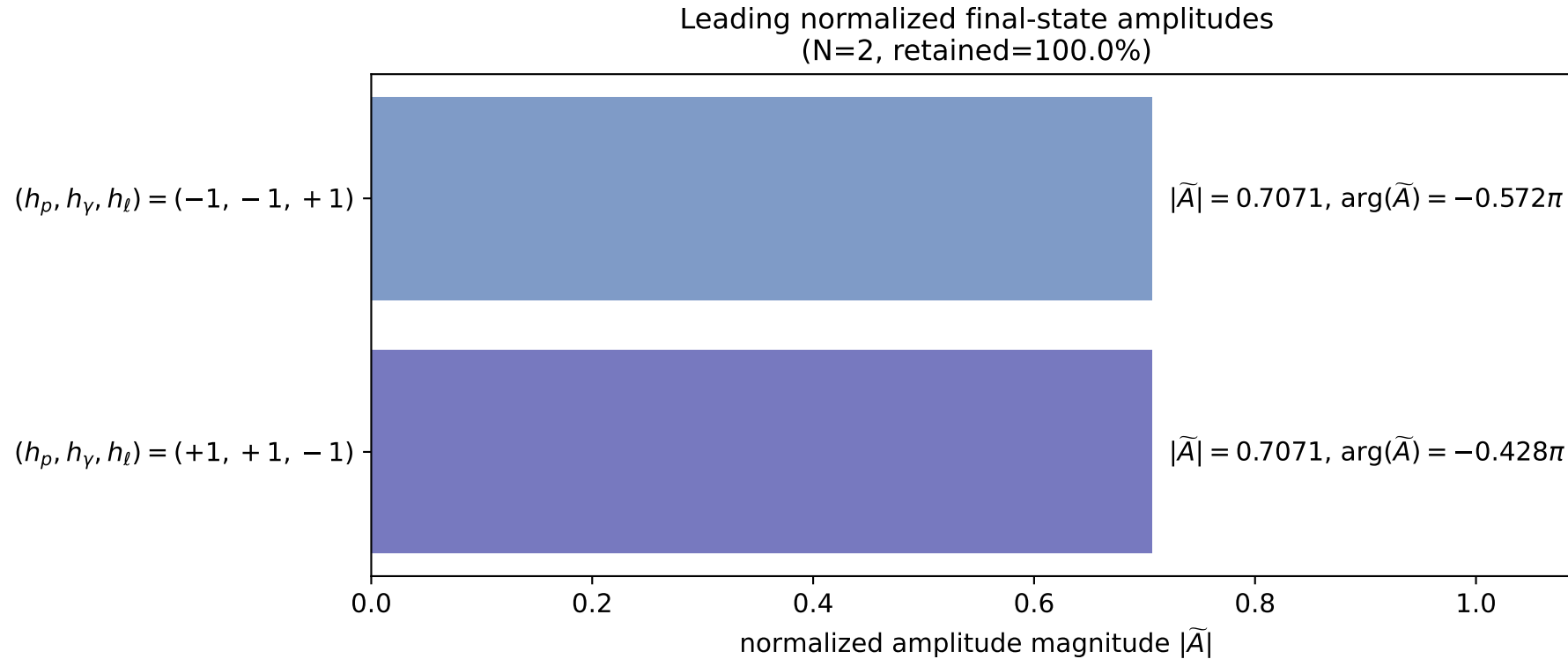


dghz_mixing_angles_polarization_03_local_minimum_97
 $d_{\text{GHZ}}=3.45768\text{e-}08$
 region: polarization_cluster_03_local_minimum_97
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0097
 lepton mixing angle: $\alpha_e=2.363027$ rad
 incoming lepton: $-0.711922|+\rangle +0.702259|-\rangle$
 proton mixing angle: $\alpha_p=0.77853959$ rad
 incoming proton: $0.71194|+\rangle +0.70224|-\rangle$

$s=19.711$, $\sqrt{s}=4.4397$, $|\vec{P}|=2.12076$, $|\vec{P}'|=0.105882$
 $\theta_{P'}=3.23647\text{e-}06$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.0751304$, $\phi_\gamma=1.34534$
 $E_\gamma=1.80081$, $\theta_{\ell'}=2.02129\text{e-}07$, $\phi_{\ell'}=3.21672$
 $Q^2=14.3782$, $x_B=0.79178$, $t=-2.16917$, $W^2=4.66097$, $y=0.964325$

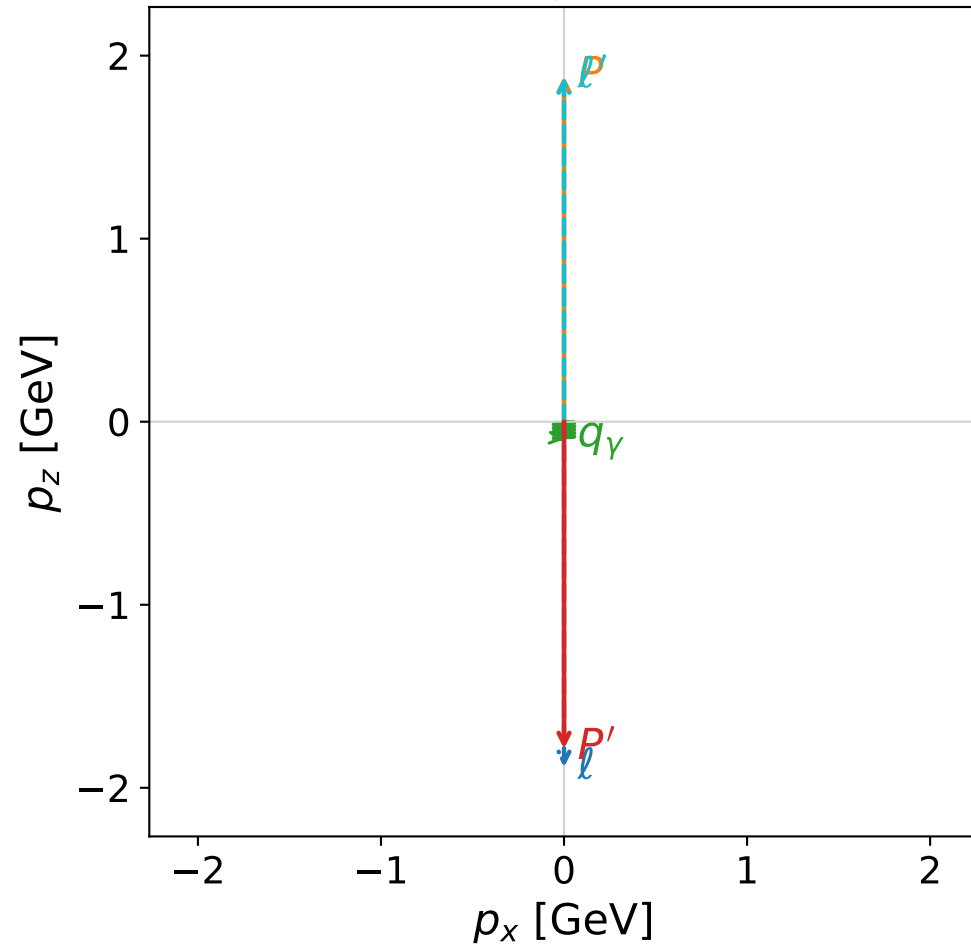
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.1208$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1208$
 P $E= 2.3189$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1208$
 ℓ' $E= 1.6949$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.6949$
 P' $E= 0.9440$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1059$
 q_γ $E= 1.8008$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8008$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.567\text{e+}21$; $I_h=-3.918\text{e-}13$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=2.980\text{e-}16$; $I_{TL}=-3.326\text{e-}10$

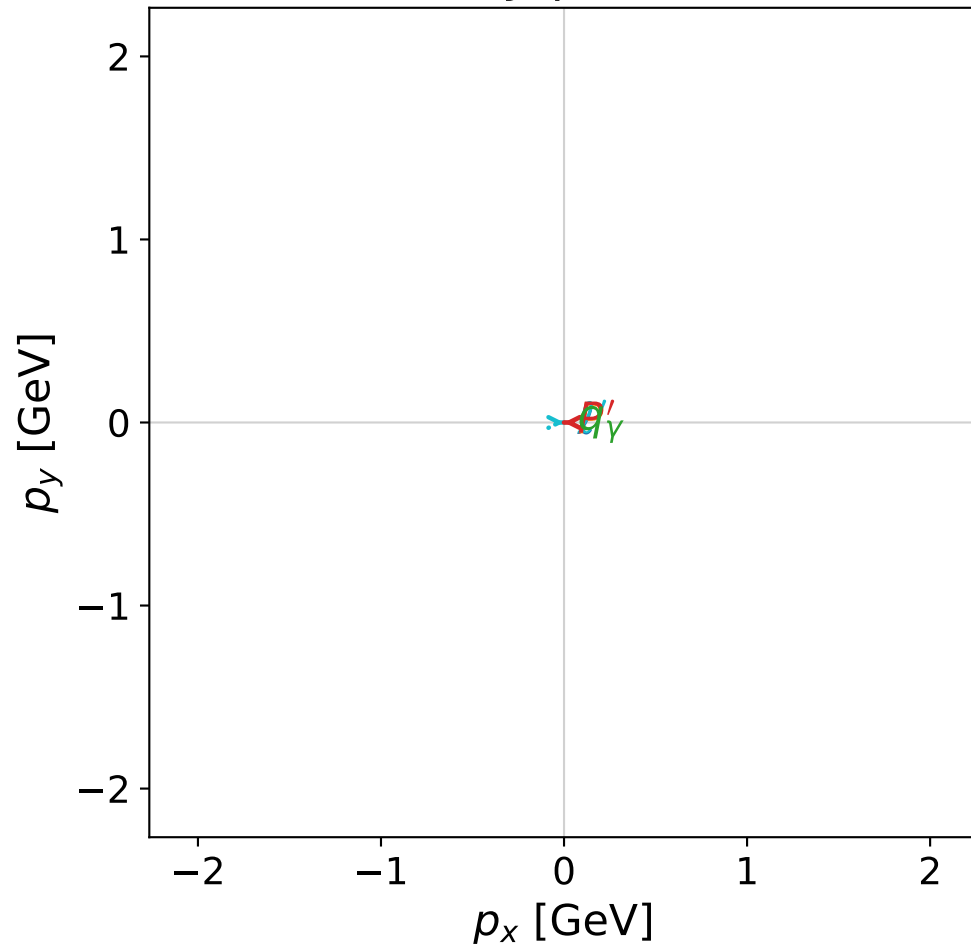


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

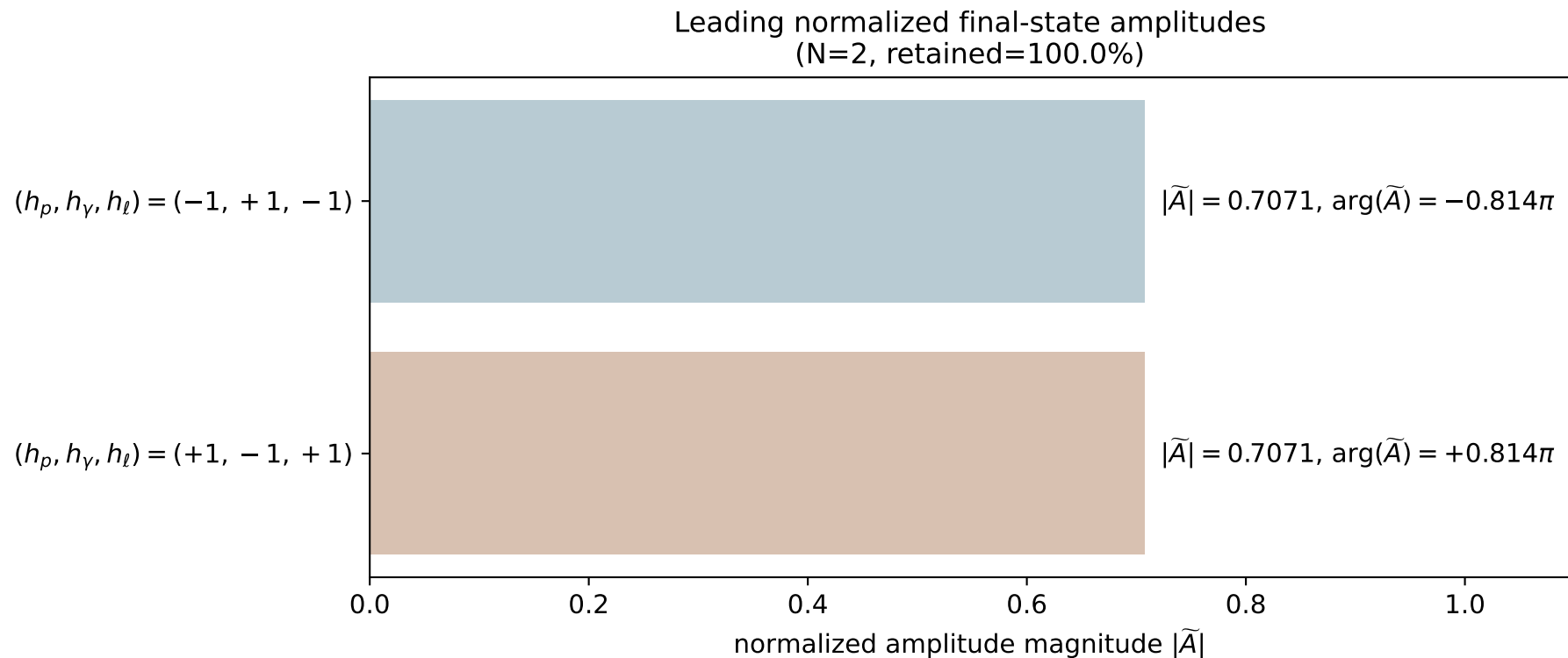


dghz_mixing_angles_polarization_03_local_minimum_102
 $d_{\text{GHZ}}=3.6745\text{e-}08$
 region: polarization_cluster_03_local_minimum_102
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0102
 lepton mixing angle: $\alpha_e=2.3754496$ rad
 incoming lepton: $-0.72059|+\rangle +0.693361|-\rangle$
 proton mixing angle: $\alpha_p=0.7661545$ rad
 incoming proton: $0.720582|+\rangle +0.693369|-\rangle$

$s=15.9699$, $\sqrt{s}=3.99624$, $|\vec{P}|=1.88803$, $|\vec{P}'|=1.78706$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.12959$, $\phi_\gamma=5.69987$
 $E_\gamma=0.0954542$, $\theta_{\ell'}=0$, $\phi_{\ell'}=0.0182279$
 $Q^2=14.217$, $x_B=0.996905$, $t=-13.4982$, $W^2=0.923985$, $y=0.945067$

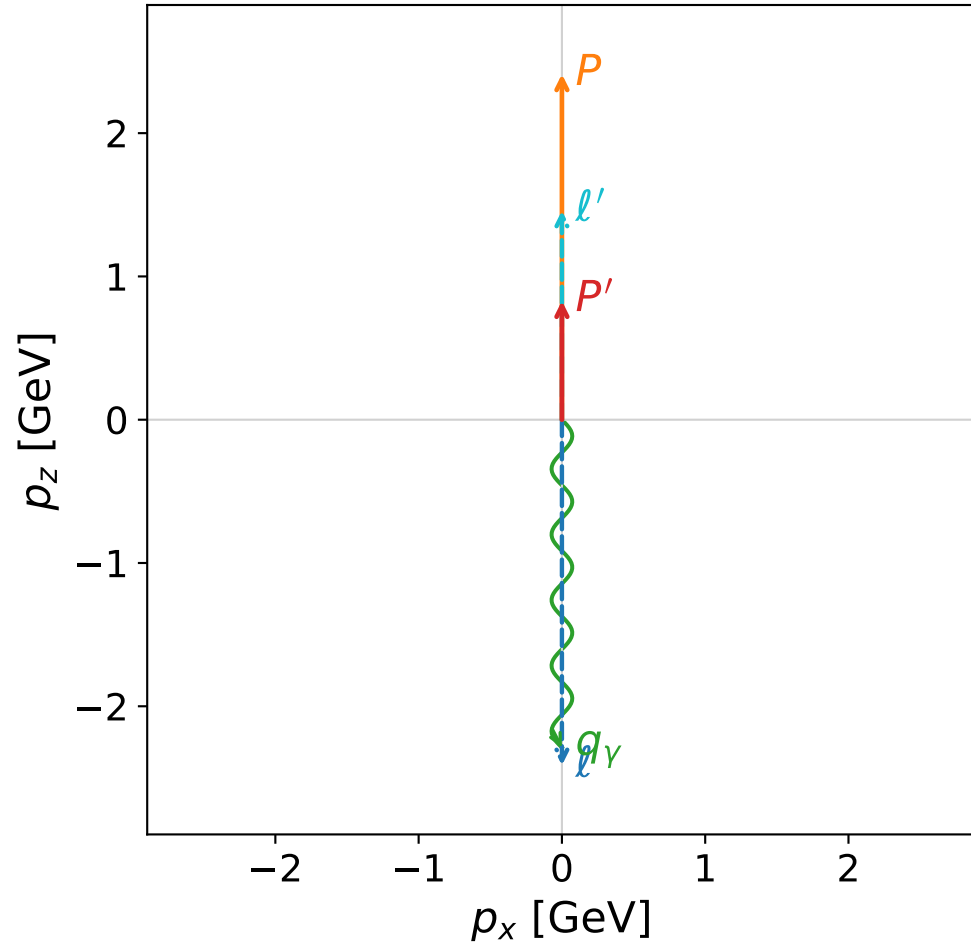
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.8880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8880$
 P $E= 2.1082$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8880$
 ℓ' $E= 1.8825$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8825$
 P' $E= 2.0183$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.7871$
 q_γ $E= 0.0955$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.0955$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=6.548\text{e+}21$; $I_h=+5.462\text{e-}13$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.763\text{e-}26$; $I_{TL}=-1.305\text{e-}18$

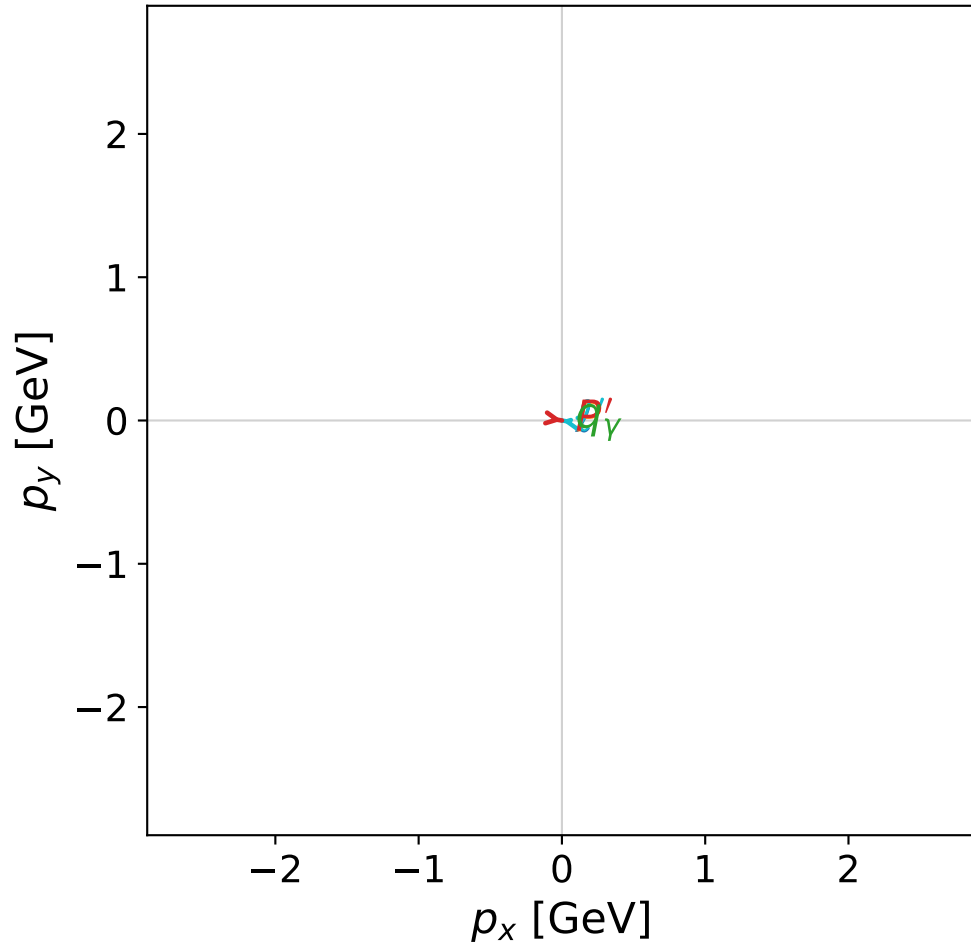


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

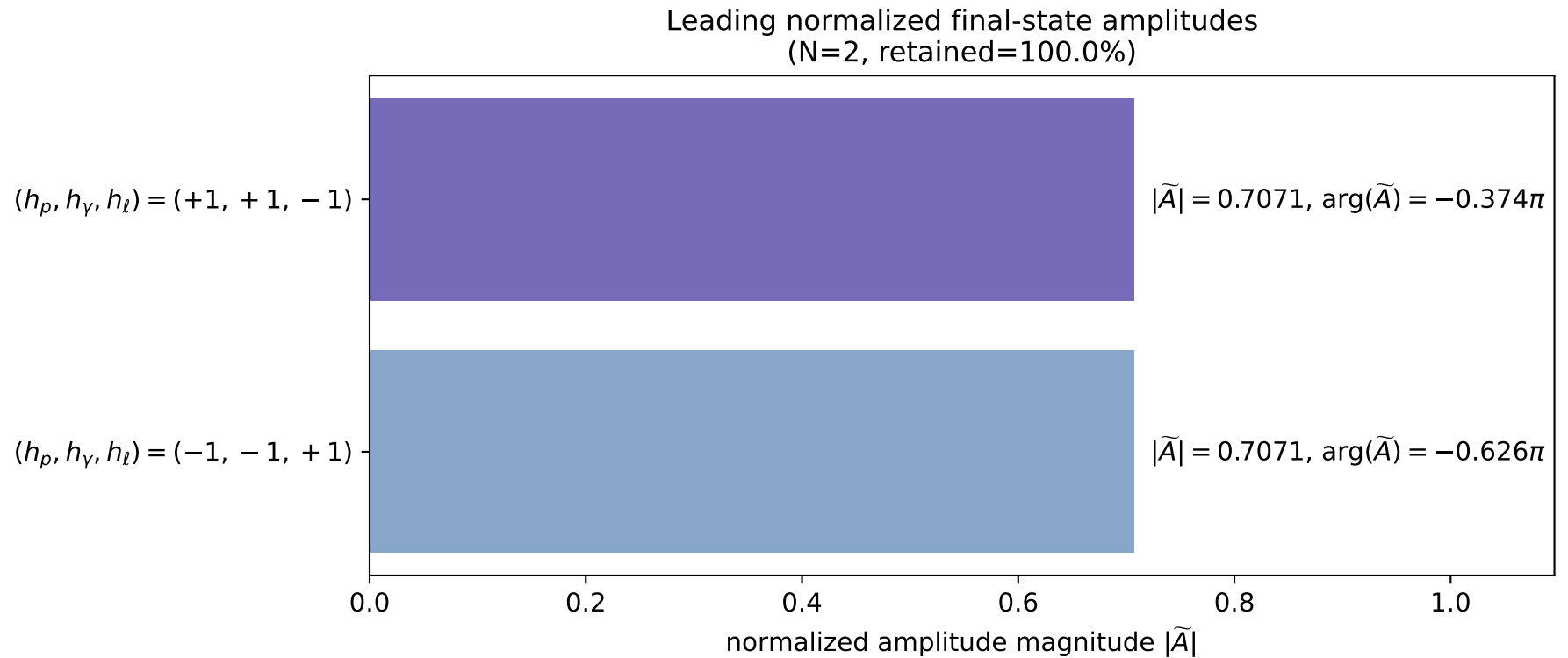


dghz_mixing_angles_polarization_03_local_minimum_103
 $d_{\{\rm GHZ\}}=3.69193\text{e-}08$
 region: polarization_cluster_03_local_minimum_103
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0103
 lepton mixing angle: $\alpha_e=2.3646323$ rad
 incoming lepton: $-0.713048|+\rangle +0.701115|-\rangle$
 proton mixing angle: $\alpha_p=0.77696696$ rad
 incoming proton: $0.713043|+\rangle +0.70112|-\rangle$

$s=24.9995$, $\sqrt{s}=4.99995$, $|\vec{P}|=2.41199$, $|\vec{P}'|=0.829186$
 $\theta_{p'}=3.04918\text{e-}05$, $\theta_\gamma=3.14159$, $\phi_{p'}=6.11999$, $\phi_\gamma=1.17541$
 $E_\gamma=2.28859$, $\theta_{l'}=1.73244\text{e-}05$, $\phi_{l'}=2.9784$
 $Q^2=14.0803$, $x_B=0.596469$, $t=-0.720361$, $W^2=10.4056$, $y=0.978705$

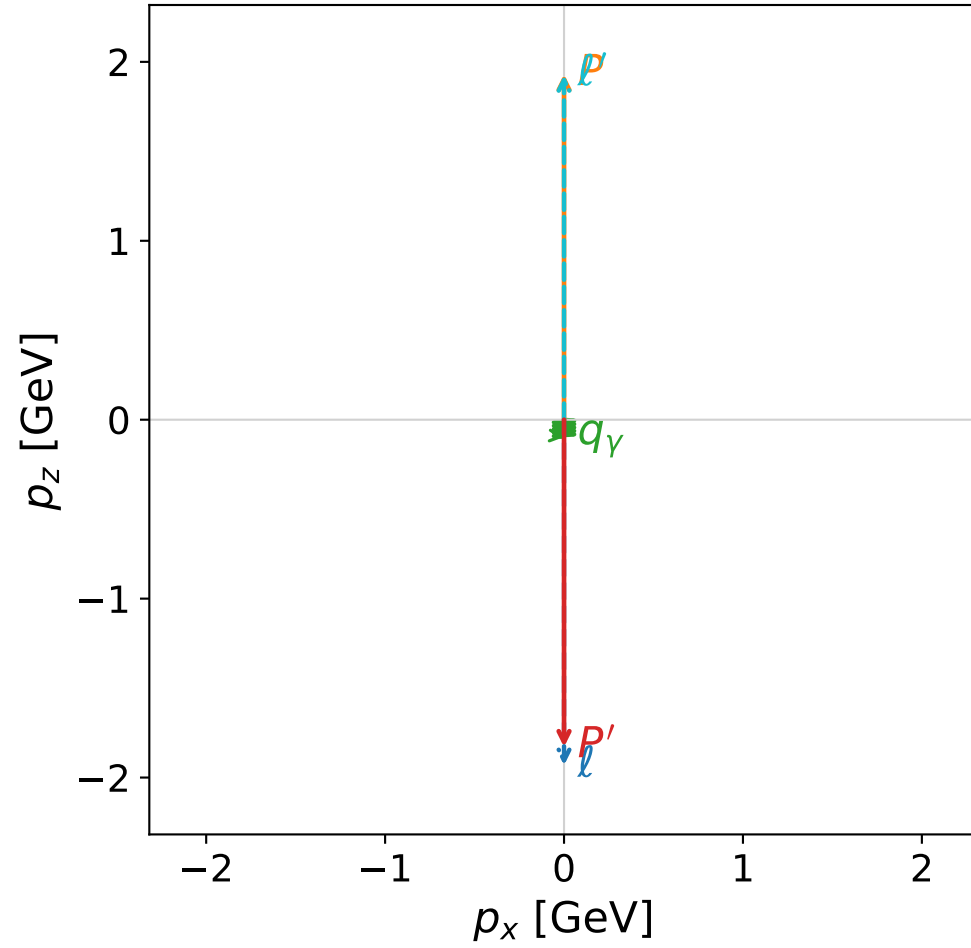
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 l' $E= 1.4594$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.4594$
 P' $E= 1.2520$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.8292$
 q_γ $E= 2.2886$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2886$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=2.398\text{e+}17$; $I_h=-3.219\text{e-}11$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_{L/T}=5.505\text{e-}12$; $I_{TL}=-5.499\text{e-}08$

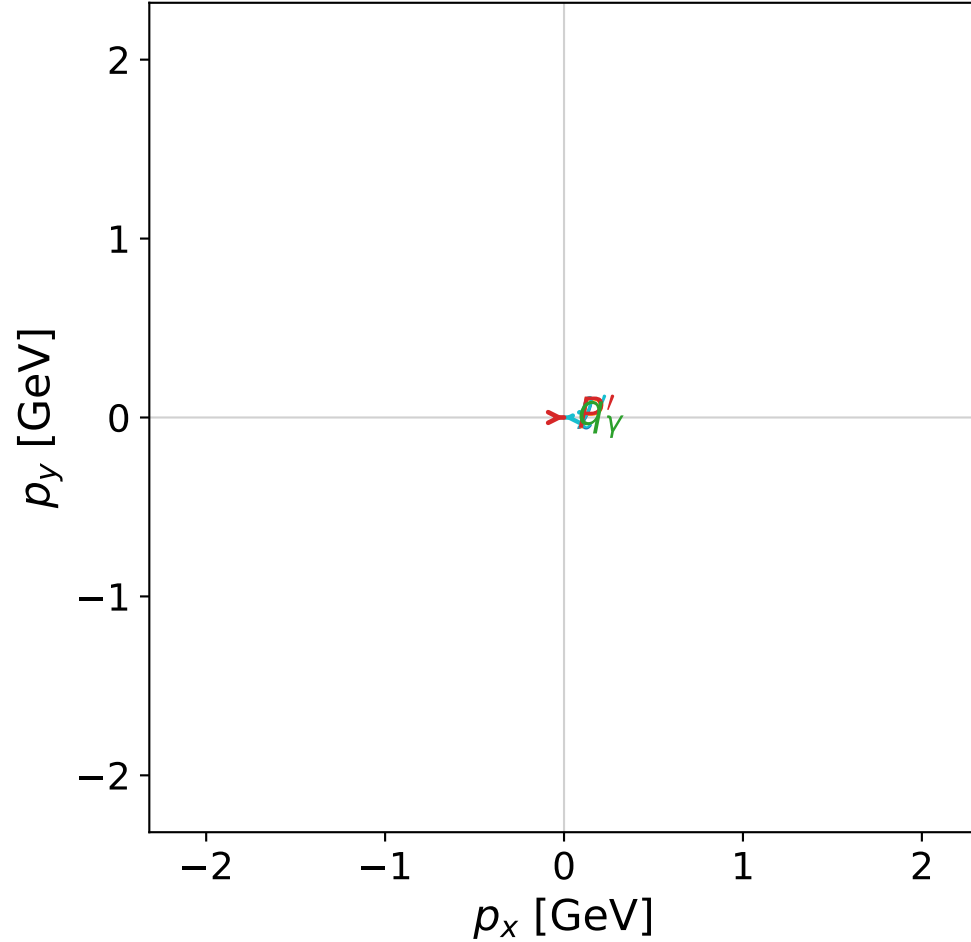


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

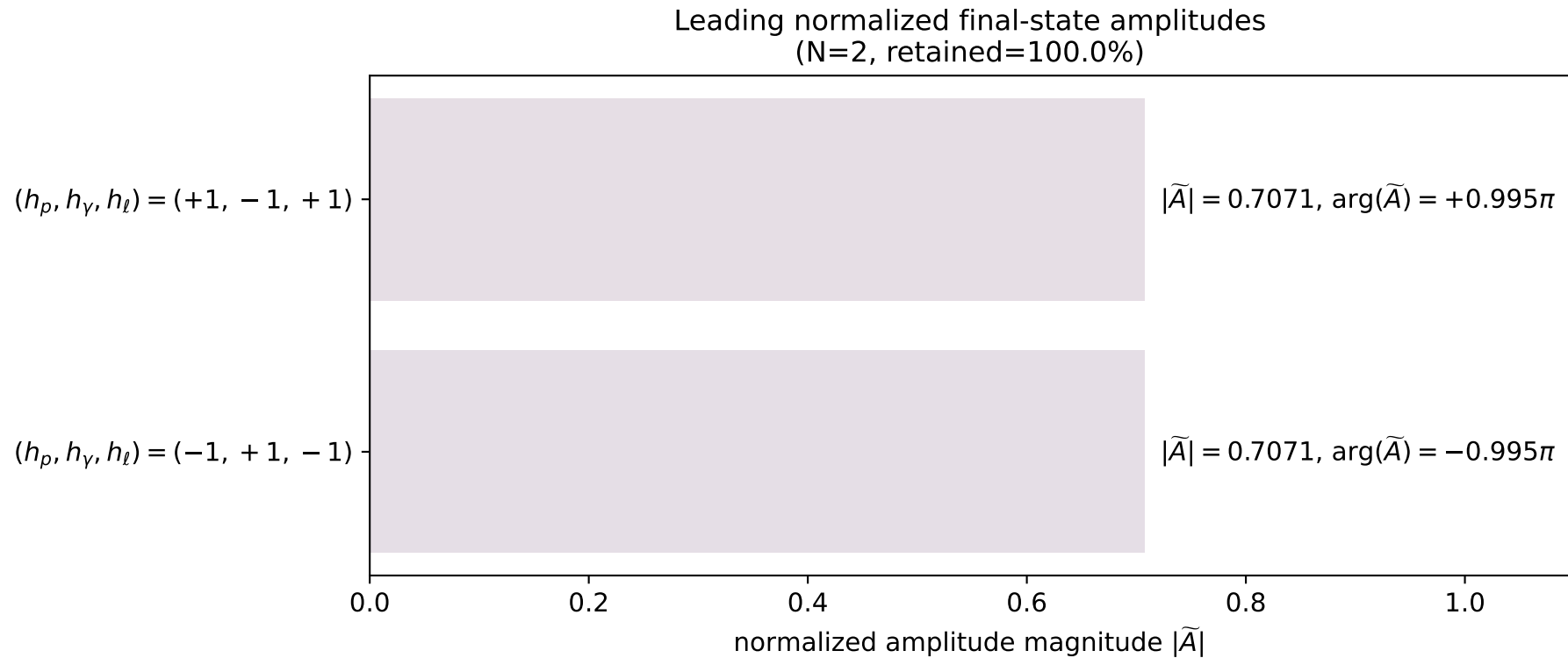


dghz_mixing_angles_polarization_03_local_minimum_106
 $d_{\text{GHZ}}=3.85907\text{e-}08$
 region: polarization_cluster_03_local_minimum_106
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0106
 lepton mixing angle: $\alpha_e=2.1441764$ rad
 incoming lepton: $-0.542475|+\rangle +0.840072|-\rangle$
 proton mixing angle: $\alpha_p=0.99739265$ rad
 incoming proton: $0.542494|+\rangle +0.840059|-\rangle$

$s=16.6364$, $\sqrt{s}=4.07877$, $|\vec{P}|=1.93153$, $|\vec{P}'|=1.82957$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=0.0149987$, $\phi_\gamma=6.26819$
 $E_\gamma=0.0966003$, $\theta_{l'}=0$, $\phi_{l'}=3.15509$
 $Q^2=14.8818$, $x_B=0.997069$, $t=-14.1375$, $W^2=0.923592$, $y=0.94726$

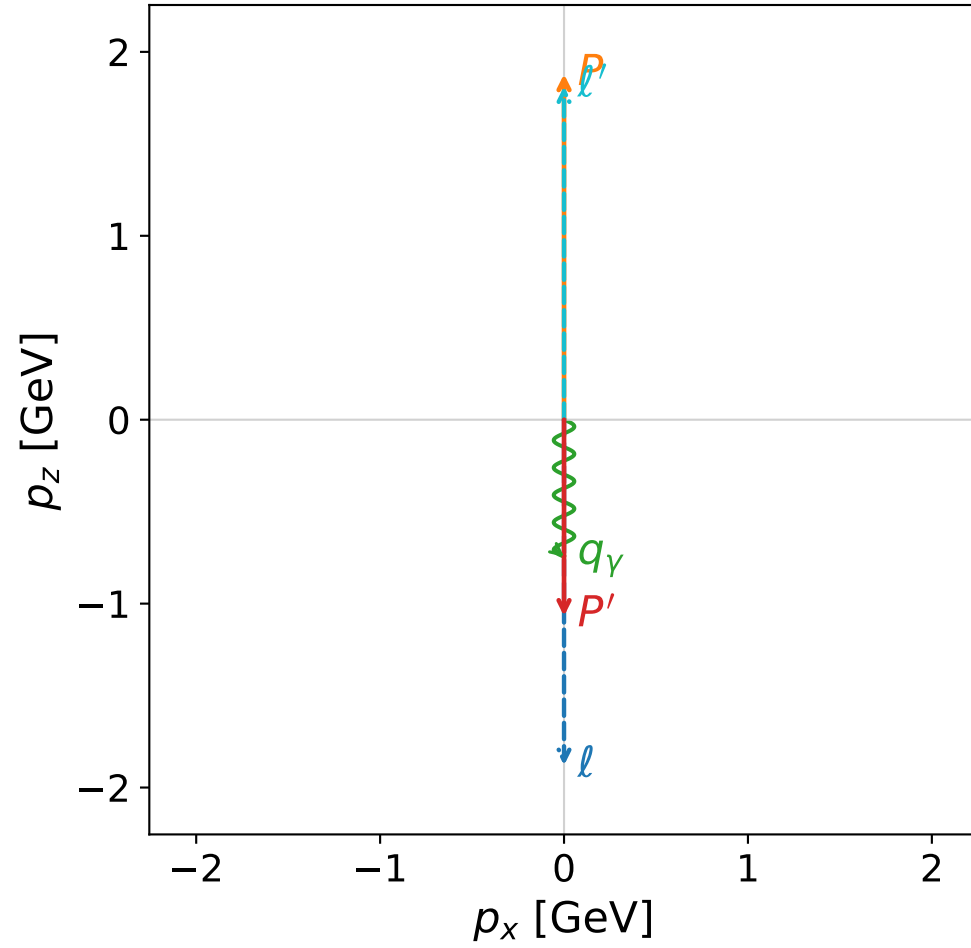
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E=1.9315$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-1.9315$
 P $E=2.1472$ $p_x=0.0000$ $p_y=0.0000$ $p_z=1.9315$
 l' $E=1.9262$ $p_x=-0.0000$ $p_y=-0.0000$ $p_z=1.9262$
 P' $E=2.0560$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-1.8296$
 q_γ $E=0.0966$ $p_x=0.0000$ $p_y=-0.0000$ $p_z=-0.0966$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=4.339\text{e}+21$; $I_h=-7.937\text{e-}12$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=3.571\text{e-}26$; $I_{TL}=+2.228\text{e-}17$

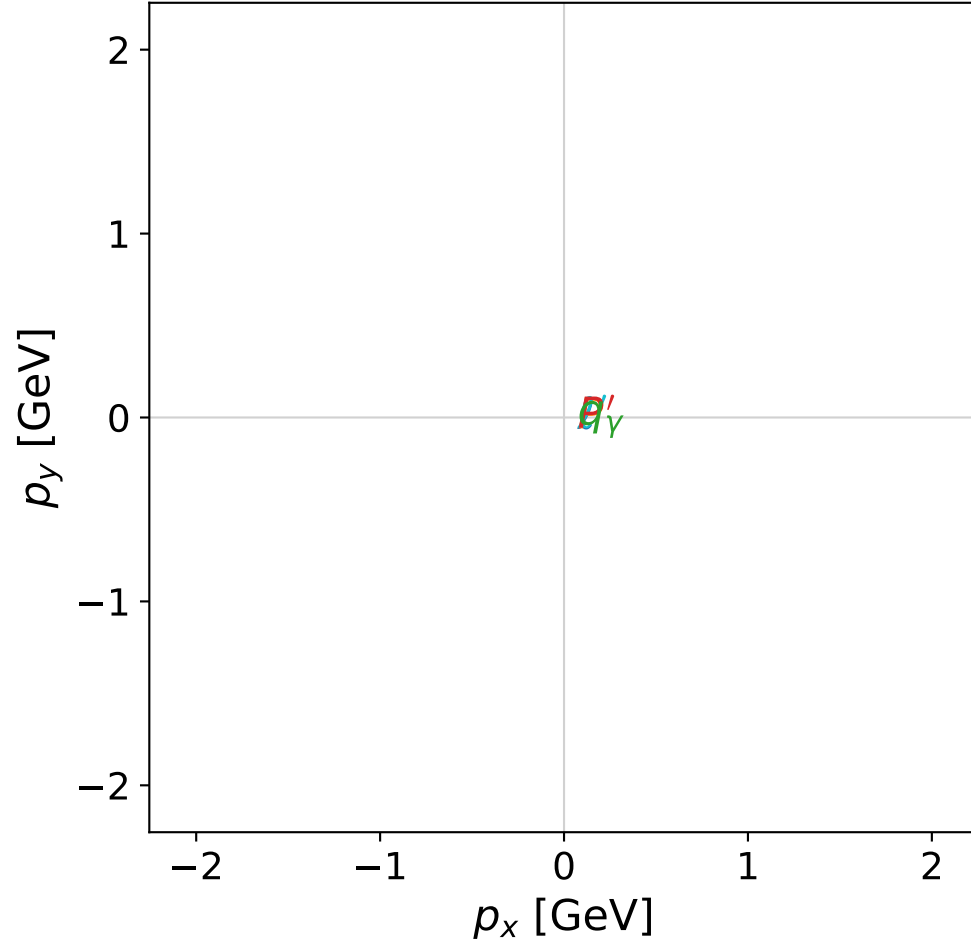


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

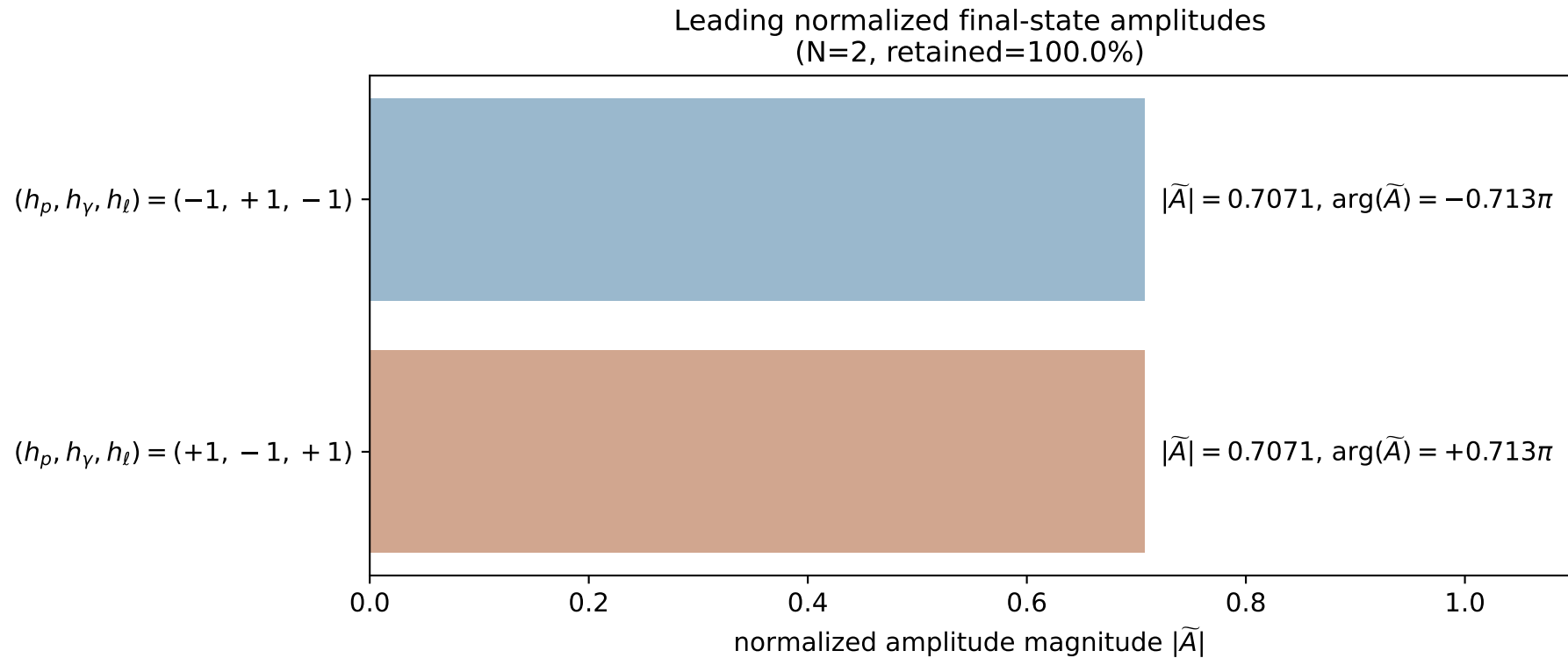


dghz_mixing_angles_polarization_03_local_minimum_107
 $d_{\text{GHZ}}=3.86512\text{e-}08$
 region: polarization_cluster_03_local_minimum_107
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0107
 lepton mixing angle: $\alpha_e=2.4205789$ rad
 incoming lepton: $-0.751137|+\rangle +0.660146|-\rangle$
 proton mixing angle: $\alpha_p=0.72102631$ rad
 incoming proton: $0.751129|+\rangle +0.660156|-\rangle$

$s=15.8347$, $\sqrt{s}=3.97928$, $|\vec{P}|=1.87909$, $|\vec{P}'|=1.06808$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=1.05624$, $\phi_\gamma=5.38261$
 $E_\gamma=0.744851$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.47839$
 $Q^2=13.6266$, $x_B=0.962801$, $t=-8.22517$, $W^2=1.40632$, $y=0.946392$

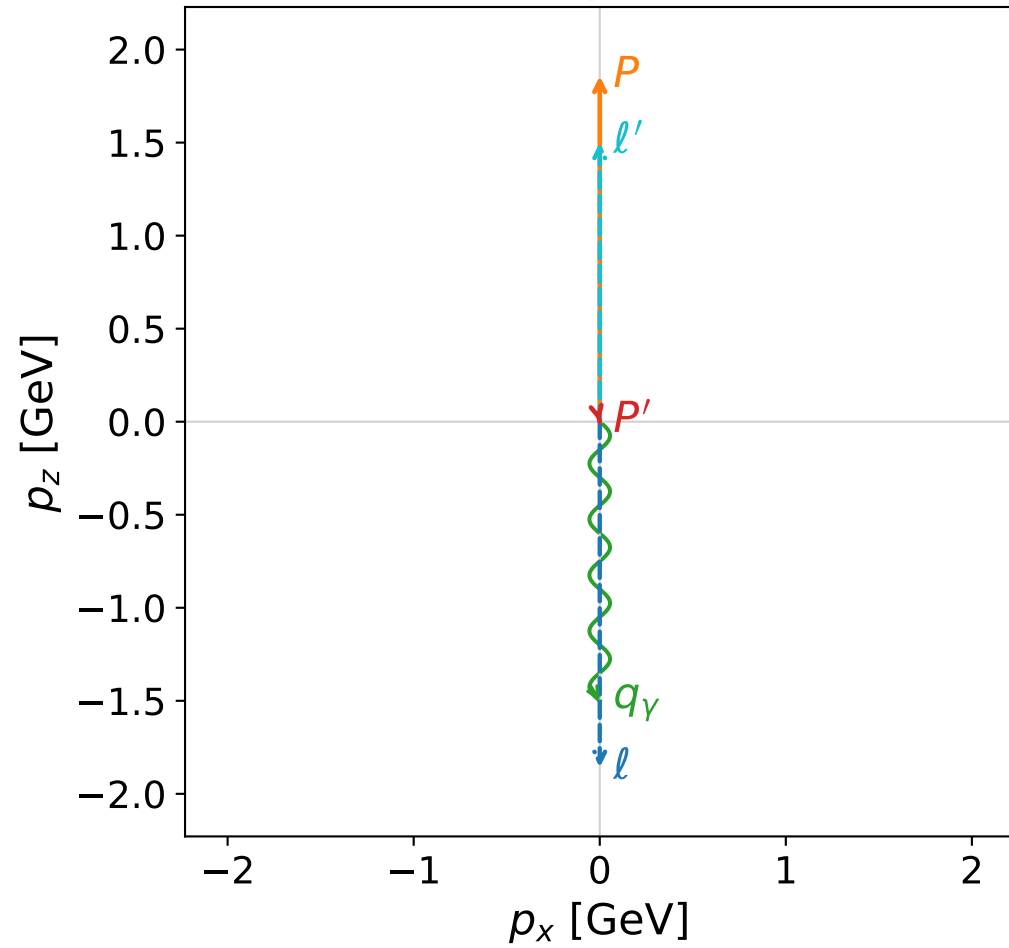
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.8791$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8791$
 P $E= 2.1002$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8791$
 ℓ' $E= 1.8129$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.8129$
 P' $E= 1.4215$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.0681$
 q_γ $E= 0.7449$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.7449$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=5.503\text{e}+23$; $I_h=+1.103\text{e-}13$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=7.558\text{e-}28$; $I_{TL}=+1.085\text{e-}18$

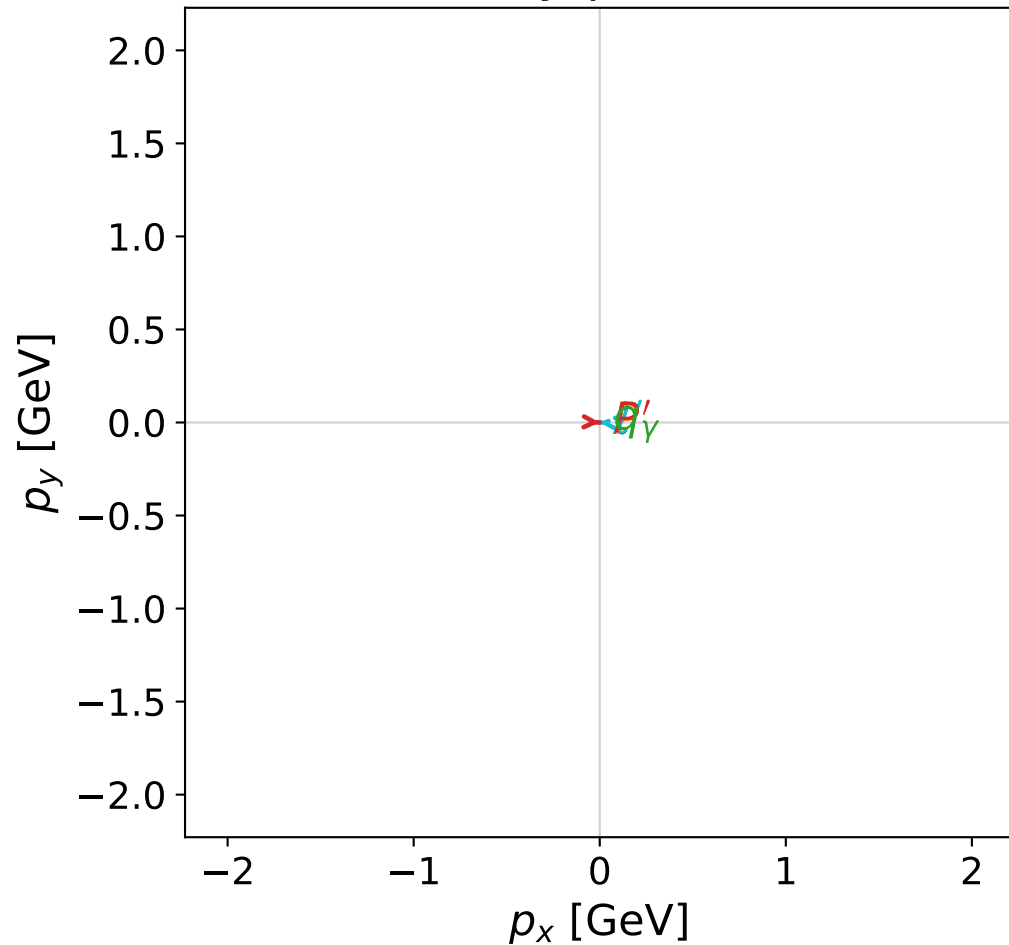


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

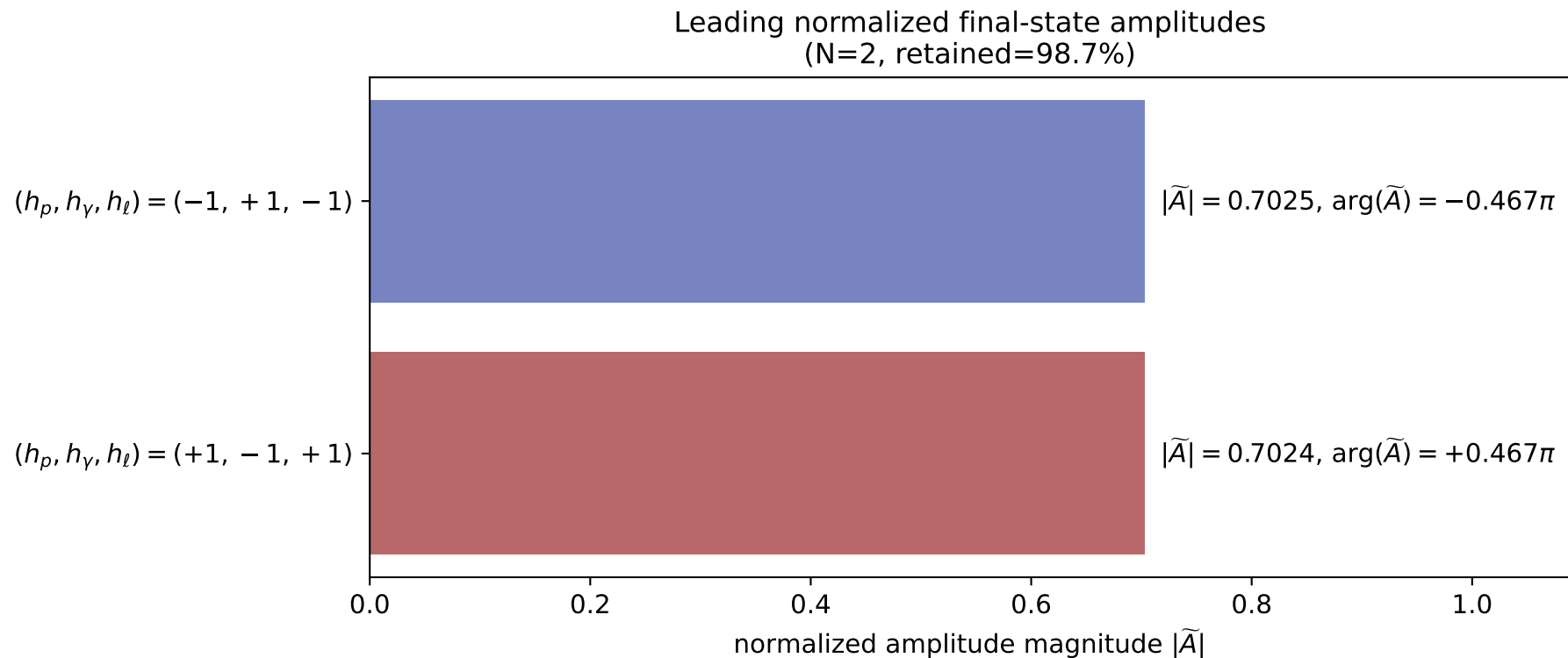


dghz_mixing_angles_polarization_03_local_minimum_118
 $d_{\text{GHZ}}=4.43159\text{e-}08$
 region: polarization_cluster_03_local_minimum_118
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0118
 lepton mixing angle: $\alpha_e=2.3993315$ rad
 incoming lepton: $-0.736942|+\rangle +0.675956|-\rangle$
 proton mixing angle: $\alpha_p=0.74228136$ rad
 incoming proton: $0.736928|+\rangle +0.675971|-\rangle$

$s=15.5089$, $\sqrt{s}=3.93814$, $|\vec{P}|=1.85736$, $|\vec{P}'|=0.000247174$
 $\theta_{p'}=2.91175$, $\theta_Y=3.14159$, $\phi_{p'}=6.24252$, $\phi_Y=4.56954$
 $E_Y=1.49995$, $\theta_{\ell'}=3.7536\text{e-}05$, $\phi_{\ell'}=3.10092$
 $Q^2=11.1456$, $x_B=0.798465$, $t=-2.14474$, $W^2=3.69303$, $y=0.954178$

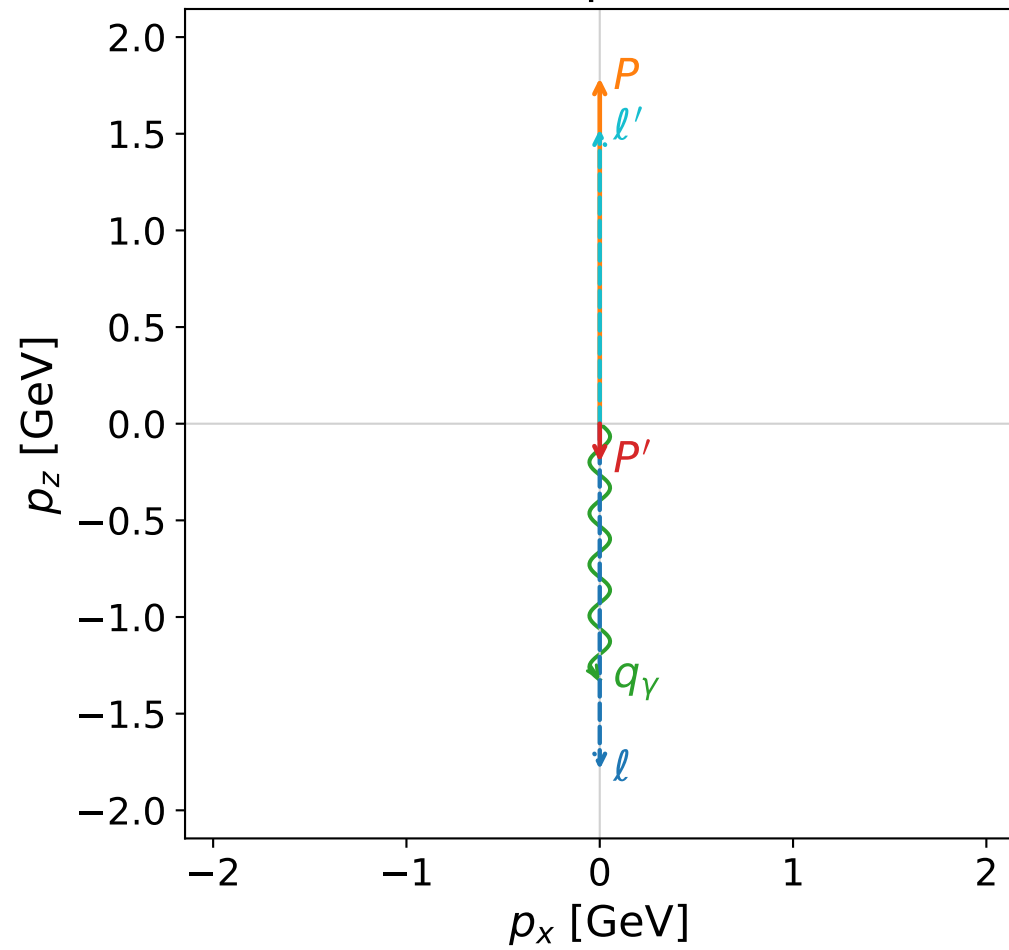
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.8574$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8574$
 P $E= 2.0808$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8574$
 ℓ' $E= 1.5002$ $p_x= -0.0001$ $p_y= 0.0000$ $p_z= 1.5002$
 P' $E= 0.9380$ $p_x= 0.0001$ $p_y= -0.0000$ $p_z= -0.0002$
 q_Y $E= 1.4999$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.4999$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=3.230\text{e+}16$; $I_h=-5.522\text{e-}10$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.227\text{e-}11$; $I_{TL}=-4.256\text{e-}07$

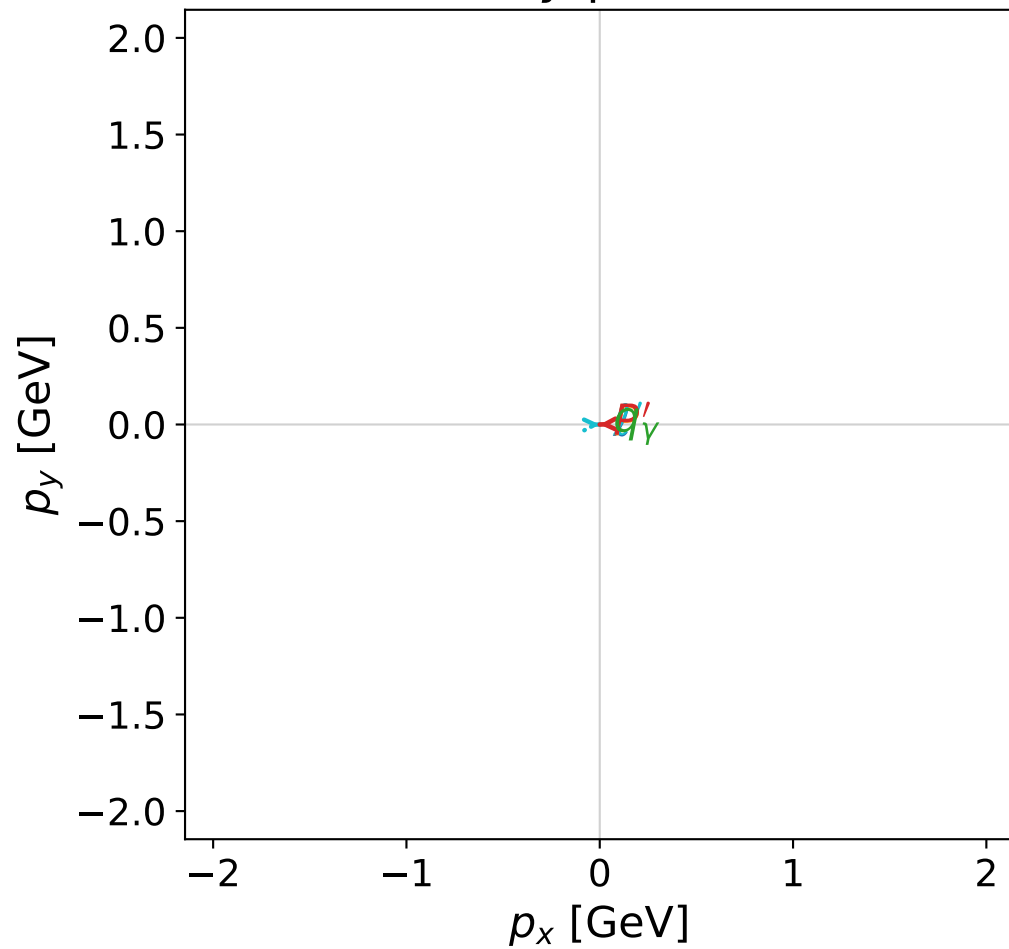


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

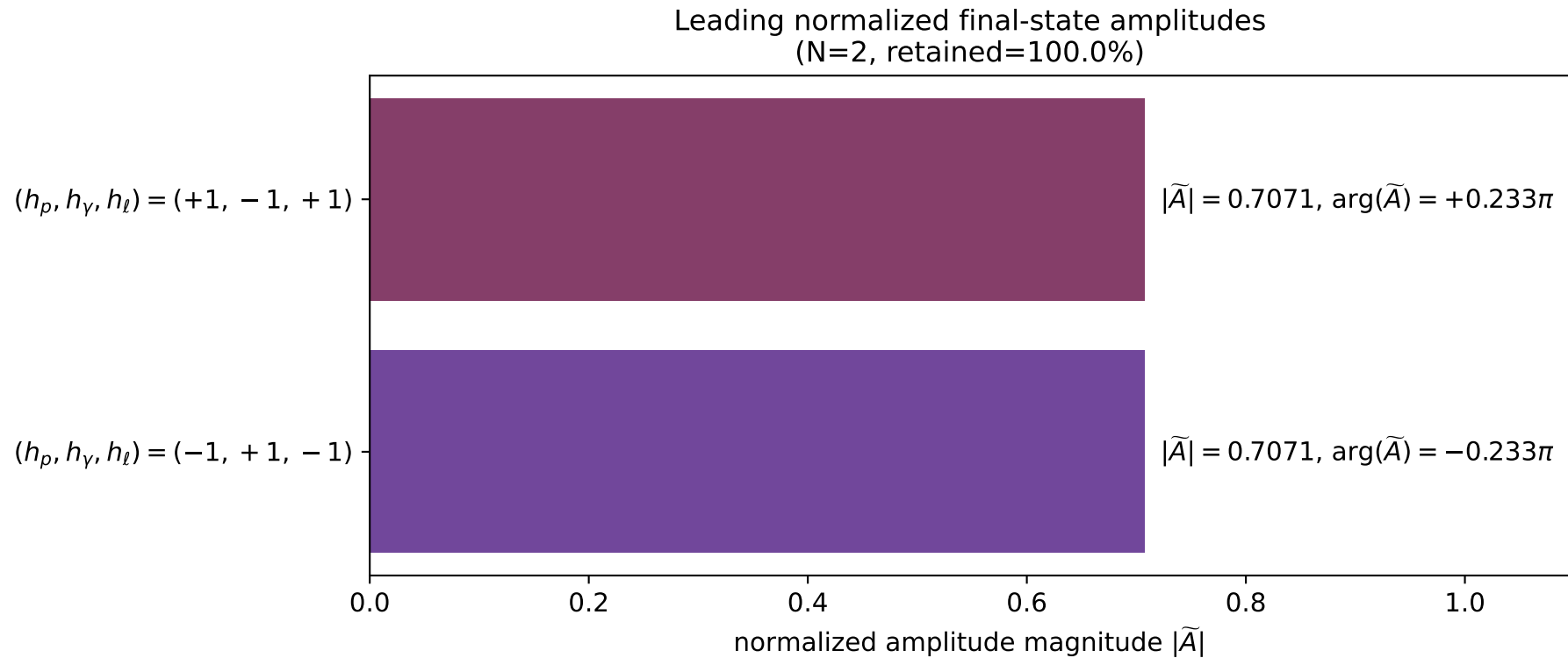


dghz_mixing_angles_polarization_03_local_minimum_124
 $d_{\{\rm GHZ\}}=4.80438\text{e-}08$
 region: polarization_cluster_03_local_minimum_124
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0124
 lepton mixing angle: $\alpha_e=2.3690498$ rad
 incoming lepton: $-0.716138|+\rangle +0.697959|-\rangle$
 proton mixing angle: $\alpha_p=0.77252117$ rad
 incoming proton: $0.716153|+\rangle +0.697943|-\rangle$

$s=14.4913$, $\sqrt{s}=3.80674$, $|\vec{P}|=1.78781$, $|\vec{P}'|=0.198316$
 $\theta_{p'}=3.14137$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.17027$, $\phi_\gamma=0.760181$
 $E_\gamma=1.32485$, $\theta_{\ell'}=2.88699\text{e-}05$, $\phi_{\ell'}=0.0286774$
 $Q^2=10.8925$, $x_B=0.843897$, $t=-2.82066$, $W^2=2.89472$, $y=0.948272$

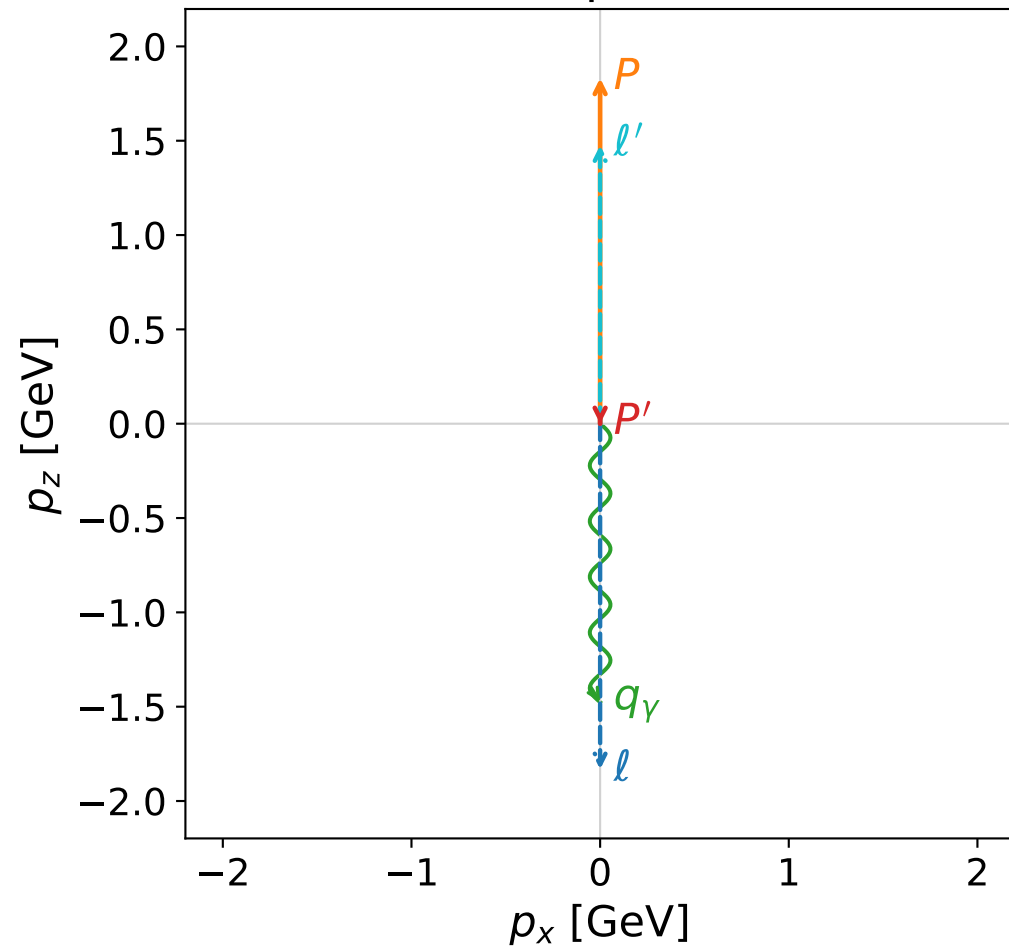
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.7878$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.7878$
 P $E= 2.0189$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7878$
 ℓ' $E= 1.5232$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.5232$
 P' $E= 0.9587$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.1983$
 q_γ $E= 1.3248$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.3248$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=4.676\text{e+}16$; $I_h=+1.434\text{e-}10$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=5.047\text{e-}12$; $I_{TL}=+8.205\text{e-}08$

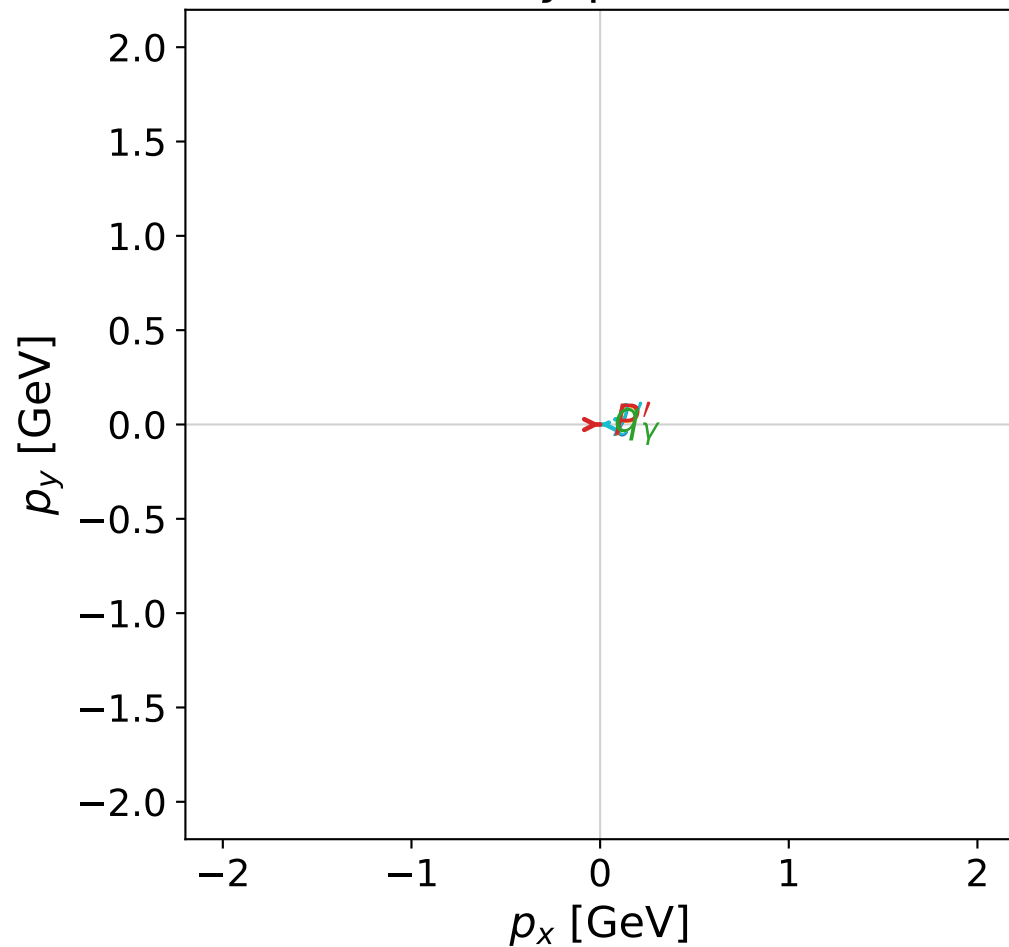


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

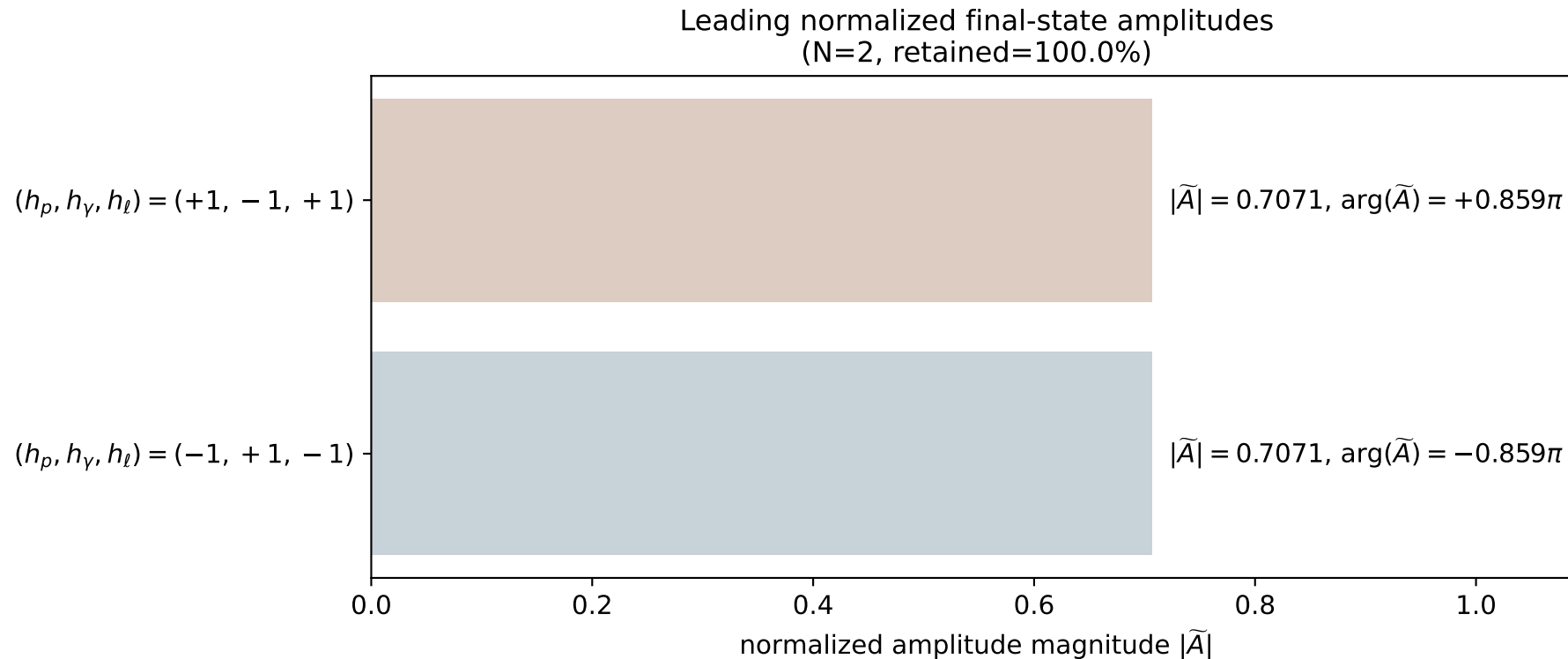


dghz_mixing_angles_polarization_03_local_minimum_128
 $d_{\text{GHZ}}=5.13615\text{e-}08$
 region: polarization_cluster_03_local_minimum_128
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0128
 lepton mixing angle: $\alpha_e=2.5301823$ rad
 incoming lepton: $-0.818839|+\rangle +0.574023|-\rangle$
 proton mixing angle: $\alpha_p=0.61140686$ rad
 incoming proton: $0.818841|+\rangle +0.57402|-\rangle$

$s=15.1318$, $\sqrt{s}=3.88997$, $|\vec{P}|=1.83189$, $|\vec{P}'|=0.000225325$
 $\theta_{P'}=3.14145$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.00233824$, $\phi_\gamma=5.84287$
 $E_\gamma=1.47587$, $\theta_{\ell'}=1.49012\text{e-}08$, $\phi_{\ell'}=3.14393$
 $Q^2=10.8162$, $x_B=0.79623$, $t=-2.10208$, $W^2=3.64791$, $y=0.953148$

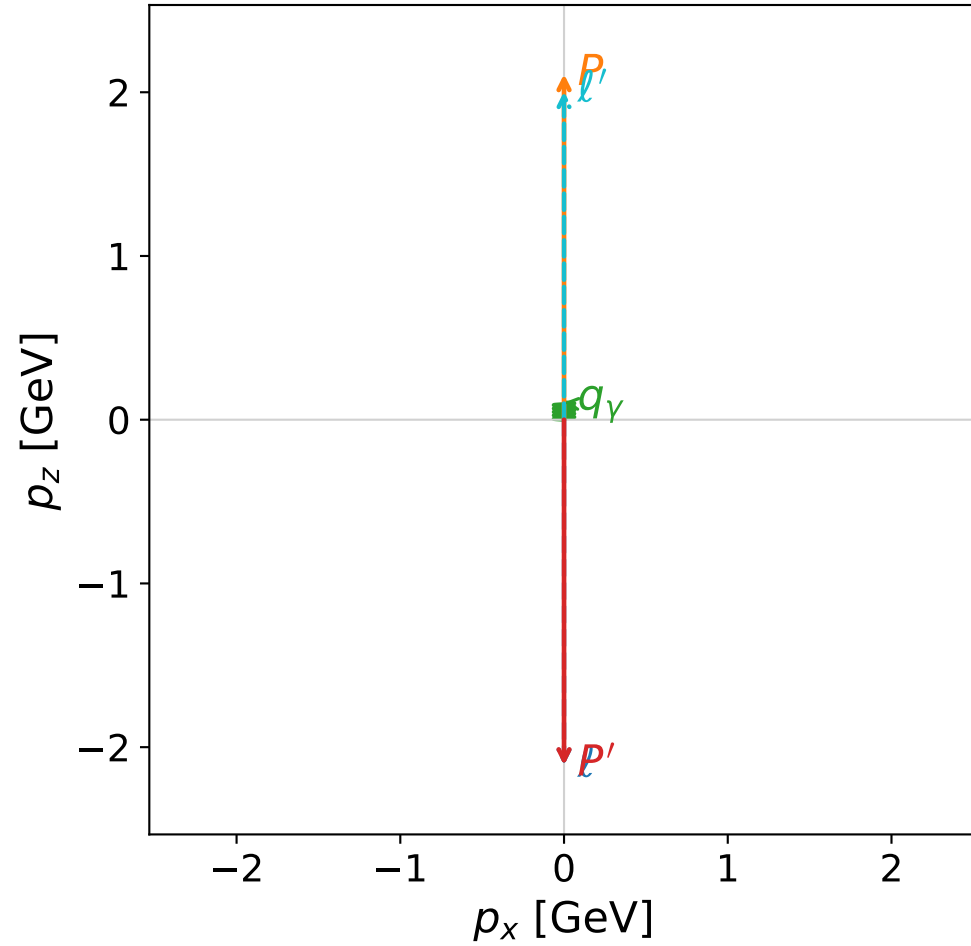
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.8319$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8319$
 P $E= 2.0581$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8319$
 ℓ' $E= 1.4761$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.4761$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0002$
 q_γ $E= 1.4759$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.4759$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.013\text{e+}23$; $I_h=-1.280\text{e-}12$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=4.709\text{e-}18$; $I_{TL}=-1.046\text{e-}09$

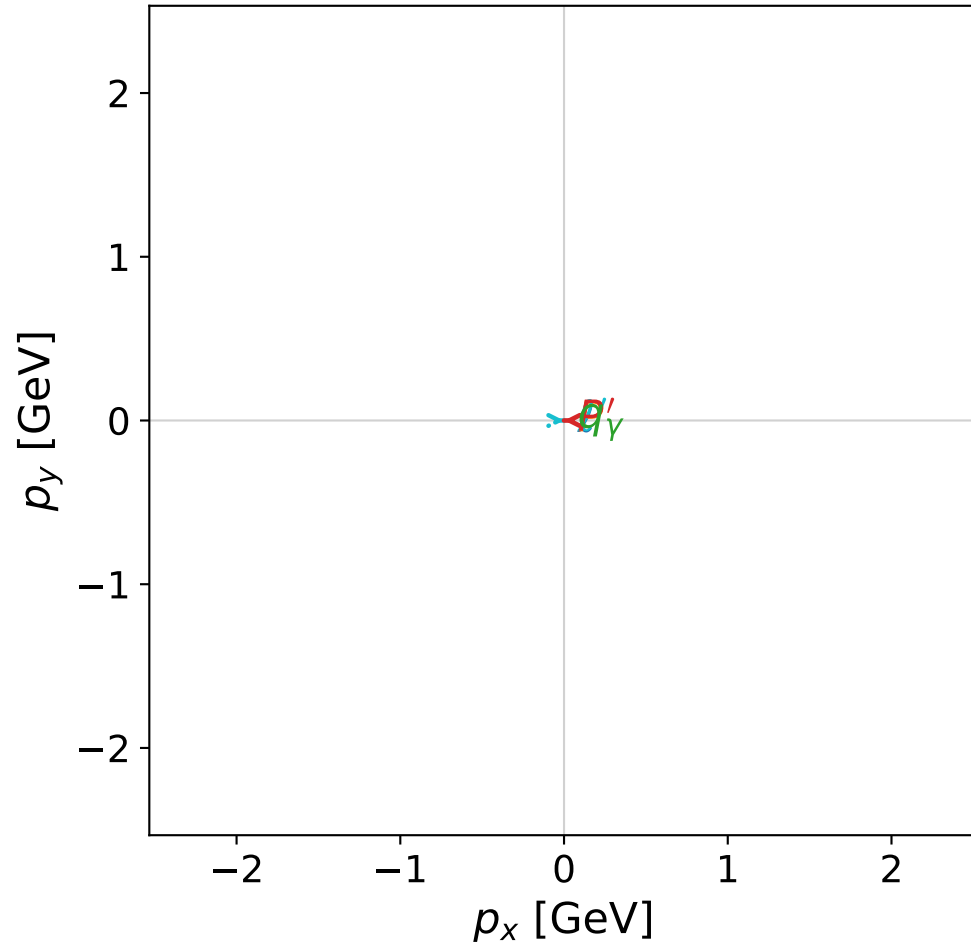


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

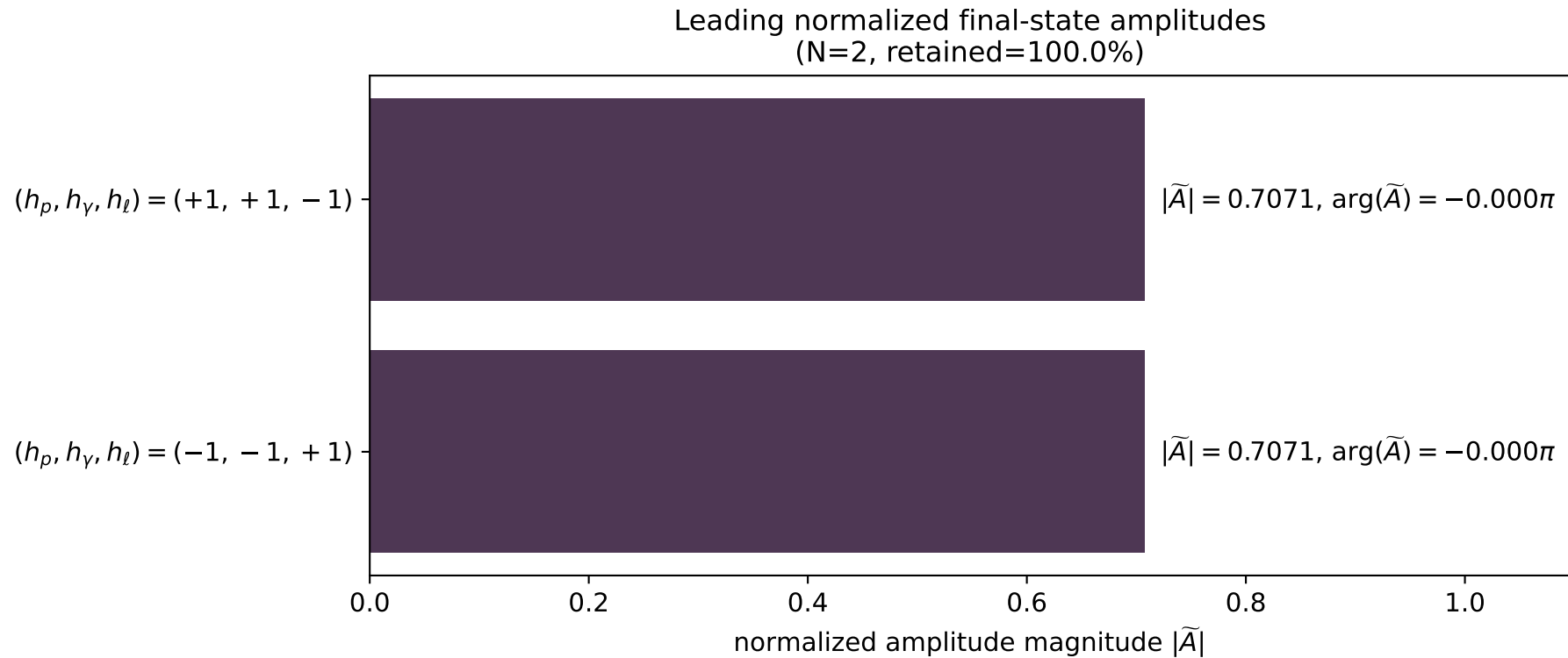


dghz_mixing_angles_polarization_03_local_minimum_131
 $d_{\text{GHZ}}=5.31748\text{e-}08$
 region: polarization_cluster_03_local_minimum_131
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0131
 lepton mixing angle: $\alpha_e=1.8802171$ rad
 incoming lepton: $-0.304507|+\rangle +0.95251|-\rangle$
 proton mixing angle: $\alpha_p=0.30941263$ rad
 incoming proton: $0.952513|+\rangle +0.304499|-\rangle$

$s=19.5413$, $\sqrt{s}=4.42055$, $|\vec{P}|=2.11076$, $|\vec{P}'|=2.11076$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=3.40678$, $\phi_\gamma=1.45431$
 $E_\gamma=0.105533$, $\theta_{l'}=0$, $\phi_{l'}=0.265183$
 $Q^2=16.9302$, $x_B=0.947768$, $t=-17.8212$, $W^2=1.81287$, $y=0.957226$

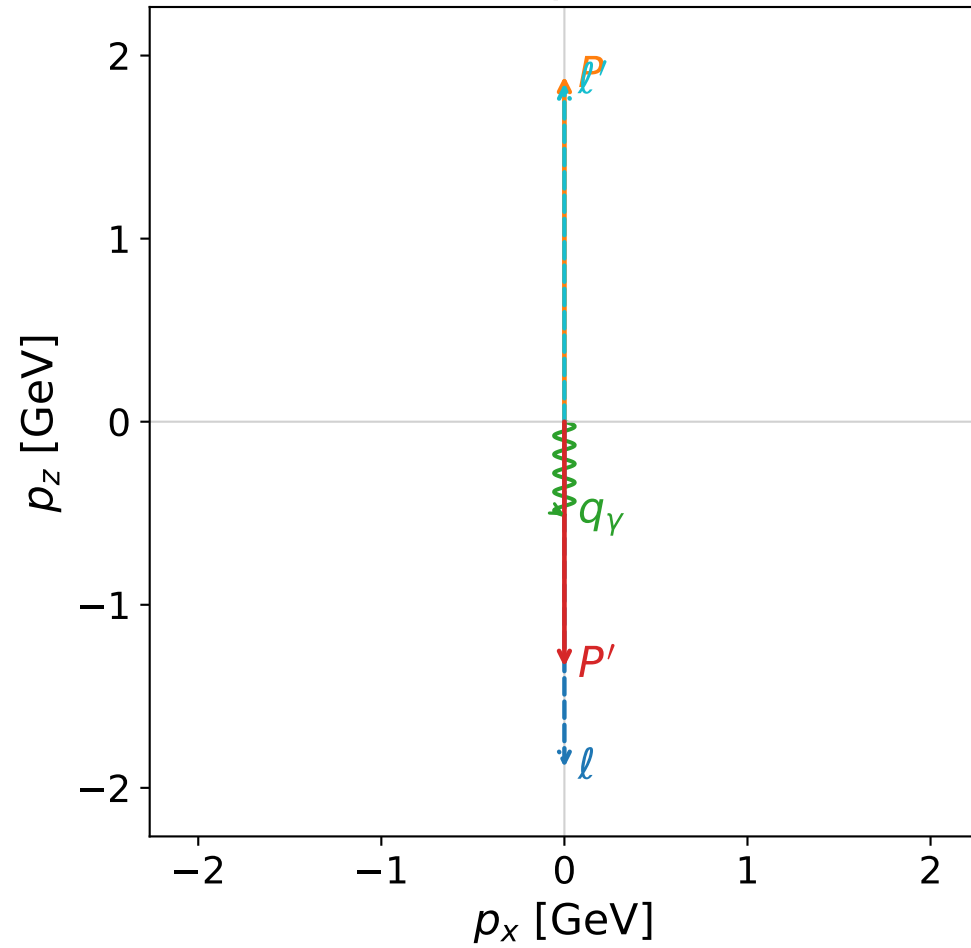
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.1108$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1108$
 P $E= 2.3098$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1108$
 ℓ' $E= 2.0052$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0052$
 P' $E= 2.3098$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.1108$
 q_γ $E= 0.1055$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1055$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=5.047\text{e}+21$; $I_h=+2.522\text{e-}13$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=9.452\text{e-}26$; $I_{TL}=+2.521\text{e-}13$

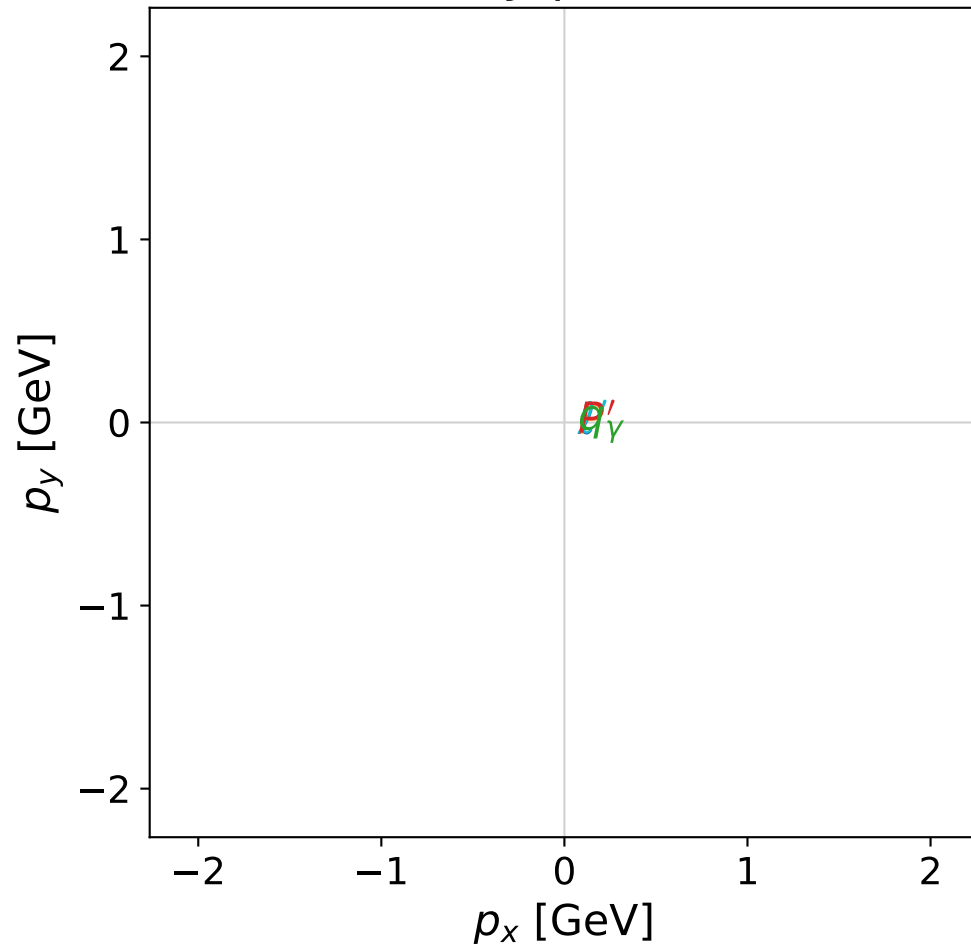


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

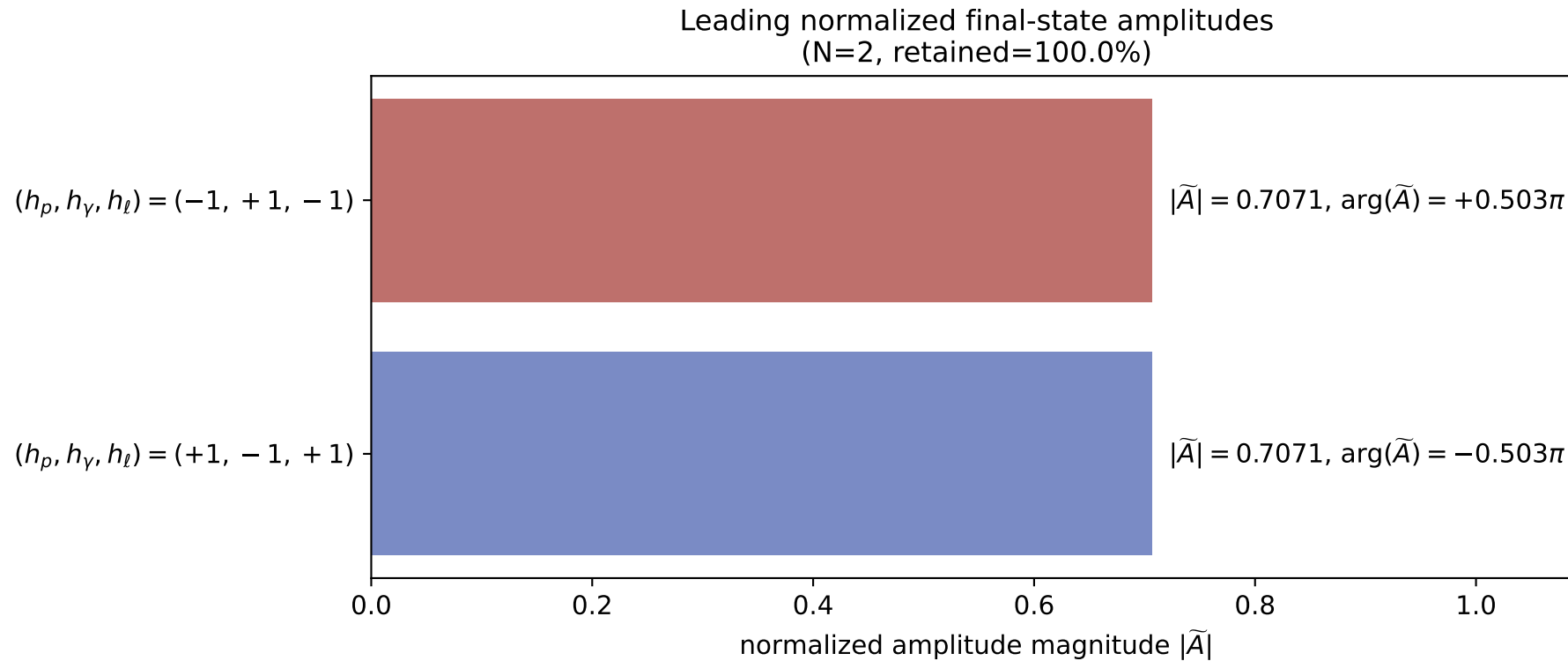


dghz_mixing_angles_polarization_03_local_minimum_132
 $d_{\text{GHZ}}=5.32794\text{e-}08$
 region: polarization_cluster_03_local_minimum_132
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0132
 lepton mixing angle: $\alpha_e=2.7526505$ rad
 incoming lepton: $-0.925311|+\rangle +0.37921|-\rangle$
 proton mixing angle: $\alpha_p=0.38894583$ rad
 incoming proton: $0.925309|+\rangle +0.379213|-\rangle$

$s=15.965$, $\sqrt{s}=3.99562$, $|\vec{P}|=1.88771$, $|\vec{P}'|=1.33879$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=5.83465$, $\phi_\gamma=1.56232$
 $E_\gamma=0.511073$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.08424$
 $Q^2=13.968$, $x_B=0.978806$, $t=-10.1864$, $W^2=1.18229$, $y=0.945994$

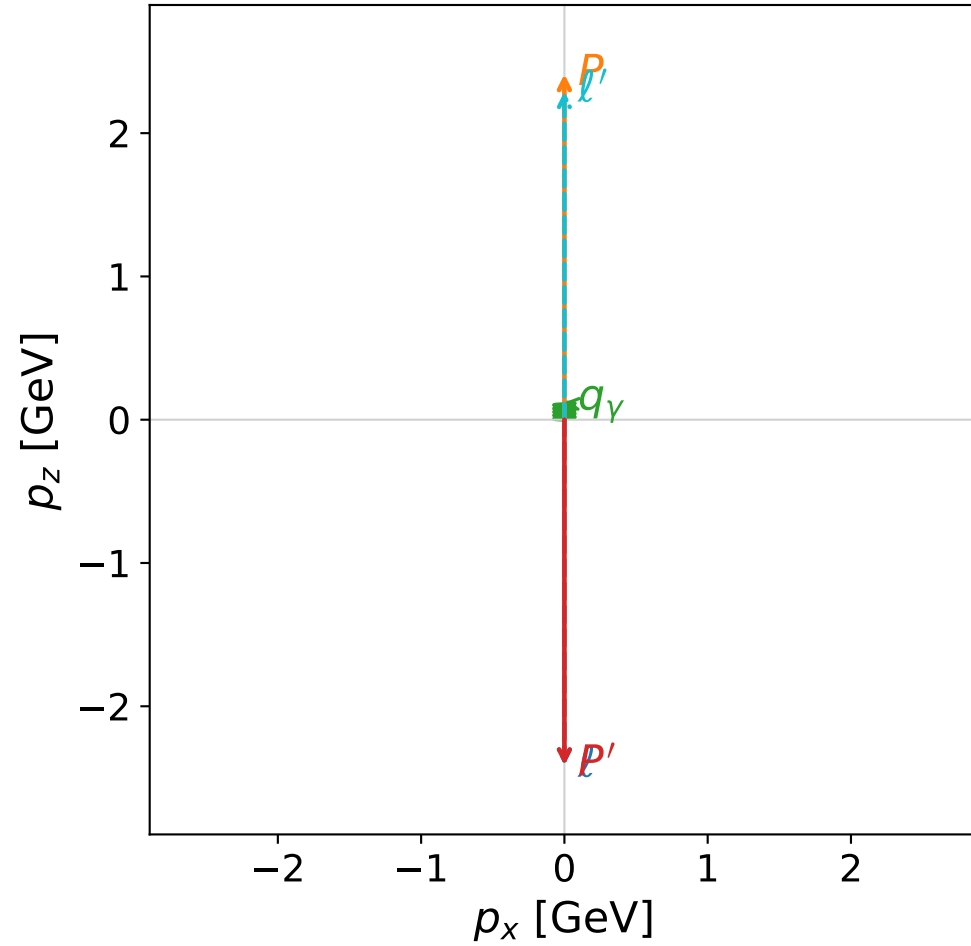
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=1.8877$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-1.8877$
 P $E=2.1079$ $p_x=0.0000$ $p_y=0.0000$ $p_z=1.8877$
 ℓ' $E=1.8499$ $p_x=-0.0000$ $p_y=0.0000$ $p_z=1.8499$
 P' $E=1.6347$ $p_x=0.0000$ $p_y=-0.0000$ $p_z=-1.3388$
 q_γ $E=0.5111$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.5111$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=7.638\text{e+}22$; $I_h=+2.098\text{e-}14$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=2.100\text{e-}29$; $I_{TL}=-5.125\text{e-}19$

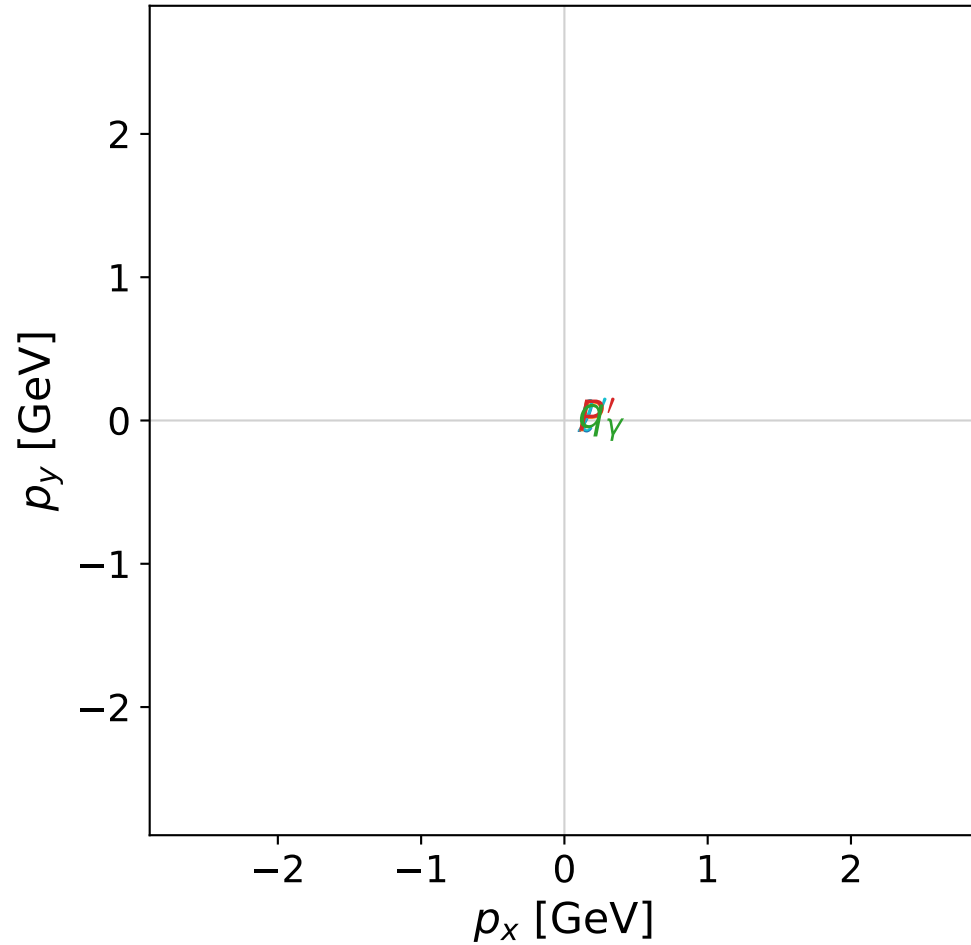


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



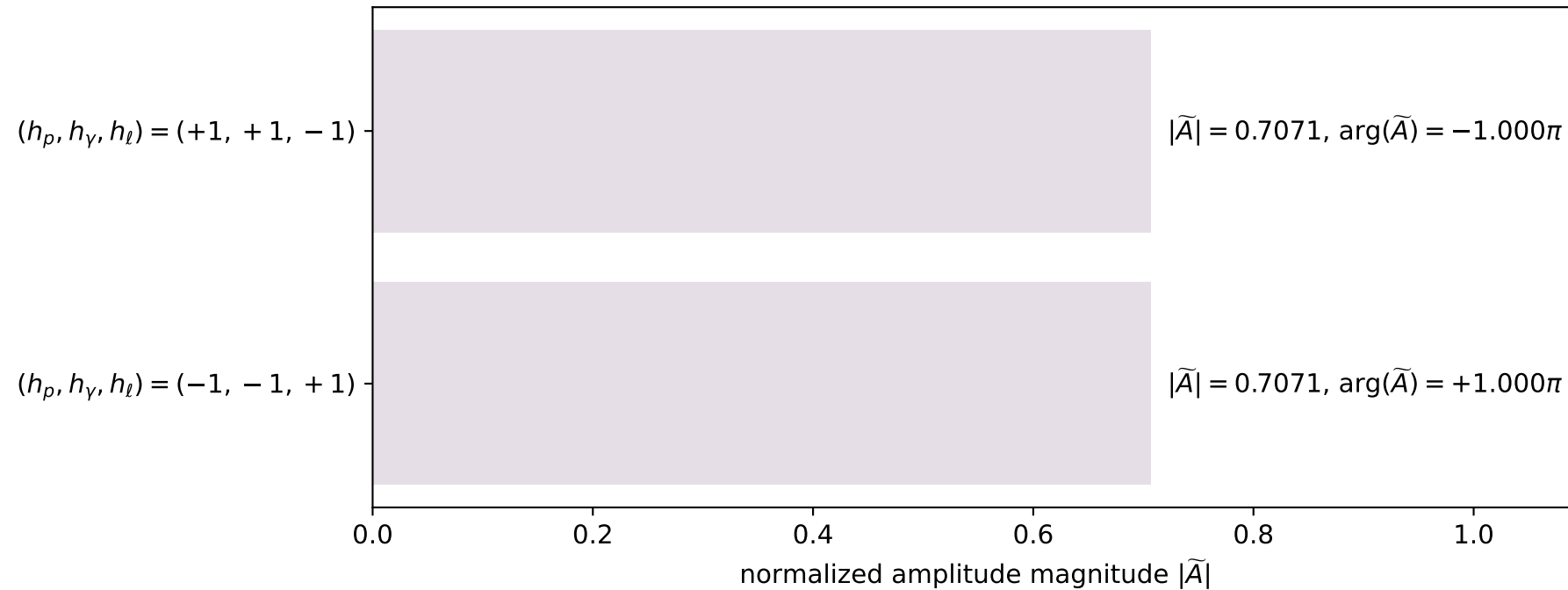
dghz_mixing_angles_polarization_03_local_minimum_134
 $d_{\text{GHZ}}=5.41283\text{e-}08$
 region: polarization_cluster_03_local_minimum_134
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0134
 lepton mixing angle: $\alpha_e=2.9154679$ rad
 incoming lepton: $-0.974543|+\rangle +0.224203|-\rangle$
 proton mixing angle: $\alpha_p=1.3446609$ rad
 incoming proton: $0.224213|+\rangle +0.97454|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.41202$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=2.5701$, $\phi_\gamma=3.68393$
 $E_\gamma=0.120596$, $\theta_{l'}=0$, $\phi_{l'}=5.71169$
 $Q^2=22.1078$, $x_B=0.948273$, $t=-23.2713$, $W^2=2.0858$, $y=0.966566$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.2914$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.2914$
 P' $E= 2.5880$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 q_γ $E= 0.1206$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.1206$

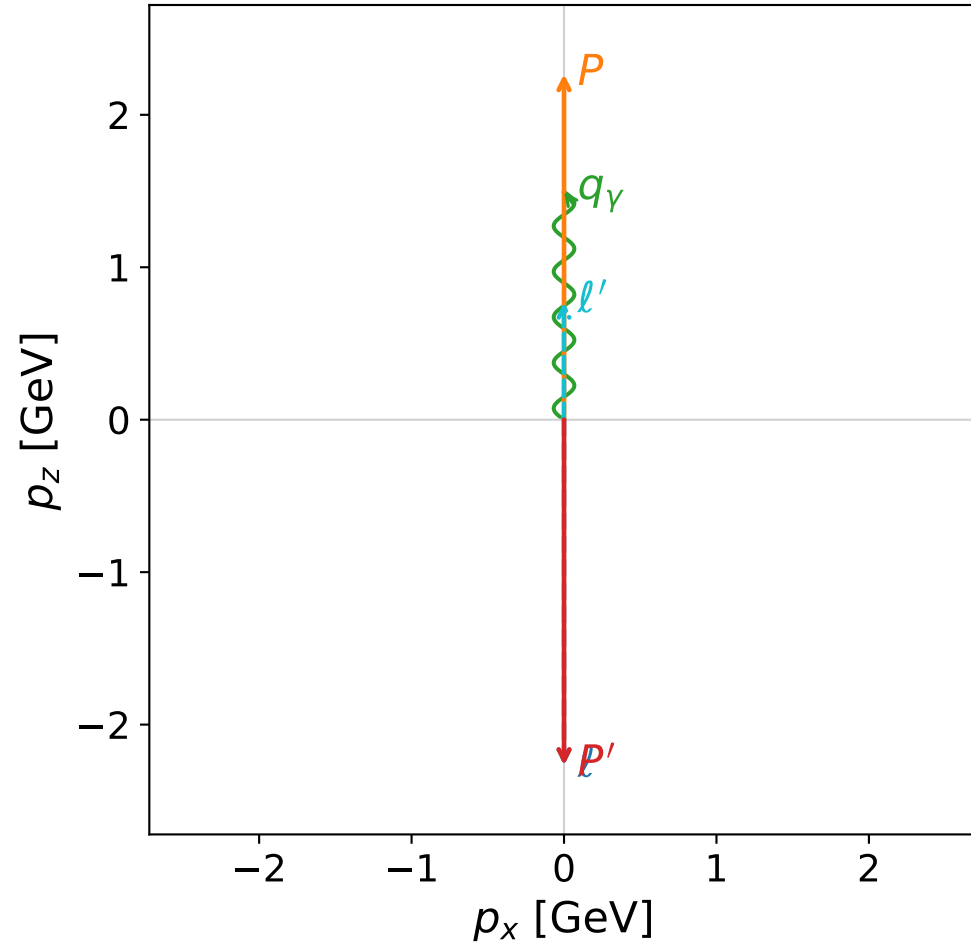
GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=3.925\text{e+}21$; $I_h=-2.607\text{e-}14$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.780\text{e-}27$; $I_{TL}=-2.585\text{e-}14$

Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

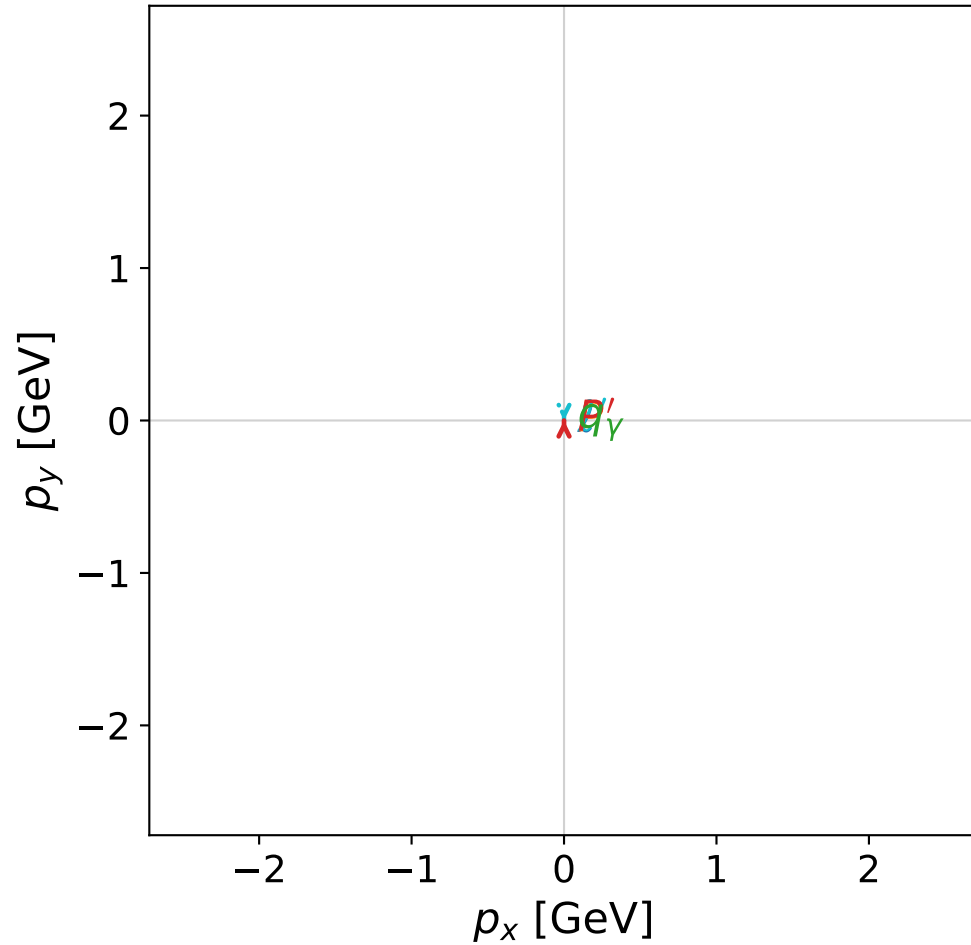


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

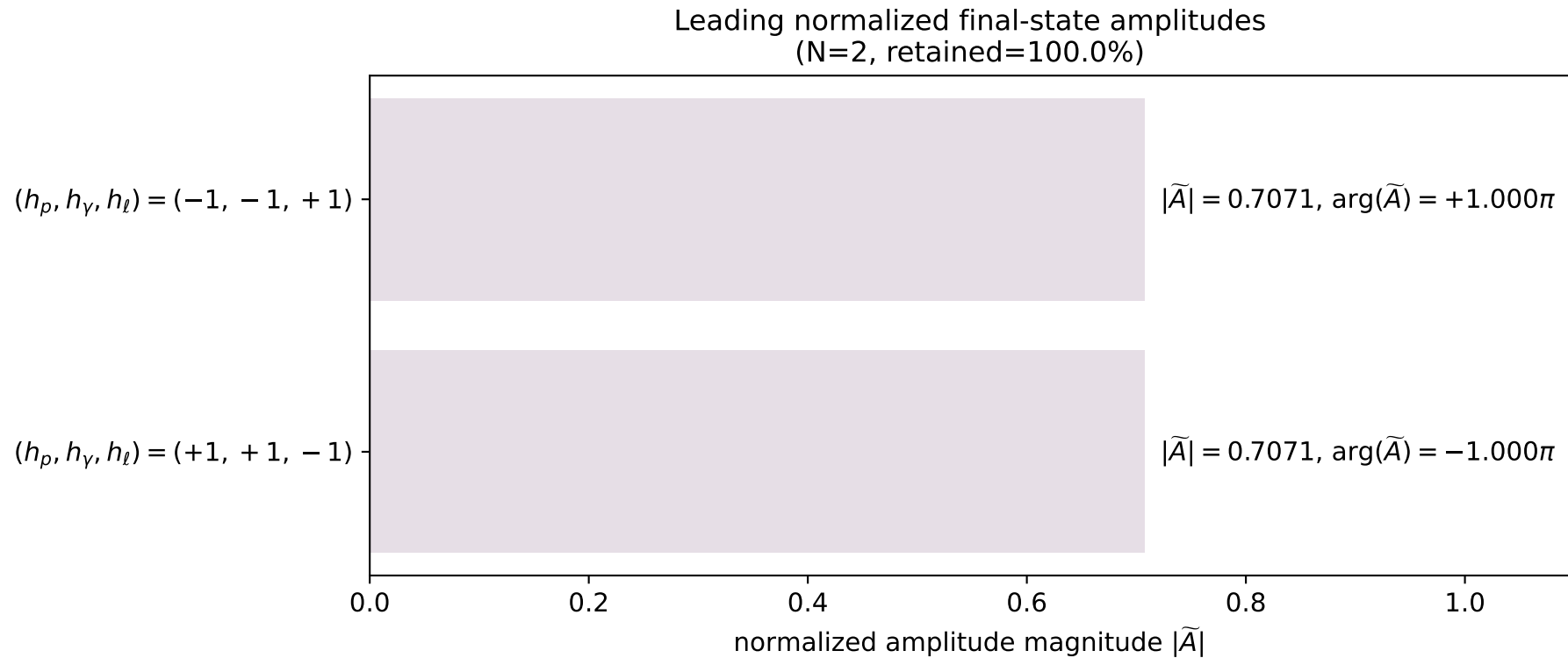


dghz_mixing_angles_polarization_03_local_minimum_143
 $d_{\text{GHZ}}=7.74877\text{e-}08$
 region: polarization_cluster_03_local_minimum_143
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0143
 lepton mixing angle: $\alpha_e=2.2187554$ rad
 incoming lepton: $-0.60356|+\rangle +0.797317|-\rangle$
 proton mixing angle: $\alpha_p=0.64796548$ rad
 incoming proton: $0.797313|+\rangle +0.603566|-\rangle$

$s=22.2813$, $\sqrt{s}=4.7203$, $|\vec{P}|=2.26695$, $|\vec{P}'|=2.26695$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=1.65437$, $\phi_\gamma=3.92312$
 $E_\gamma=1.49471$, $\theta_{l'}=0$, $\phi_{l'}=4.79596$
 $Q^2=7.00254$, $x_B=0.331661$, $t=-20.5563$, $W^2=14.9908$, $y=0.986548$

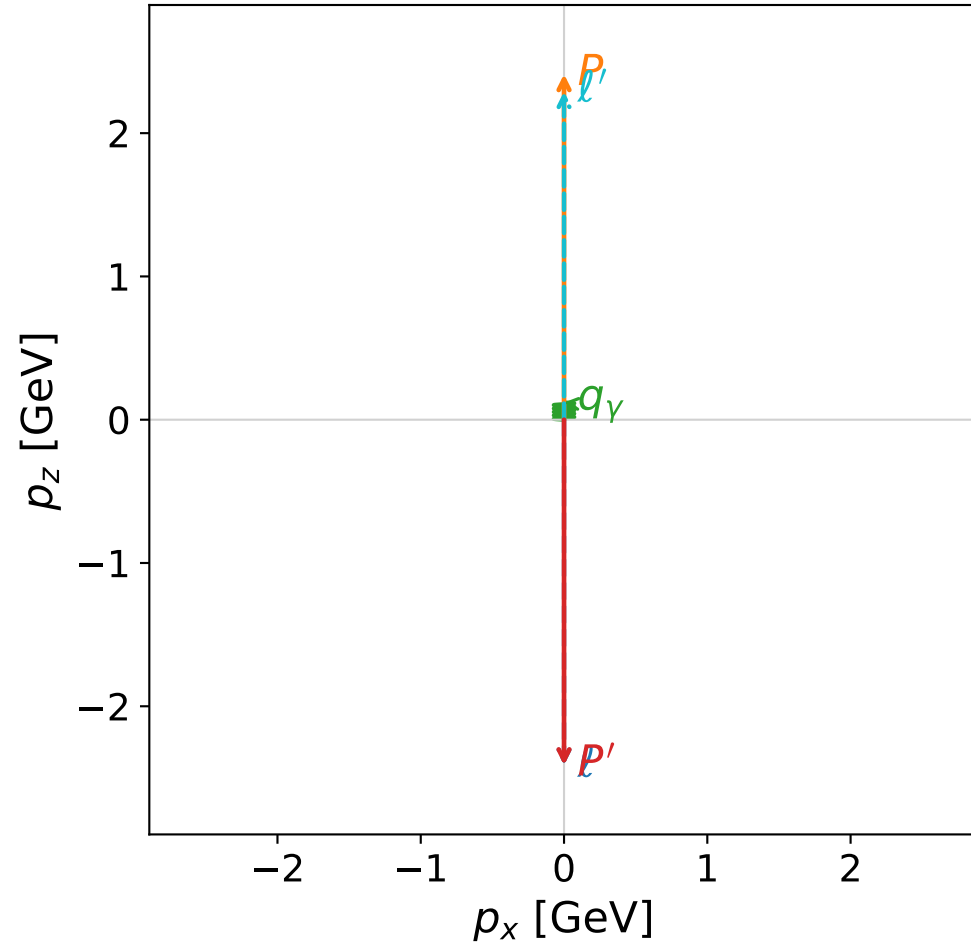
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2670$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2670$
 P $E= 2.4533$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2670$
 l' $E= 0.7722$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.7722$
 P' $E= 2.4533$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.2670$
 q_γ $E= 1.4947$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.4947$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.320\text{e}+24$; $I_h=-9.443\text{e-}15$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=4.813\text{e-}29$; $I_{TL}=-9.443\text{e-}15$

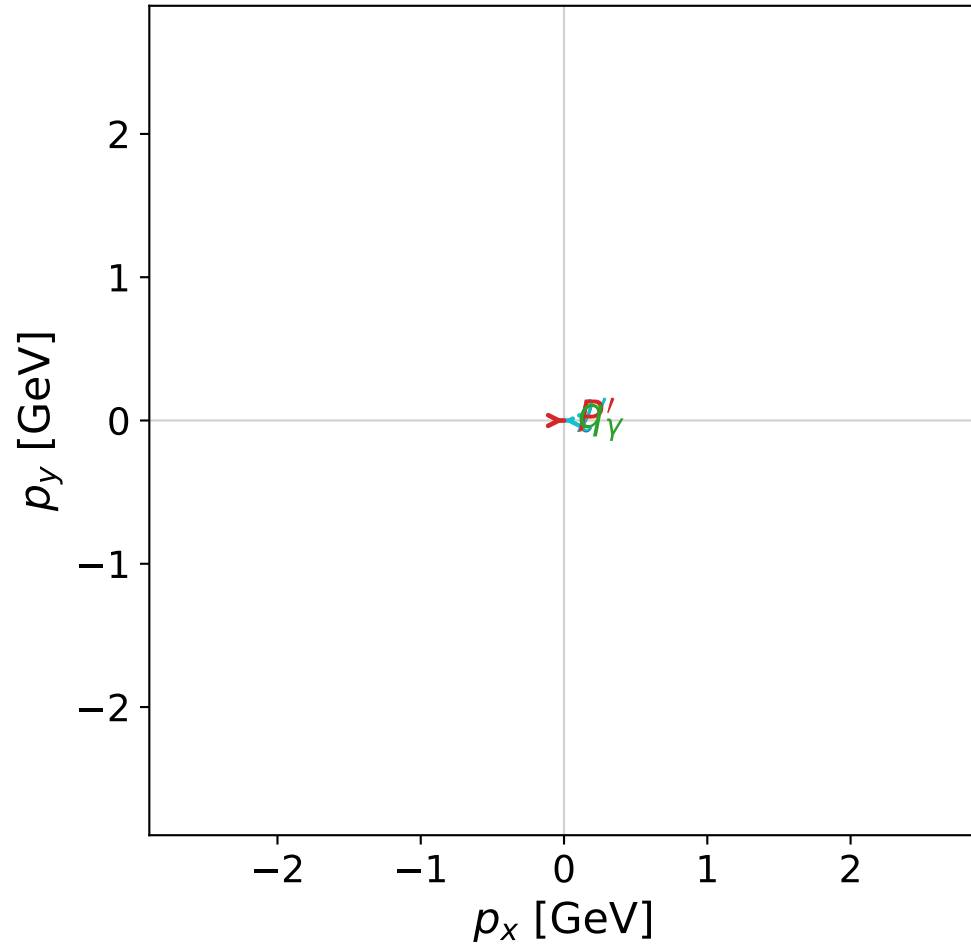


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

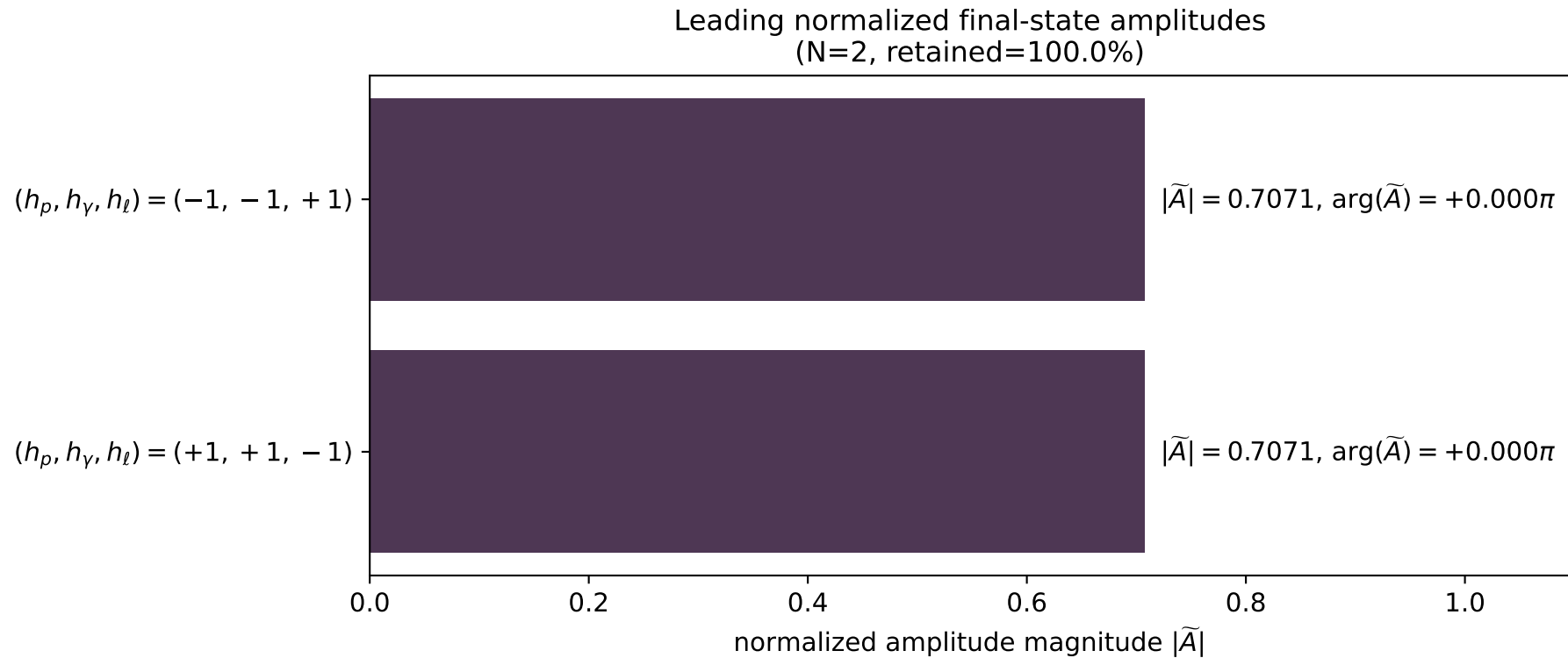


dghz_mixing_angles_polarization_03_local_minimum_154
 $d_{\text{GHZ}}=9.73504\text{e-}08$
 region: polarization_cluster_03_local_minimum_154
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0154
 lepton mixing angle: $\alpha_e=3.0190559$ rad
 incoming lepton: $-0.992502|+\rangle +0.12223|-\rangle$
 proton mixing angle: $\alpha_p=1.4482597$ rad
 incoming proton: $0.12223|+\rangle +0.992502|-\rangle$

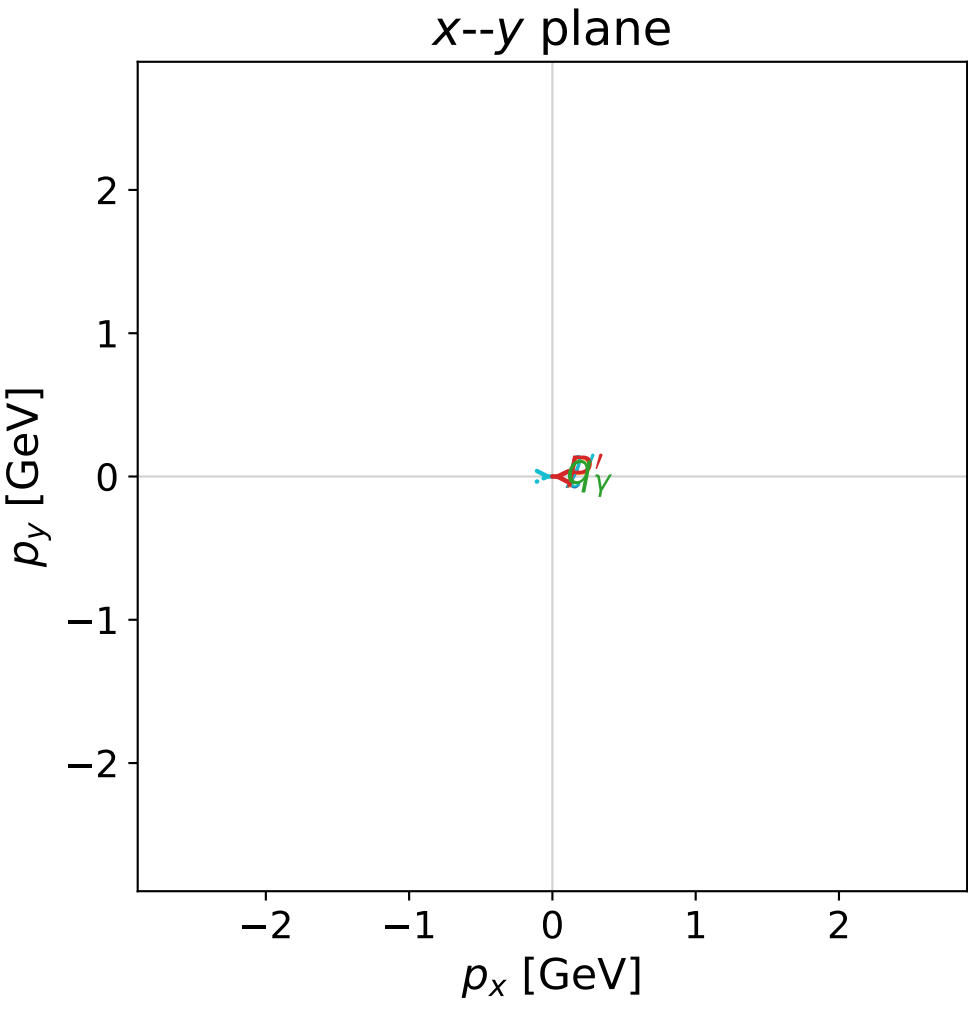
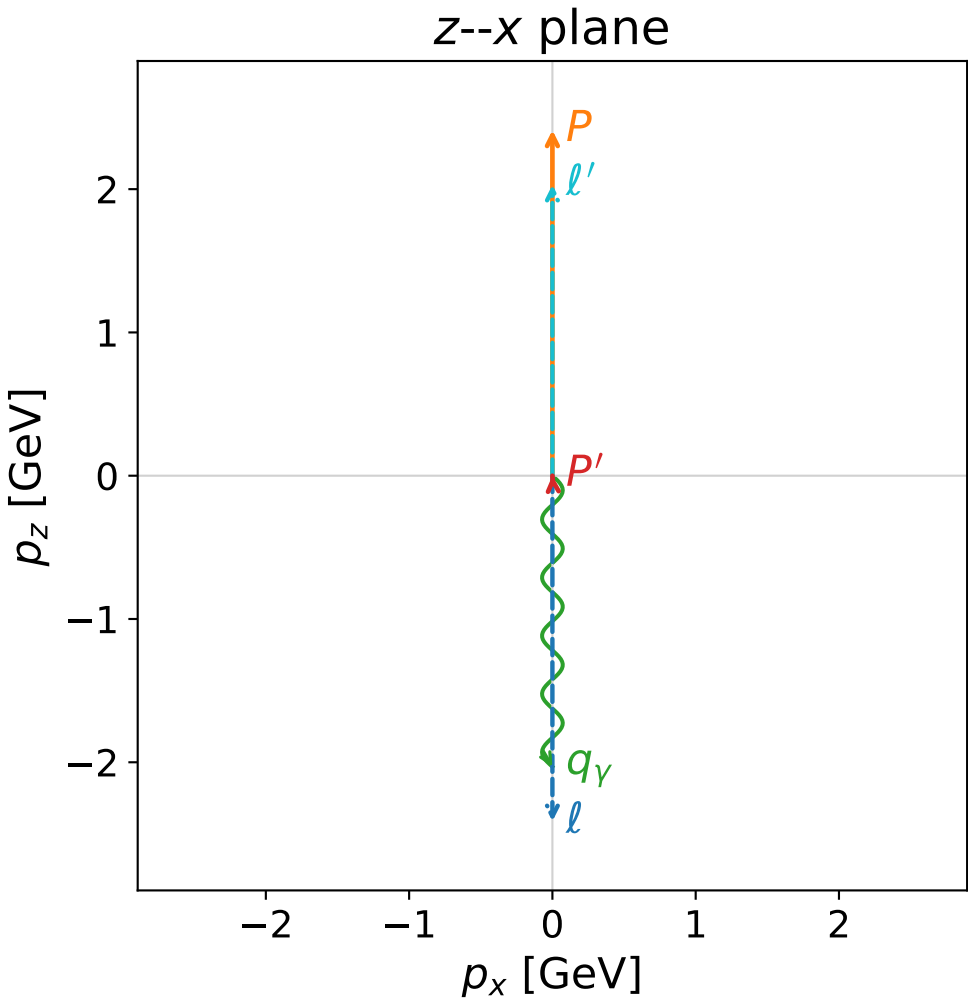
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.41202$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=0.116076$, $\phi_\gamma=6.10382$
 $E_\gamma=0.120666$, $\theta_{l'}=0$, $\phi_{l'}=3.25767$
 $Q^2=22.1071$, $x_B=0.948242$, $t=-23.2713$, $W^2=2.08651$, $y=0.966567$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.2913$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.2913$
 P' $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 q_γ $E= 0.1207$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.1207$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=2.735\text{e}+21$; $I_h=+1.876\text{e-}12$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=2.989\text{e-}23$; $I_{TL}=+1.876\text{e-}12$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

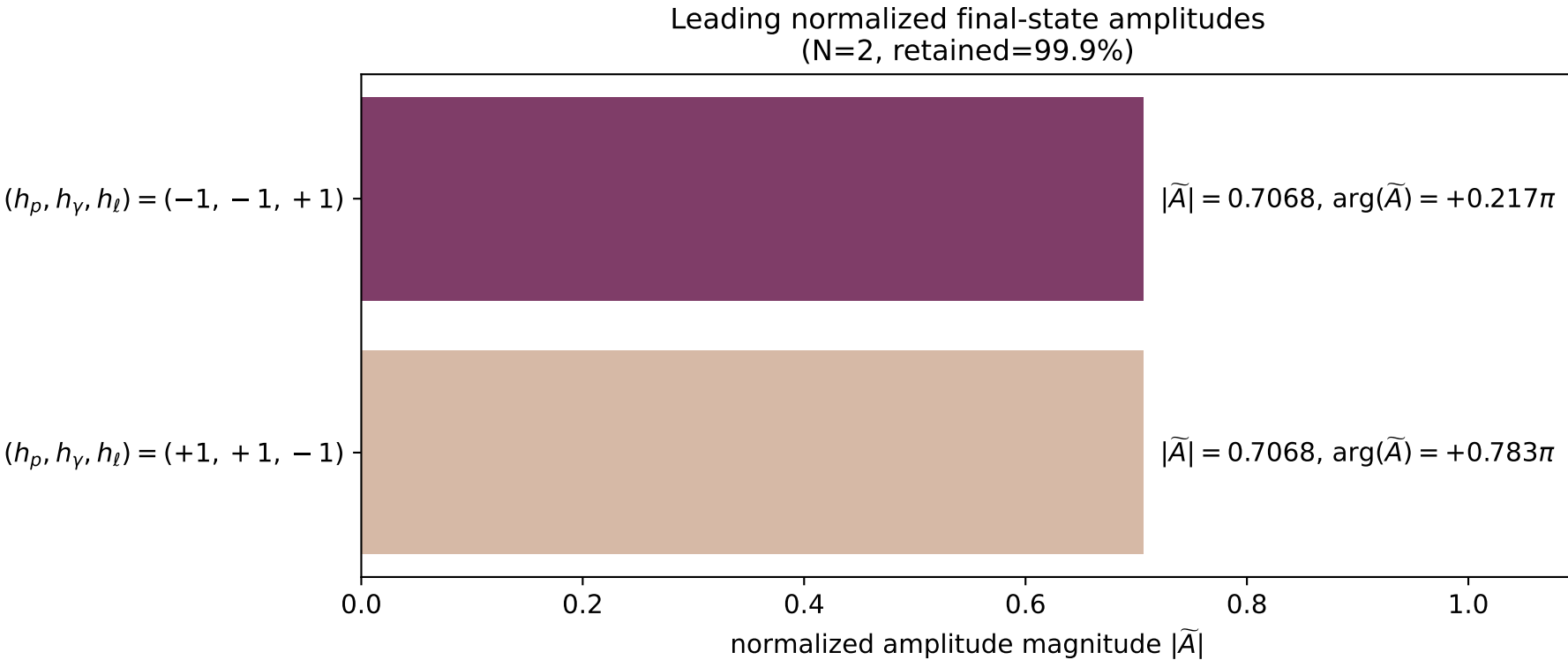


dghz_mixing_angles_polarization_03_local_minimum_156
d_{\rm GHZ}=9.98211e-08
region: polarization_cluster_03_local_minimum_156
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0156
lepton mixing angle: alpha_e=3.0075642 rad
incoming lepton: -0.991032|+> +0.133628|->
proton mixing angle: alpha_p=0.13401798 rad
incoming proton: 0.991033|+> +0.133617|->

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.000632829$
 $\theta_{P'}=0.0587284$, $\theta_Y=3.14159$, $\phi_{P'}=3.12895$, $\phi_Y=3.82447$
 $E_Y=2.03132$, $\theta_{\ell'}=1.82912\text{e-}05$, $\phi_{\ell'}=6.27054$
 $Q^2=19.5922$, $x_B=0.837076$, $t=-3.09232$, $W^2=4.69316$, $y=0.97037$

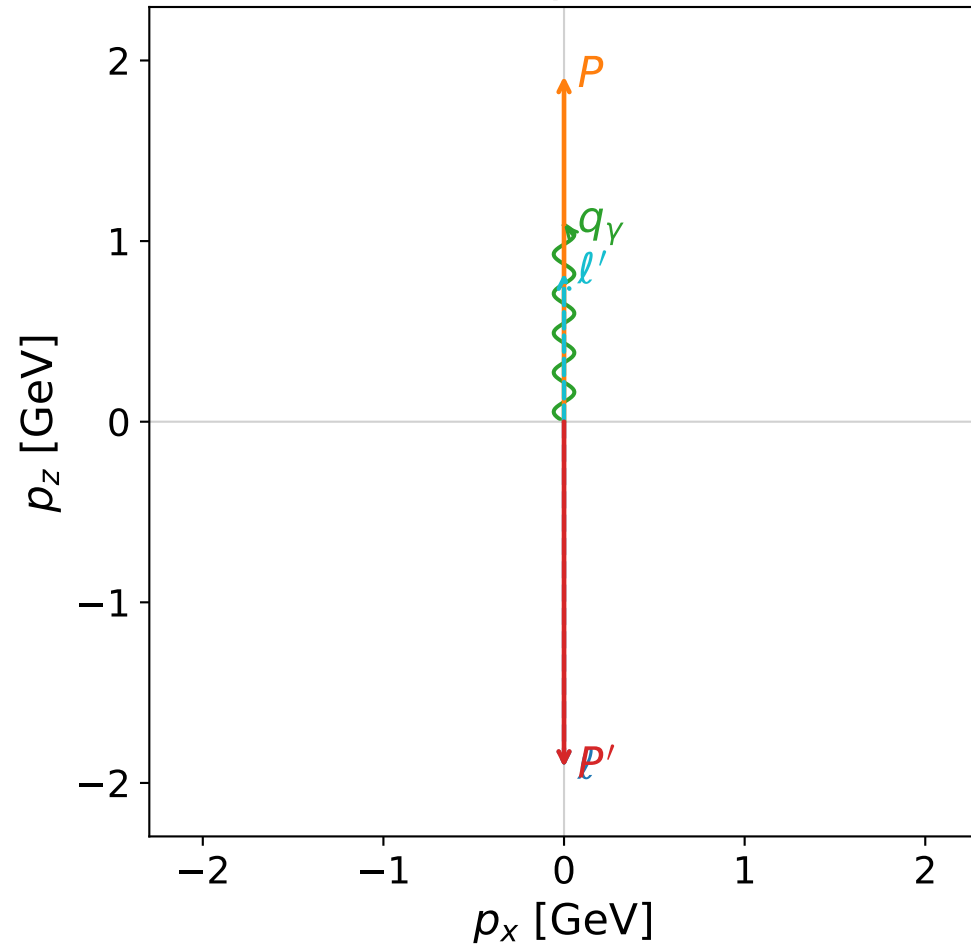
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0307$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.0307$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0006$
 q_Y $E= 2.0313$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.0313$

GHZ mechanism: helicity flipping (full H G C)
helicity powers: flip 100.000%, conserve 0.000%
R_flip/conserve=1.350e+16; I_h=+8.376e-09
virtual-photon powers: T 100.000%, L 0.000%
R_L/T=1.783e-11; I_TL=+5.758e-06

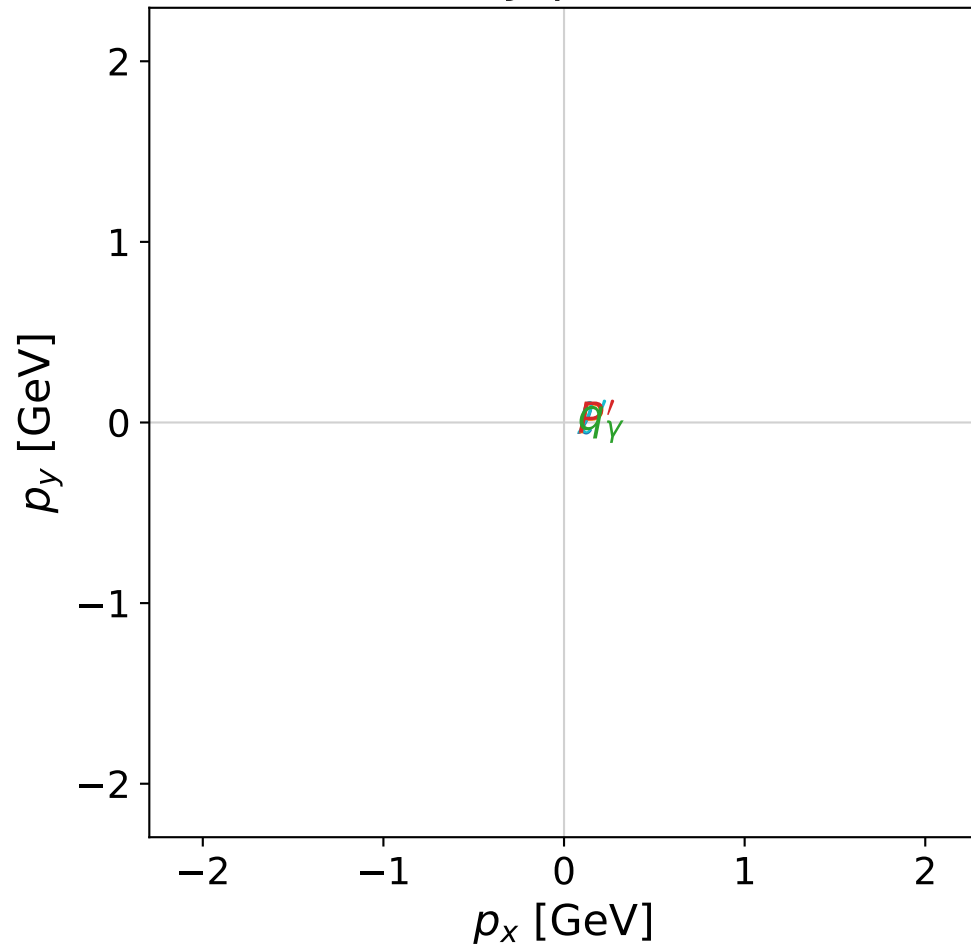


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

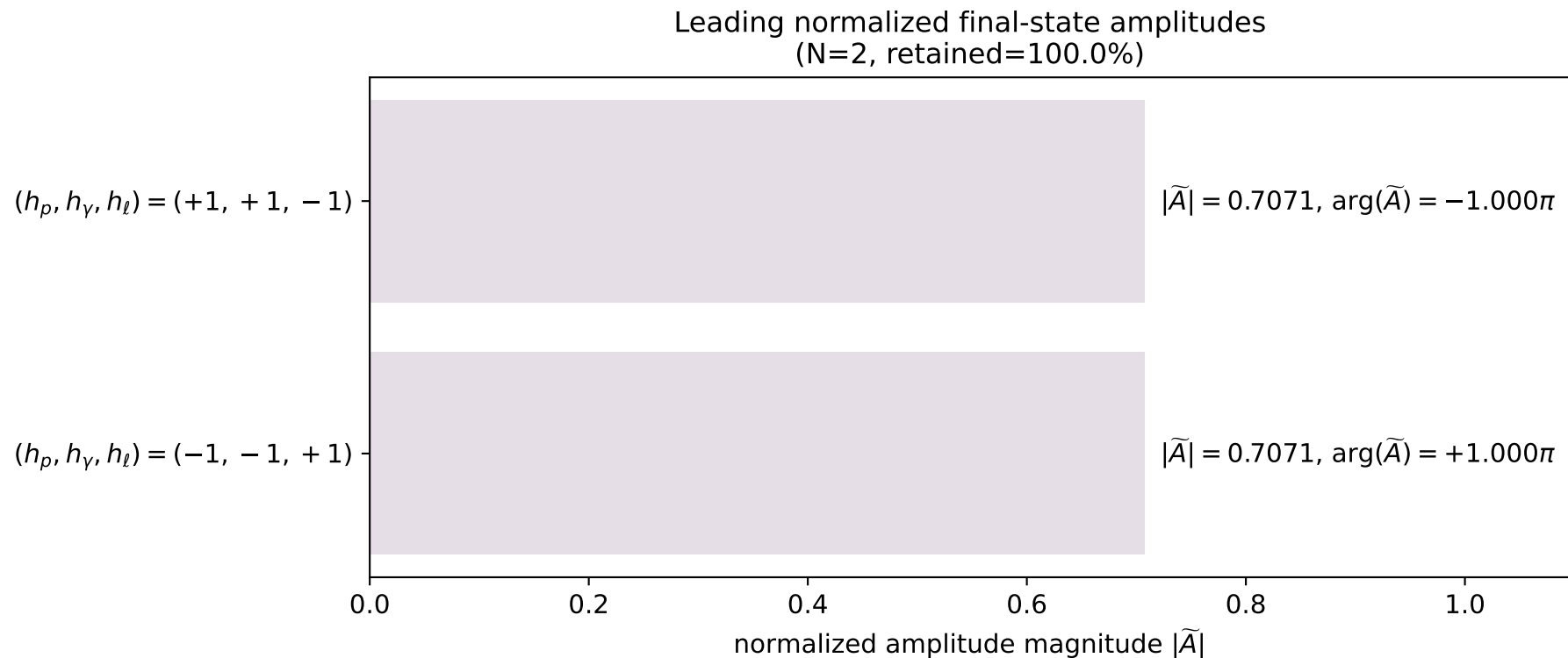


dghz_mixing_angles_polarization_03_local_minimum_158
 $d_{\text{GHZ}}=1.03904\text{e-}07$
 region: polarization_cluster_03_local_minimum_158
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0158
 lepton mixing angle: $\alpha_e=2.0331734$ rad
 incoming lepton: $-0.446077|+\rangle +0.894995|-\rangle$
 proton mixing angle: $\alpha_p=0.46236846$ rad
 incoming proton: $0.894999|+\rangle +0.446069|-\rangle$

$s=16.3566$, $\sqrt{s}=4.04434$, $|\vec{P}|=1.91339$, $|\vec{P}'|=1.91339$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=2.40943$, $\phi_\gamma=3.29633$
 $E_\gamma=1.09092$, $\theta_{l'}=0$, $\phi_{l'}=5.55102$
 $Q^2=6.29483$, $x_B=0.416353$, $t=-14.6443$, $W^2=9.70397$, $y=0.976878$

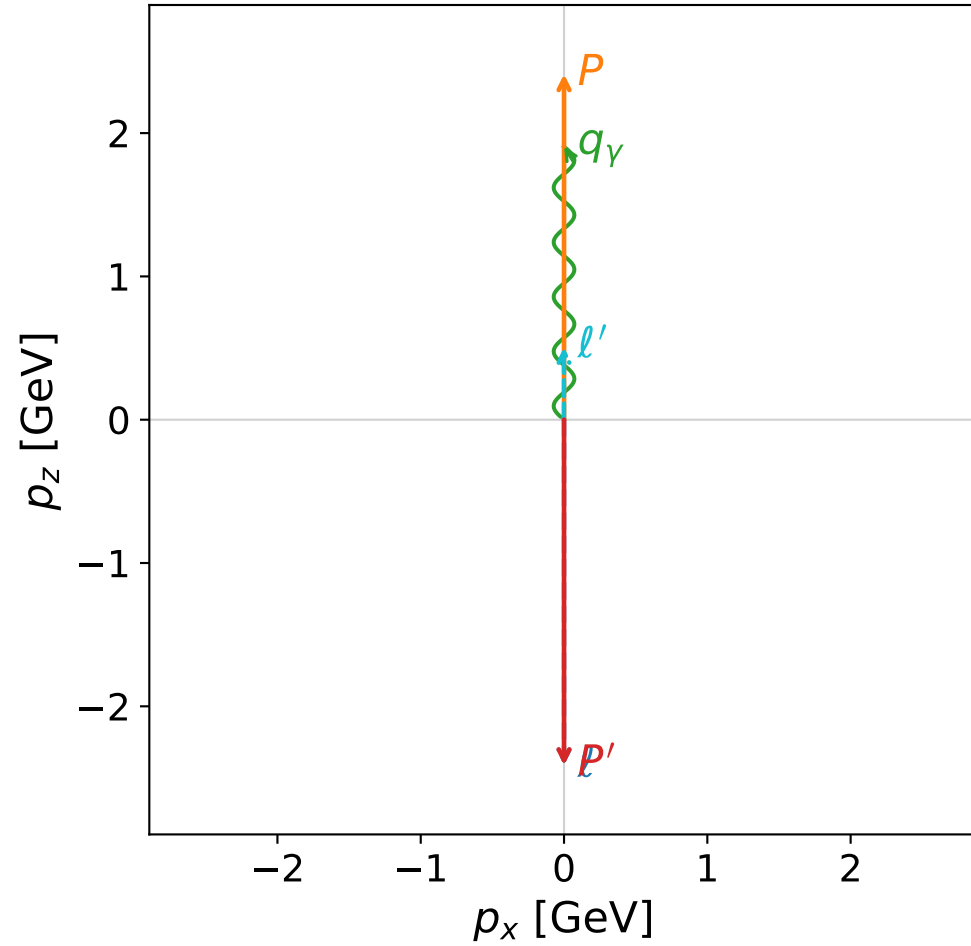
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.9134$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9134$
 P $E= 2.1309$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9134$
 l' $E= 0.8225$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.8225$
 P' $E= 2.1309$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.9134$
 q_γ $E= 1.0909$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.0909$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.306\text{e+}24$; $I_h=-3.532\text{e-}15$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=9.781\text{e-}30$; $I_{TL}=-3.534\text{e-}15$

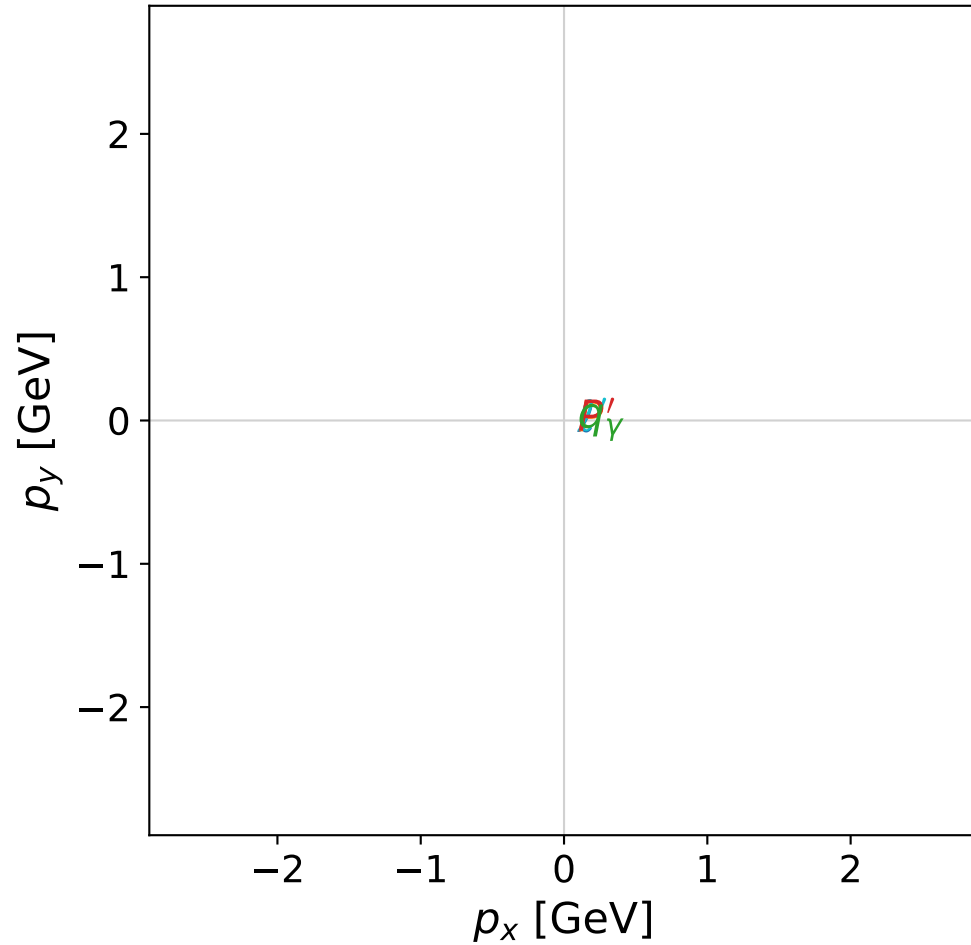


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

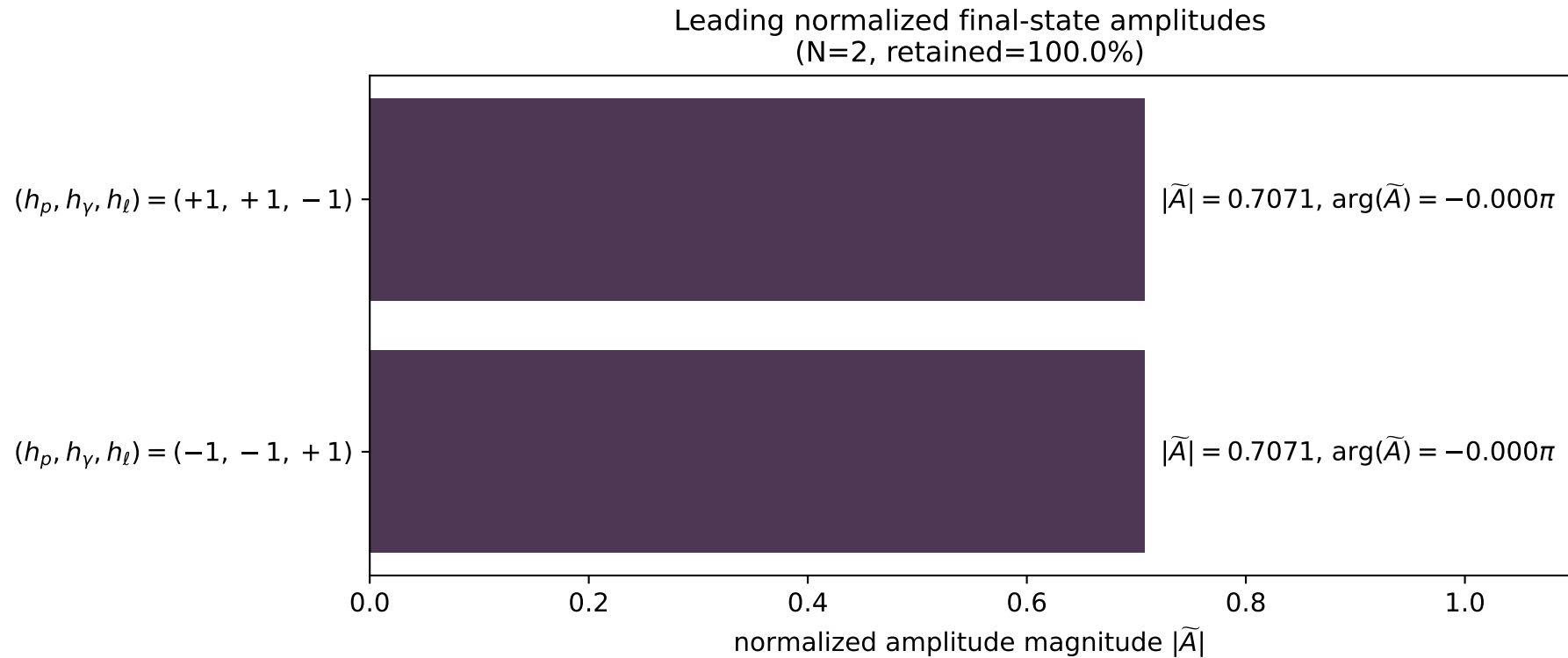


dghz_mixing_angles_polarization_03_local_minimum_161
 $d_{\text{GHZ}}=1.13122\text{e-}07$
 region: polarization_cluster_03_local_minimum_161
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0161
 lepton mixing angle: $\alpha_e=2.5224319$ rad
 incoming lepton: $-0.814366|+\rangle +0.580352|-\rangle$
 proton mixing angle: $\alpha_p=0.95162257$ rad
 incoming proton: $0.580362|+\rangle +0.814358|-\rangle$

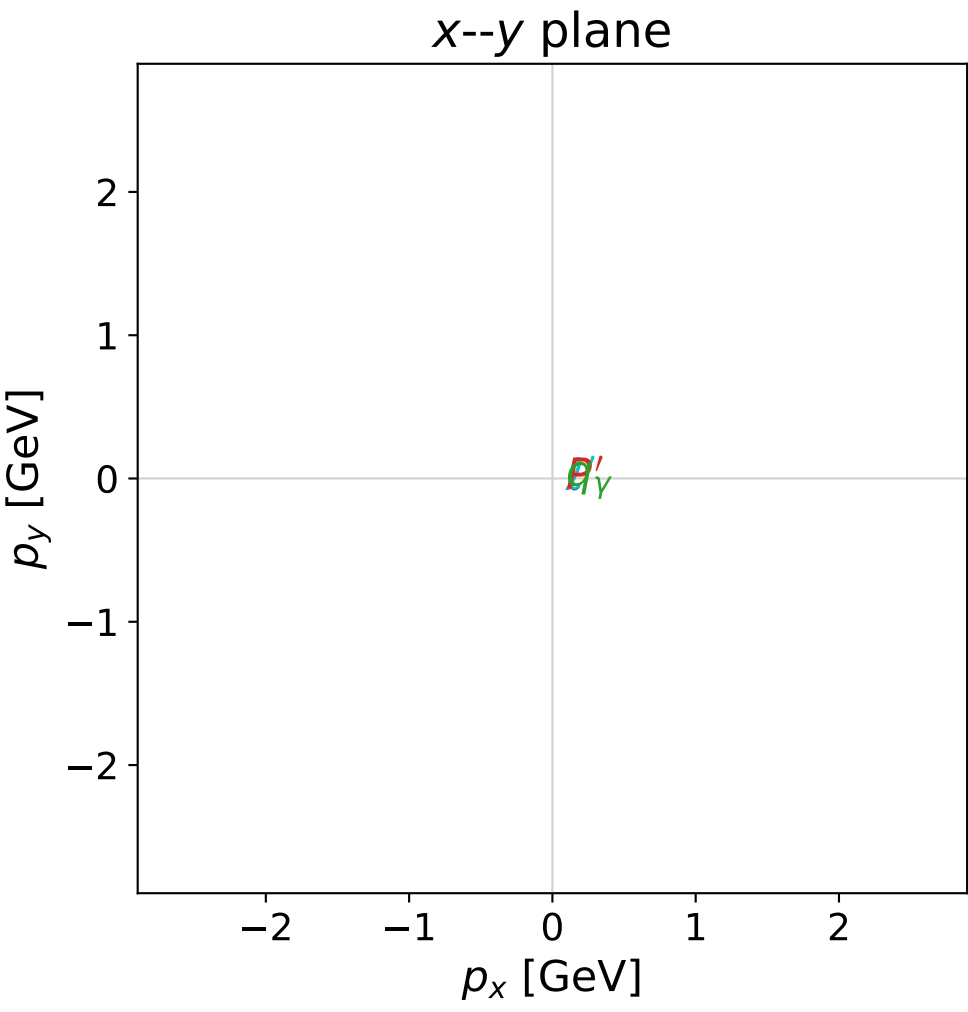
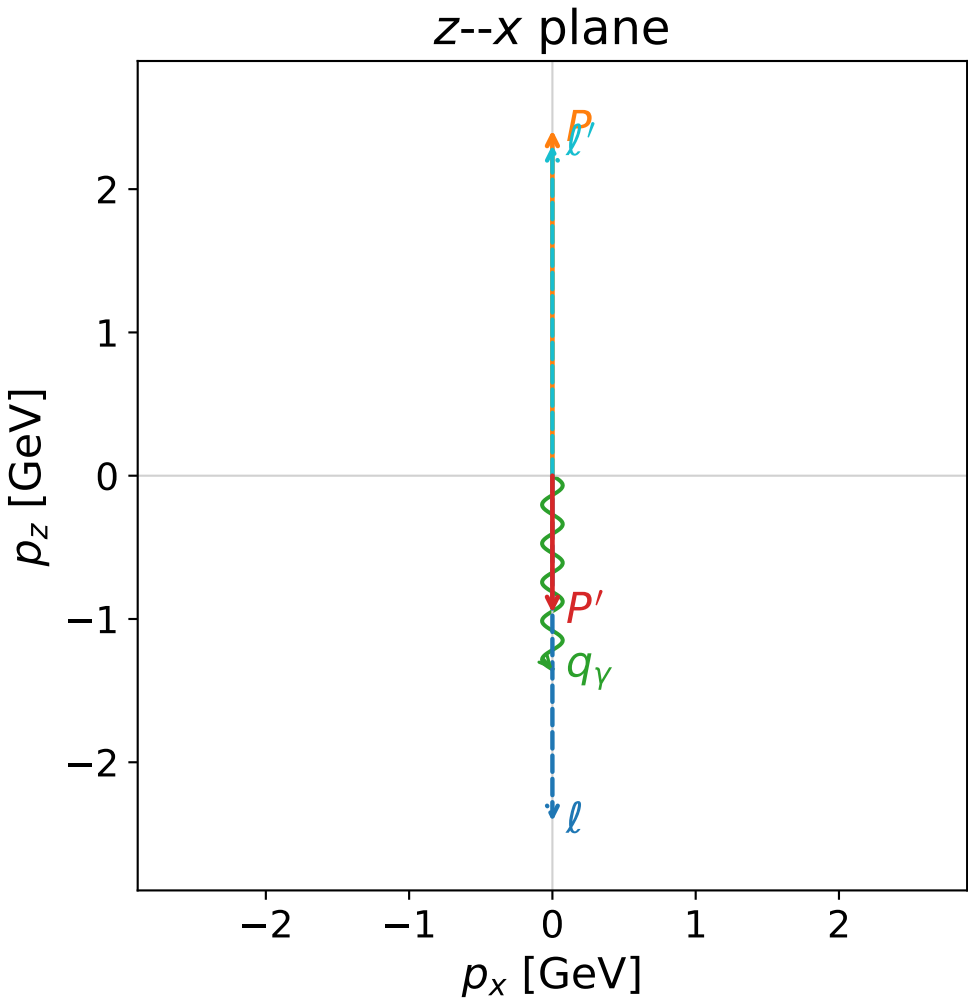
$s=24.9928$, $\sqrt{s}=4.99928$, $|\vec{P}|=2.41164$, $|\vec{P}'|=2.41164$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=5.60969$, $\phi_\gamma=5.70666$
 $E_\gamma=1.90533$, $\theta_{\ell'}=0$, $\phi_{\ell'}=2.4681$
 $Q^2=4.88418$, $x_B=0.204062$, $t=-23.2641$, $W^2=19.9304$, $y=0.992609$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4116$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4116$
 P $E= 2.5876$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4116$
 ℓ' $E= 0.5063$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.5063$
 P' $E= 2.5876$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.4116$
 q_γ $E= 1.9053$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.9053$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.788\text{e+}24$; $I_h=+1.292\text{e-}14$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=9.344\text{e-}29$; $I_{TL}=+1.292\text{e-}14$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

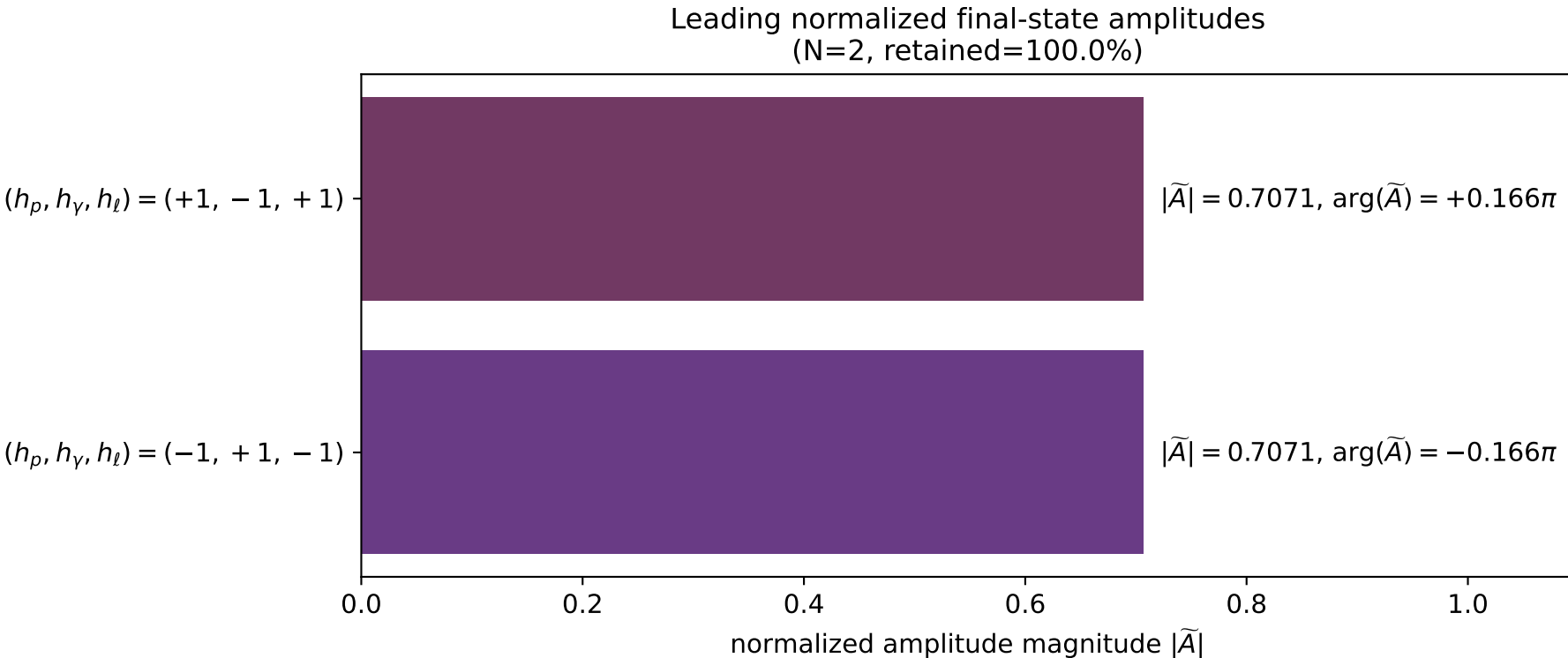


dghz_mixing_angles_polarization_03_local_minimum_174
d_{\rm GHZ}=2.67083e-07
region: polarization_cluster_03_local_minimum_174
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0174
lepton mixing angle: alpha_e=3.0976361 rad
incoming lepton: -0.999034|+> +0.0439424|->
proton mixing angle: alpha_p=0.043954198 rad
incoming proton: 0.999034|+> +0.04394|->

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.956774$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.14456$, $\phi_\gamma=3.66323$
 $E_\gamma=1.35168$, $\theta_{\ell'}=0$, $\phi_{\ell'}=0.307656$
 $Q^2=22.2721$, $x_B=0.955566$, $t=-9.79096$, $W^2=1.9155$, $y=0.966317$

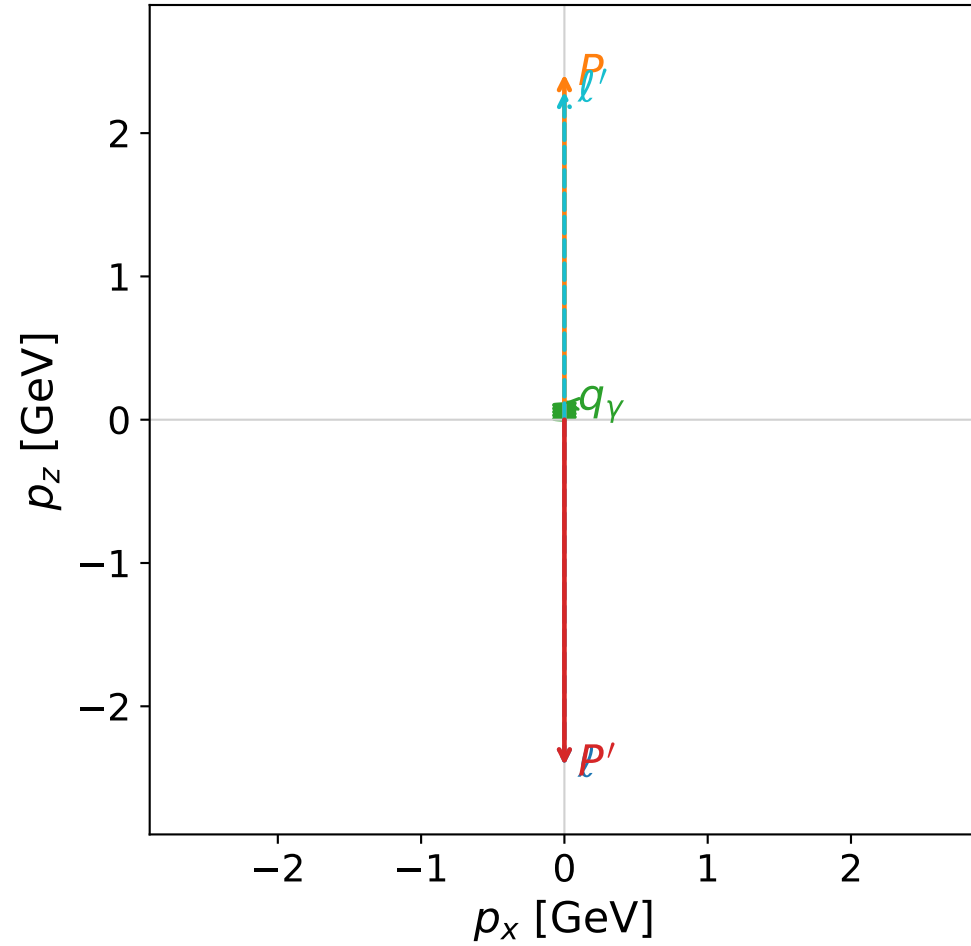
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.3085$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3085$
 P' $E= 1.3399$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.9568$
 q_γ $E= 1.3517$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.3517$

GHZ mechanism: helicity flipping (full H G C)
helicity powers: flip 100.000%, conserve 0.000%
R_flip/conserve=3.047e+21; I_h=-8.922e-12
virtual-photon powers: T 100.000%, L 0.000%
R_L/T=1.656e-27; I_TL=-3.858e-17

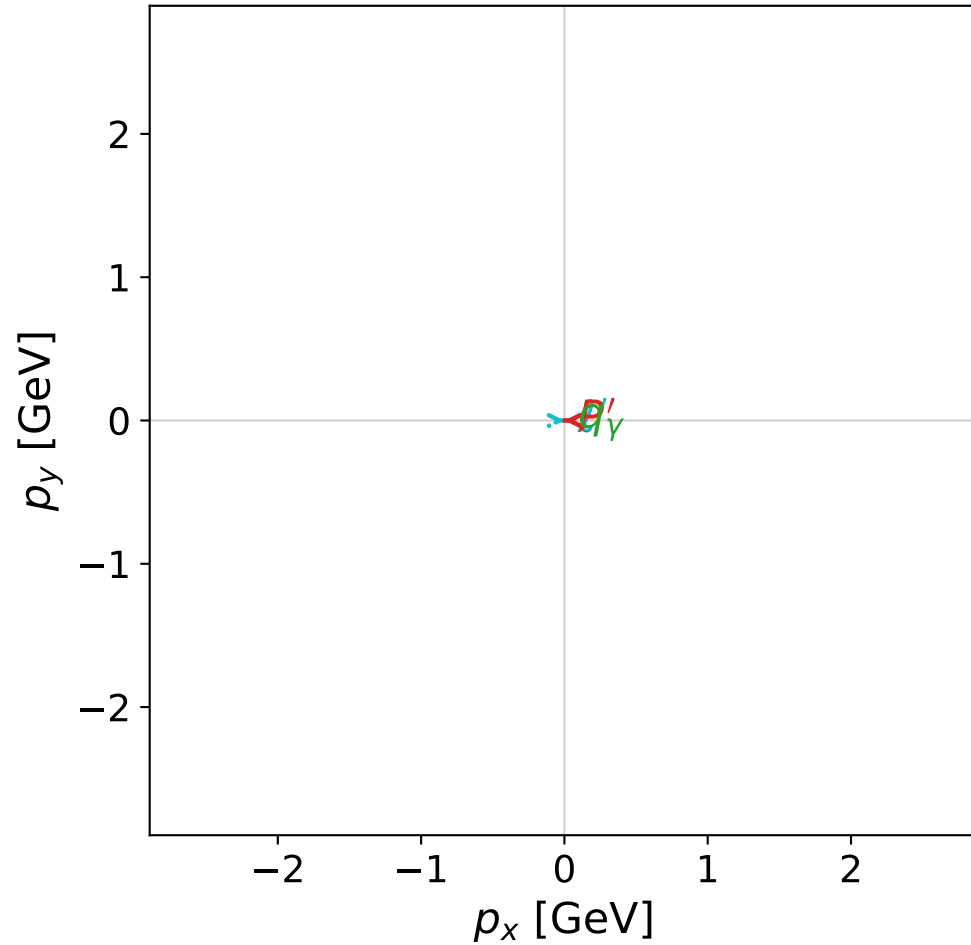


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



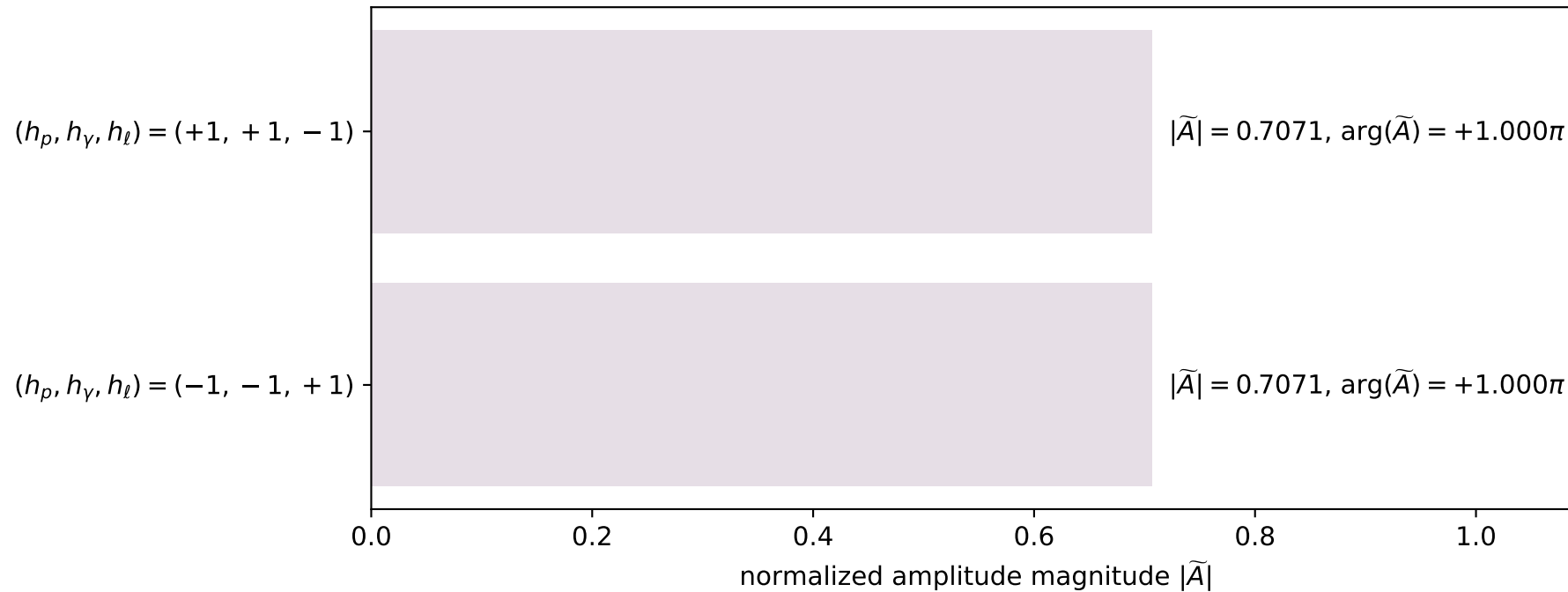
dghz_mixing_angles_polarization_03_local_minimum_178
 $d_{\text{GHZ}}=4.23054\text{e-}07$
 region: polarization_cluster_03_local_minimum_178
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0178
 lepton mixing angle: $\alpha_e=3.1136594$ rad
 incoming lepton: $-0.99961|+\rangle +0.0279296|-\rangle$
 proton mixing angle: $\alpha_p=1.5428613$ rad
 incoming proton: $0.0279314|+\rangle +0.99961|-\rangle$

$s=24.9961$, $\sqrt{s}=4.99961$, $|\vec{P}|=2.41181$, $|\vec{P}'|=2.41181$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=3.46255$, $\phi_\gamma=3.37801$
 $E_\gamma=0.120586$, $\theta_{l'}=0$, $\phi_{l'}=0.320954$
 $Q^2=22.104$, $x_B=0.948272$, $t=-23.2674$, $W^2=2.08561$, $y=0.966561$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4118$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4118$
 P $E= 2.5878$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4118$
 ℓ' $E= 2.2912$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2912$
 P' $E= 2.5878$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.4118$
 q_γ $E= 0.1206$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.1206$

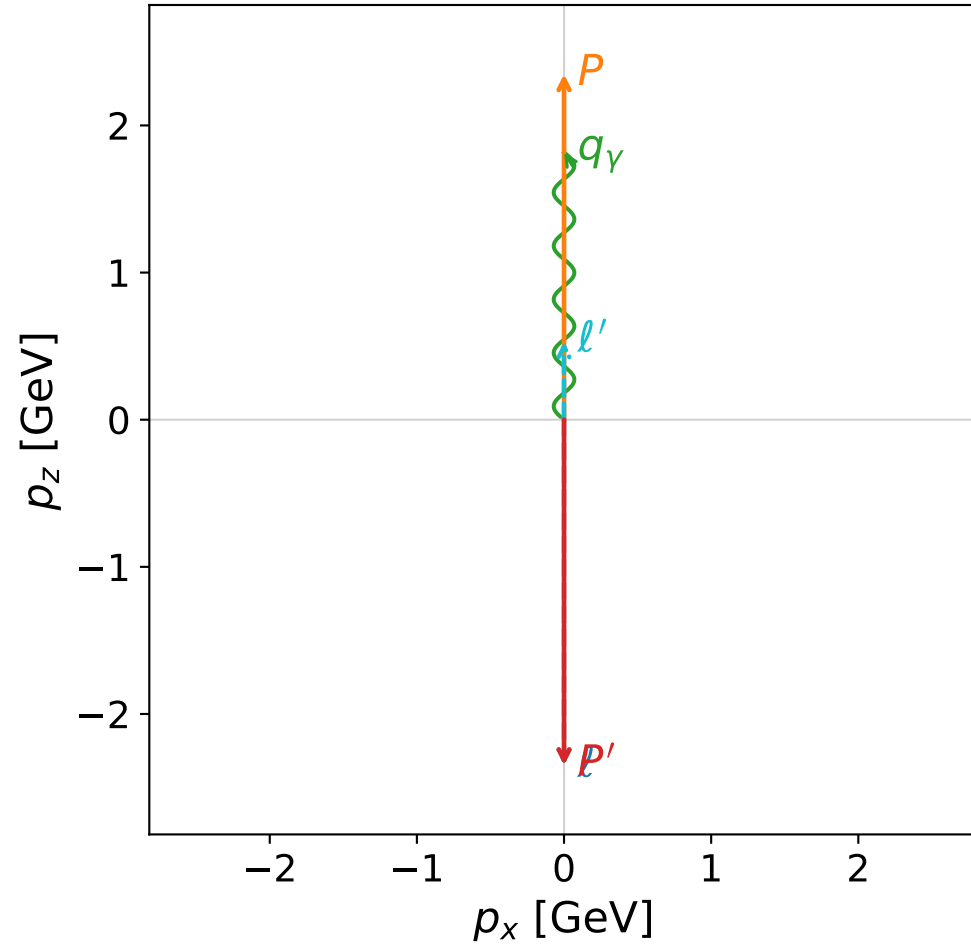
GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=3.203\text{e+}21$; $I_h=-2.550\text{e-}13$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.043\text{e-}23$; $I_{TL}=-2.529\text{e-}13$

Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

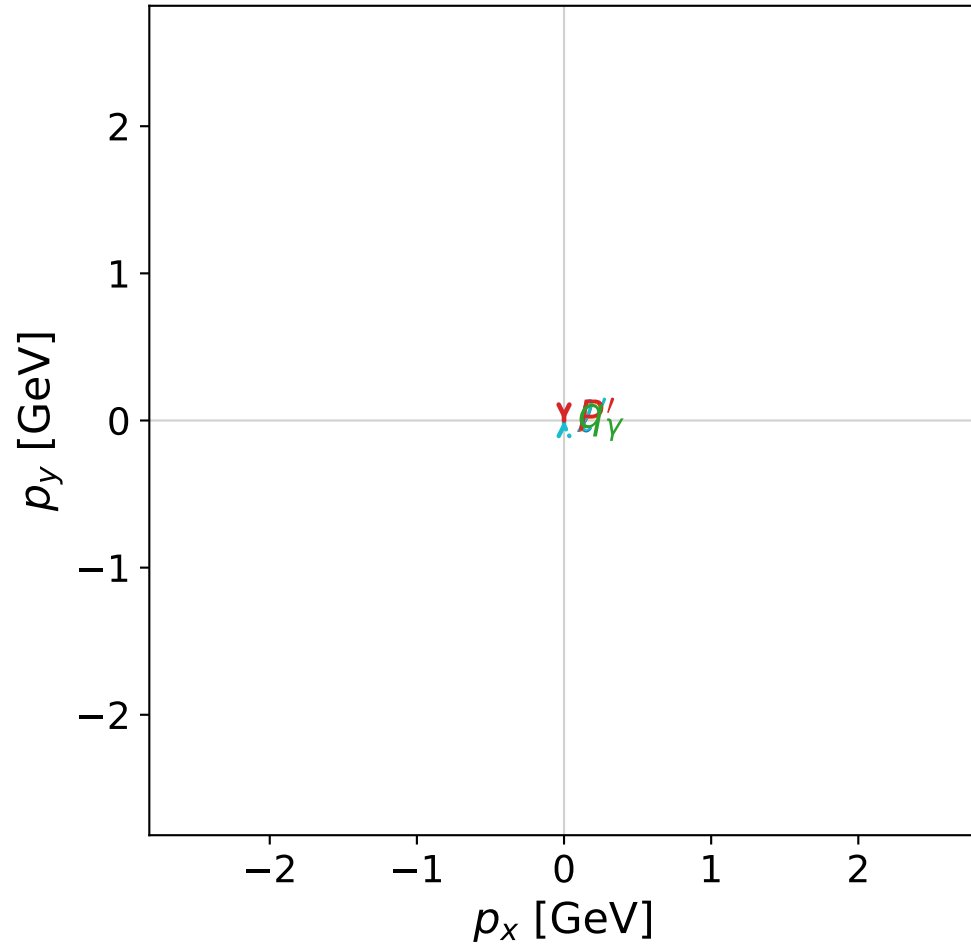


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



dghz_mixing_angles_polarization_03_local_minimum_181
 $d_{\text{GHZ}}=5.4582\text{e-}07$
 region: polarization_cluster_03_local_minimum_181
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0181
 lepton mixing angle: $\alpha_e=3.0452158$ rad
 incoming lepton: $-0.995359|+\rangle +0.0962277|-\rangle$
 proton mixing angle: $\alpha_p=1.4744086$ rad
 incoming proton: $0.0962386|+\rangle +0.995358|-\rangle$

$s=23.7939$, $\sqrt{s}=4.8779$, $|\vec{P}|=2.34876$, $|\vec{P}'|=2.34876$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=4.56992$, $\phi_\gamma=2.75915$
 $E_\gamma=1.81698$, $\theta_{\ell'}=0$, $\phi_{\ell'}=1.42833$
 $Q^2=4.99613$, $x_B=0.219879$, $t=-22.0667$, $W^2=18.6059$, $y=0.991628$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3488$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3488$
 P $E= 2.5291$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3488$
 ℓ' $E= 0.5318$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.5318$
 P' $E= 2.5291$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.3488$
 q_γ $E= 1.8170$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.8170$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.797\text{e+}24$; $I_h=+6.821\text{e-}16$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=6.341\text{e-}30$; $I_{TL}=+6.857\text{e-}16$

Leading normalized final-state amplitudes
(N=2, retained=100.0%)

